

Integrated Science

Basic 9

Complete BECE Preparation
Examination Papers & Marking Schemes

Beacon Educational Consult

2026 Edition

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Preface

This comprehensive Integrated Science textbook for Basic 9 has been carefully prepared to support Junior High School students in their preparation for the Basic Education Certificate Examination (BECE).

The book contains:

- Multiple choice questions with complete answer keys
- Theory questions with detailed marking schemes
- Curriculum-aligned content following the 2019 Standards-Based Curriculum
- Practice papers from BECE Mock examinations

Each question has been verified for accuracy and aligned with the appropriate curriculum objectives. The marking schemes provide detailed explanations to help students understand the reasoning behind each answer.

We hope this book will be a valuable resource in your academic journey.

— Beacon Educational Consult

Accra, Ghana

2026

BEACON EDUCATIONAL SERIES

Integrated Science for Basic 9

A Comprehensive Question Bank
with Marking Schemes

Curriculum-Aligned Practice Papers
for Basic 9 (JHS 3) Students and Teachers

Beacon Educational Consult
2026 Edition

Integrated Science for Basic 9
First Edition, 2026

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This book is designed to support the Ghana Education Service (GES) curriculum for Junior High School. It is intended as a supplementary resource for teachers and students.

The questions in this book are original practice materials created for educational purposes. They are not official BECE or GES examination questions.

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PREFACE

This book has been developed to support Junior High School Basic 9 (JHS 3) students and teachers in their study of Integrated Science, as prescribed by the Ghana Education Service (GES) and the National Council for Curriculum and Assessment (NaCCA).

The book contains a comprehensive collection of practice questions organised according to the curriculum's strand and sub-strand structure. Each paper includes both multiple-choice questions (Section A) and structured theory questions (Section B), reflecting the format of BECE examinations.

Key Features

- Curriculum-Aligned Content — All questions mapped to specific content standards.
- Comprehensive Coverage — Topic-based papers covering all major strands.
- Detailed Marking Schemes — Complete answer keys with method marks and marking guidance.
- Professional Diagrams — Clear, scientifically accurate illustrations.
- Cognitive Demand — Questions spanning all levels of Bloom's taxonomy.

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ANSWER KEY AND MARKING SCHEMES

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Paper 19: Understanding the Environment	
Paper 20: Waste Management	

Paper 1: Agriculture

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A simple agricultural tool is best described as
 - A. any machine driven by a tractor engine
 - B. a hand tool used on the farm that works without an engine
 - C. a chemical used for killing weeds on the farm
 - D. a building in which farm produce is stored
2. The farm tool used mainly for clearing weeds and cutting bush is the
 - A. garden line
 - B. wheelbarrow
 - C. watering can
 - D. cutlass
3. A hoe is used chiefly for
 - A. measuring the spacing between crops
 - B. cutting down tall trees
 - C. carrying water to the nursery
 - D. making mounds, ridges and seedbeds and for weeding
4. The tool used for digging and turning hard soil in a garden is the
 - A. garden fork
 - B. pair of secateurs
 - C. hand trowel
 - D. watering can
5. A hand trowel is most useful for
 - A. carrying manure to the field
 - B. spraying insecticide on crops
 - C. lifting and transplanting seedlings in the nursery
 - D. felling a mango tree
6. Which tool would a farmer use to carry manure from the compost heap to the field?
 - A. A rake
 - B. A pair of secateurs
 - C. A dibber
 - D. A wheelbarrow
7. A rake is used on the garden to
 - A. level the seedbed and gather dry leaves and rubbish
 - B. make deep holes for planting yam
 - C. cut the branches of fruit trees
 - D. pump water from a well
8. Which of these tools would be the MOST suitable for pruning the branches of an orange tree?
 - A. A watering can
 - B. A pair of secateurs
 - C. A hoe
 - D. A garden rake
9. The blade of a cutlass is made of steel chiefly because steel is
 - A. hard and strong, and can be sharpened to keep a cutting edge
 - B. cheap and rots quickly in the soil
 - C. light and soft, so that it bends easily

D. a poor conductor of heat

11. Which of the following materials would be the LEAST suitable for making the blade of a hoe?

- A. Mild steel
- B. Hardened steel
- C. Dry wood
- D. Scrap iron from a vehicle spring

10. The handle of a hoe is usually made of wood because wood is

- A. light, fairly strong and does not become too hot or too cold to hold
- B. harder than the metal blade
- C. heavier than metal and so steadies the tool
- D. able to conduct electricity safely

- 13.** Old vehicle springs and scrap metal are often used by village blacksmiths because they
- A. are cheap, easy to obtain and made of steel that can be heated and shaped
 - B. need no heating before they are shaped
 - C. are softer than any new metal
 - D. cannot rust under any condition
- 15.** A piece of old motor tyre is sometimes nailed on to a tool handle in order to
- A. give a firm grip that does not slip in a sweaty hand
 - B. make the tool much heavier
 - C. prevent the metal part from rusting
 - D. sharpen the cutting edge of the tool
- 17.** The part of a cutlass that passes into the handle is called the
- A. ferrule
 - B. cutting edge
 - C. socket
 - D. tang
- 19.** A metal ring (ferrule) is fitted round the end of a tool handle in order to
- A. make the handle smooth to hold
 - B. keep the blade sharp for longer
 - C. prevent the wood from splitting when the tool is used
 - D. join two blades together
- 21.** Two pieces of metal in a farm tool may be joined permanently by
- A. painting over the joint
 - B. welding them together
 - C. tying them with raffia
 - D. pressing them together by hand
- 12.** A class making a rake uses iron nails for the teeth mainly because the nails are
- A. soft, so that they can bend round stones
 - B. able to rot away in the soil after use
 - C. hard, pointed and strong enough to comb the soil without bending easily
 - D. lighter than any other material available
- 14.** Bamboo is sometimes used for the handles of light farm tools because it is
- A. a good conductor of heat
 - B. straight, light and strong for its weight
 - C. the only material that does not break
 - D. heavier than wood
- 16.** Which pair of materials would a pupil choose to make a simple garden rake from things found around the school?
- A. A clay block and palm fronds
 - B. A wooden bar and iron nails
 - C. A glass rod and cotton thread
 - D. A plastic sheet and dry grass
- 18.** Rivets are used on a cutlass to
- A. prevent the blade from rusting
 - B. balance the weight of the blade
 - C. sharpen the cutting edge of the blade
 - D. hold the two pieces of the handle firmly on to the tang
- 20.** The blade of a hoe is fixed to its handle by means of
- A. an eye or socket into which the handle is driven and wedged
 - B. paint applied to both parts
 - C. glue spread on the handle only
 - D. a length of string tied round the blade
- 22.** A pupil binds the handle of a home-made rake to its head with wire and then adds a short wooden brace. The brace is added mainly to
- A. increase the number of teeth

- B. stop the handle from twisting and make the joint stronger
- C. make the rake heavier to use
- D. protect the hands from the wire

24. A blacksmith heats a piece of scrap steel until it is red hot before shaping it because the hot metal

- A. becomes heavier and easier to hold
- B. becomes soft enough to be beaten into shape
- C. cannot burn the blacksmith's hands
- D. changes into a different metal

23. The FIRST step in making a simple farm tool should be to

- A. heat the metal in the forge
- B. sharpen the cutting edge
- C. decide on the tool needed and make a drawing with the measurements
- D. fix the handle to the blade

- 25.** The beating of hot metal into the required shape on an anvil is called
- seasoning
 - tempering
 - galvanising
 - forging
- 27.** Wood for a tool handle is seasoned (dried) before use so that the handle
- can be bent into any shape
 - becomes heavier in the hand
 - burns easily when it is worn out
 - does not shrink, crack or become loose later
- 29.** The file and the grinding stone are used in the making of a cutlass mainly to
- paint the finished blade
 - join the handle to the tang
 - smooth the metal and put a sharp edge on the blade
 - heat the metal before forging
- 31.** Which of the following is a CONSTRAINT likely to face a school that wants to make its own farm tools?
- Plenty of scrap metal in the town
 - Too many pupils willing to help
 - A long holiday during the term
 - Lack of money, tools and skilled hands for the work
- 33.** When a file or a grinding stone is being used, the eyes should be protected because
- small pieces of metal and grit fly off the work
 - the file gives out a bright light
 - the noise can damage the eyes
 - the metal gives off a poisonous gas
- 26.** Hardening and tempering of a finished blade is carried out in order to
- make the blade soft so that it bends in use
 - give the blade a bright colour
 - make the blade hard enough to keep its edge without being too brittle
 - add more metal to the blade
- 28.** In which order should the following steps be carried out in making a hoe?
- I, II, III, IV
 - IV, II, III, I
 - III, I, IV, II
 - II, IV, I, III
- 30.** Before a newly made tool is given out for use, it should be
- left in water for a week
 - painted and hung up without testing
 - heated once more in the forge
 - tested on the work it is meant to do and checked for loose or sharp parts
- 32.** A safety precaution that must be taken when hot metal is being beaten is to
- work quickly in a dark corner
 - wear thick gloves, an apron and eye protection, and use long tongs
 - cool the metal by blowing on it with the mouth
 - hold the hot metal with the bare hand

- 35.** A pupil finds that the only metal available for a blade is a soft strip of aluminium sheet. The BEST decision is to
- A. look for a piece of steel, since aluminium is too soft to keep a cutting edge
 - B. use the aluminium because it is lighter than steel
 - C. abandon the project altogether
 - D. use the aluminium and sharpen it every few minutes
- 37.** The metal parts of tools that are to be stored for a long time should be
- A. buried in the soil of the garden
 - B. left outside in the rain to cool
 - C. rubbed with oil or grease and kept in a dry store
 - D. washed with salt water and left wet
- 39.** A class made ten rakes and found that the teeth of three of them bent on the first day. The BEST improvement to the design would be to
- A. paint the teeth a bright colour
 - B. use a lighter piece of wood for the head
 - C. use thicker, harder nails and set them closer to the head
 - D. make the handle much longer
- 34.** Tools with sharp edges should be carried
- A. over the shoulder with the edge upwards
 - B. by holding the sharp edge in the palm
 - C. with the sharp edge pointed downwards and away from the body
 - D. in the pocket of the overall
- 36.** Farm tools are cleaned and dried after use mainly to
- A. prevent them from rusting and to keep them in good working order
 - B. make them heavier for the next job
 - C. sharpen their cutting edges automatically
 - D. change the metal into a stronger one
- 38.** A wooden handle that has become rough and splintered should be
- A. burnt at the rough end
 - B. wrapped in a wet cloth
 - C. left as it is, since the roughness gives grip
 - D. smoothed with sandpaper and treated with oil
- 40.** In a report on the tools it has made, a class should include
- A. only a drawing of the finished tool
 - B. the materials used, the steps followed, the difficulties met and how the tools performed when tested
 - C. only the price of the materials bought
 - D. only the names of the pupils who worked

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a table or a labelled sketch is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 2: Animal Feed and Nutrition

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Animal feed is best described as
 - A. any material eaten by farm animals that supplies them with nutrients
 - B. any grass that grows on the farm land
 - C. the medicine given to sick animals
 - D. the water given to animals in the morning
2. A mixture of feed ingredients prepared to meet the daily needs of an animal is called a
 - A. supplement
 - B. roughage
 - C. vaccine
 - D. ration
3. Which of the following feed ingredients is used mainly as a source of energy?
 - A. Fish meal
 - B. Maize grain
 - C. Common salt
 - D. Oyster shell
4. Groundnut cake, soya bean meal and fish meal are added to poultry feed mainly to supply
 - A. water
 - B. protein
 - C. fibre
 - D. carbohydrate
5. Oyster shell and bone meal are included in a layer ration chiefly to supply
 - A. protein for growth of muscle
 - B. energy for movement
 - C. vitamin C for health
 - D. calcium for the formation of strong egg shells and bones
6. Premix is added to a home-mixed ration in a small quantity because it supplies
 - A. most of the energy in the ration
 - B. the bulk that fills the stomach
 - C. most of the protein in the ration
 - D. vitamins and trace minerals, which are needed in very small amounts
7. When mixing feed by hand on a clean floor, the farmer should
 - A. pour all the ingredients into one heap and leave them
 - B. spread the ingredient of the largest quantity first and add the smaller ones on top, then mix thoroughly
 - C. mix the ingredients with plenty of water before use
 - D. add the premix first and the maize last without mixing
8. Before mixing premix into a large quantity of feed, the farmer should first
 - A. add the premix to the drinking water only
 - B. dissolve the premix in a bucket of water
 - C. mix the premix with a small quantity of maize meal and then add it to the rest
 - D. roast the premix over a fire

- 9.** Mixed feed should be stored in a cool, dry room on pallets MAINLY to
- make the feed heavier
 - prevent dampness, which causes mould to grow in the feed
 - help the feed to ferment before use
 - allow rats to reach the feed easily
- 11.** An animal with a simple, one-chambered stomach is described as a
- herbivore of four stomachs
 - monogastric animal
 - ruminant
 - polygastric animal
- 13.** The largest compartment of the stomach of a ruminant is the
- abomasum
 - omasum
 - rumen
 - reticulum
- 15.** Ruminants are able to feed on grass and other fibrous material because
- they swallow stones that grind the grass
 - their saliva contains an enzyme that digests cellulose
 - they chew the grass into powder with their front teeth
 - micro-organisms in the rumen break down the fibre (cellulose) for them
- 17.** Feed such as maize, wheat bran and groundnut cake, which is rich in nutrients and low in fibre, is described as
- forage
 - roughage
 - concentrate
 - bulk feed
- 19.** Which of the following would be the MOST suitable feed for a flock of local fowls kept in a deep litter house?
- Fresh elephant grass only
 - Dry maize stalks only
 - A balanced poultry mash with
- 10.** Cattle, sheep and goats are grouped together as
- monogastrics
 - poultry
 - pets
 - ruminants
- 12.** Which of the following animals is a monogastric farm animal?
- Goat
 - Sheep
 - Pig
 - Cow
- 14.** The compartment of the ruminant stomach in which digestive juices are produced, like the stomach of a human being, is the
- abomasum
 - omasum
 - reticulum
 - rumen
- 16.** Grass, hay, silage and crop residues are examples of
- roughages
 - concentrates
 - mineral supplements
 - vitamin premixes
- 18.** A fowl has no teeth. The part of its digestive system that grinds the grain is the
- gizzard
 - liver
 - cloaca
 - crop

clean water

D. Palm oil mixed with sand

21. A farmer notices white and green mould on a bag of feed. He should

- A. reject the bag, because mouldy feed contains poisons that can kill the animals
- B. dry the feed in the sun and feed it to the young animals
- C. mix it with fresh feed and use it quickly
- D. give it to the laying birds only

20. The BEST reason for keeping a daily record of the feed given on a poultry farm is that the record

- A. is required before the birds can drink water
- B. makes the farm look busy
- C. shows how much feed the birds use and helps the farmer work out his costs and profit
- D. shows the colour of the feed used

- 23.** A pupil wishes to compare the feed taken by a goat and by a pig on a school farm. The BEST method is to
- guess the amount from the size of each animal
 - weigh the feed given to each animal each day and record it in a table
 - ask the animals' owner only once
 - count the number of times each animal drinks water
- 25.** A farmer who must change his birds from one brand of feed to another should
- give the two feeds in equal amounts for one year
 - mix the new feed with the old one and increase it gradually over several days
 - stop feeding the birds for two days before the change
 - change to the new feed suddenly in one day
- 27.** The first milk produced by a cow after calving, which protects the calf against disease, is called
- whey
 - butter
 - silage
 - colostrum
- 29.** A ration given to birds that have started laying must be rich in
- fat only
 - sand and grit only
 - calcium and protein
 - fibre only
- 31.** The small quantity of concentrate given to piglets while they are still suckling is known as
- creep feed
 - finisher feed
 - dry feed
 - silage
- 22.** Which of the following is the BEST guide when selecting feed for a farm animal?
- The day of the week on which it is bought
 - The distance of the shop from the farm
 - The colour of the sack the feed comes in
 - The class of the animal and its stage of growth
- 24.** Clean drinking water must be available to farm animals at all times because water
- replaces the protein in the feed
 - makes the feed cheaper to buy
 - is needed for digestion, for transport of nutrients and for the production of milk and eggs
 - prevents the animals from eating
- 26.** Animals are commonly grouped by stage of growth as
- cheap, costly and very costly animals
 - small, medium and imported animals
 - young, growing and mature animals
 - wild, tame and trained animals
- 28.** The feed given to day-old chicks up to about four weeks is the
- finisher mash
 - cattle lick
 - chick starter mash
 - layer mash
- 30.** Feed for young, growing animals must contain more protein than feed for mature animals because protein is needed for
- keeping the animal warm at night
 - colouring the skin of the animal
 - replacing the water lost in urine
 - the building of new body tissue as the animal grows

- 33.** A fattening bull and a lactating (milking) cow are given different rations because
- animals at different stages and levels of production need different amounts of nutrients
 - the two animals belong to different species
 - only one of them is a ruminant
 - feed is given according to the colour of the animal
- 35.** Feeding troughs and drinkers should be washed regularly MAINLY to
- make the animals eat more slowly
 - prevent the spread of disease among the animals
 - increase the weight of the feed
 - make the feed taste sweeter
- 37.** A poultry farmer should provide enough feeder space for all the birds so that
- every bird, including the weaker ones, gets its share of the feed
 - the strongest birds can eat all the feed
 - the feed can be stored in the house
 - the birds can sleep in the feeders
- 39.** A school farm has plenty of fresh grass but the goats are still thin at the end of the dry season. The BEST advice is to
- stop feeding the goats for a few days
 - sell all the goats at once
 - give the goats only water until the rains come
 - add a concentrate supplement and a mineral lick to the dry forage available
- 32.** Which of the following changes should a farmer make as his broilers move from the starter stage to the finisher stage?
- Feed the birds only maize stalks
 - Give the birds only water for the last two weeks
 - Give the birds the chick mash again
 - Change to a finisher ration, which has more energy and slightly less protein
- 34.** A farmer feeds his growing kids the same low-protein feed he gives his mature goats. The MOST likely result is that the kids will
- grow faster than normal
 - grow slowly and remain small and weak
 - become fat within one week
 - produce milk earlier than expected
- 36.** In the hot season, farm animals are best fed
- in the cool hours of the early morning and the evening
 - at noon when the sun is hottest
 - only when they are seen to be thin
 - only once every three days
- 38.** Which of the following is the BEST sign that the feeding programme on a poultry farm is working well?
- The feeders are washed once a year
 - The birds are growing steadily, are active and lay well
 - The birds crowd at the door of the house
 - The feed store is always full
- 40.** Which of the following practices on a livestock farm shows the BEST use of knowledge of feeding?
- Grouping the animals by stage of growth and giving each group a suitable ration and clean water
 - Giving all the animals the same feed at all stages
 - Feeding the animals only when feed is cheap

- D. Leaving the animals to find all their own feed throughout the year

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps in any calculation and give the correct unit with every answer.

Question 7

- (a) Use a pencil and a ruler where a table or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 3: Circulatory

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The liquid part of blood in which the cells float is called
 - A. platelet fluid
 - B. plasma
 - C. serum sugar
 - D. lymph gland
2. Which of the following is NOT a component of human blood?
 - A. Red blood cells
 - B. Plasma
 - C. Chlorophyll
 - D. Platelets
3. The red colouring matter in red blood cells, which carries oxygen, is
 - A. chlorophyll
 - B. haemophilia
 - C. insulin
 - D. haemoglobin
4. Red blood cells are well suited to their work because they
 - A. are biconcave in shape and contain haemoglobin but no nucleus
 - B. have long tails for swimming in plasma
 - C. are the largest cells in the body
 - D. contain a large nucleus and chloroplasts
5. The chief function of white blood cells is to
 - A. defend the body against disease-causing organisms
 - B. give the blood its red colour
 - C. make the blood clot at a wound
 - D. carry oxygen to the tissues
6. Platelets are important in the blood because they
 - A. digest the food in the small intestine
 - B. carry carbon dioxide to the lungs
 - C. produce the hormones of the body
 - D. help the blood to clot when a blood vessel is cut
7. Which of the following is a function of blood plasma?
 - A. Producing new bones in the body
 - B. Breaking down food in the stomach
 - C. Filtering air before it enters the lungs
 - D. Transporting digested food, hormones and waste products around the body
8. A person who has too few red blood cells or too little haemoglobin is said to be suffering from
 - A. anaemia
 - B. hypertension
 - C. diabetes
 - D. arthritis
9. Which of the following foods would BEST help a person to build up the haemoglobin in the blood?
 - A. Boiled plain water
 - B. Refined cooking oil
 - C. Pure white sugar
 - D. Kontomire and liver, which are

rich in iron

- 10.** The organ that pumps blood round the body is the
- A. liver
 - B. lung

- 11.** The human heart has
- A. two chambers
 - B. three chambers
 - C. six chambers
 - D. four chambers
- 13.** The wall of the left ventricle is thicker than that of the right ventricle because the left ventricle
- A. holds a much larger volume of blood
 - B. contains no muscle at all
 - C. pumps blood to the whole body, which needs a greater force
 - D. pumps blood only to the lungs
- 15.** Which blood vessel has valves along its length to prevent the backward flow of blood?
- A. Artery
 - B. Vein
 - C. Capillary
 - D. Aorta
- 17.** The wall of a capillary is only one cell thick. This is an advantage because it
- A. allows materials to pass quickly between the blood and the body cells
 - B. makes the capillary very strong
 - C. prevents blood from flowing in it
 - D. stores blood for a long time
- 19.** The large artery that carries blood from the left ventricle to the rest of the body is the
- A. pulmonary vein
 - B. aorta
 - C. jugular vein
 - D. vena cava
- 21.** Blood is carried from the heart to the lungs by the
- A. pulmonary vein
 - B. aorta
 - C. vena cava
 - D. pulmonary artery
- 23.** The chamber of the heart that receives deoxygenated blood from the body is the
- A. left ventricle
 - C. heart
 - D. kidney
- 12.** The upper chambers of the heart are the
- A. arteries
 - B. atria
 - C. valves
 - D. ventricles
- 14.** Blood vessels that carry blood away from the heart are called
- A. veins
 - B. arteries
 - C. capillaries
 - D. ducts
- 16.** Exchange of gases, food and waste between the blood and the body cells takes place in the
- A. heart chambers
 - B. veins
 - C. capillaries
 - D. arteries
- 18.** The valves between the atria and the ventricles prevent blood from
- A. reaching the lungs for oxygen
 - B. flowing back into the atria when the ventricles contract
 - C. mixing with the plasma
 - D. entering the heart at all
- 20.** The blood of a human being passes through the heart twice in one complete round of the body. This is described as
- A. double circulation
 - B. open circulation
 - C. closed respiration
 - D. single circulation
- 22.** Oxygenated blood returns from the lungs to the heart through the
- A. pulmonary vein
 - B. renal artery
 - C. pulmonary artery
 - D. vena cava

- B. left atrium
- C. right atrium
- D. right ventricle

- 25.** A nurse found that a pupil's heart beat 76 times in one minute. This measurement is the pupil's
- breathing volume
 - pulse rate
 - blood group
 - body temperature
- 27.** The instrument used to measure blood pressure is the
- stethoscope only
 - sphygmomanometer
 - thermometer
 - hydrometer
- 29.** Which of the following habits increases a person's risk of high blood pressure?
- Walking briskly for thirty minutes each day
 - Sleeping for seven to eight hours a night
 - Eating plenty of fruits and vegetables
 - Eating very salty, fatty foods and taking little exercise
- 31.** A stroke occurs when
- the kidney makes too much urine
 - the lungs fill with air during exercise
 - the blood supply to part of the brain is blocked or a vessel in the brain bursts
 - the stomach produces too much acid
- 33.** Sickle cell disease affects the circulatory system chiefly because the red blood cells
- become sickle shaped, carry less oxygen and block small blood vessels
 - stop the heart from beating at all
 - turn into white blood cells
 - grow larger and carry more oxygen
- 24.** The pulse felt at the wrist is caused by
- the wave of pressure passing along the artery each time the heart beats
 - blood flowing backwards in a vein
 - the opening of the valves of the veins
 - air moving in the blood vessels
- 26.** Blood pressure is best described as
- the speed at which a person breathes
 - the volume of blood in the body
 - the number of red blood cells in the blood
 - the force with which the blood presses against the walls of the blood vessels
- 28.** A person whose blood pressure stays much higher than normal is said to have
- hypertension
 - haemophilia
 - anaemia
 - hypotension
- 30.** Fatty material deposited on the inner wall of an artery is dangerous because it
- widens the artery and lowers the blood pressure
 - narrows the artery, so the heart must work harder and a clot may block it
 - increases the number of white blood cells
 - turns the blood into plasma
- 32.** Which of the following is the BEST advice for an adult who has just been told that his blood pressure is high?
- Take only herbal drinks sold on the street
 - Avoid going to the clinic until he feels pain
 - Reduce salt and fatty foods, exercise regularly and take the medicine prescribed by the clinic
 - Stop eating completely for one week

- 35.** The word equation for aerobic respiration is
- A. glucose + carbon dioxide → oxygen + energy
 - B. oxygen + water → glucose + carbon dioxide
 - C. glucose + oxygen → carbon dioxide + water + energy
 - D. carbon dioxide + water → glucose + oxygen
- 37.** Gaseous exchange in the lungs takes place in the
- A. trachea
 - B. ribs
 - C. bronchi
 - D. air sacs (alveoli)
- 39.** The circulatory system and the respiratory system work together because the blood
- A. changes carbon dioxide into oxygen in the heart
 - B. stores oxygen in the ribs until it is needed
 - C. carries the oxygen taken in at the lungs to the cells and brings carbon dioxide back to the lungs
 - D. produces the air that enters the lungs
- 34.** Respiration in the cells of the body is best described as the
- A. taking in of air through the nose only
 - B. breakdown of glucose using oxygen to release energy, carbon dioxide and water
 - C. building up of glucose from carbon dioxide and water
 - D. removal of urine from the body
- 36.** Breathing differs from respiration in that breathing
- A. takes place only in the cells of the body
 - B. releases energy while respiration does not
 - C. is the movement of air into and out of the lungs, while respiration releases energy in the cells
 - D. means exactly the same thing as respiration
- 38.** During gaseous exchange in the lungs,
- A. only carbon dioxide moves, in both directions
 - B. oxygen passes from the blood into the air sacs
 - C. no gas moves across the wall of the air sac
 - D. oxygen passes from the air sacs into the blood and carbon dioxide passes from the blood into the air sacs
- 40.** During hard exercise a footballer at the Accra Sports Stadium breathes faster and his heart beats faster. This happens MAINLY because his muscles need
- A. more oxygen and glucose, and the carbon dioxide produced must be removed quickly
 - B. more water to cool the bones
 - C. less blood than when he is resting
 - D. to stop respiring for a short time

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Spell the names of the parts of the body correctly; marks are lost where the name written means a different part.

Question 7

- (a) Use a pencil and a ruler where a table or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 4: Climate Change and C

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Climate is BEST described as
 - A. the condition of the atmosphere of a place on one day
 - B. the average weather conditions of a place over many years
 - C. the temperature recorded at noon on a given day
 - D. the amount of rain that falls in a single month
2. Which of the following is a difference between weather and climate?
 - A. Weather changes from day to day, while climate is the average over many years
 - B. Weather affects plants, while climate affects only animals
 - C. Weather is found in towns, while climate is found in villages
 - D. Weather is measured only at night, climate only by day
3. Climate change means
 - A. the movement of clouds across the sky
 - B. a lasting change in the average weather pattern of a place or of the earth
 - C. the change from morning sunshine to afternoon rain
 - D. the change of seasons within one year
4. Which of the following is evidence of climate change in Ghana?
 - A. The occurrence of harmattan in December
 - B. The sun rising in the east every morning
 - C. Rainy seasons that begin later and are less reliable than before
 - D. The change from day to night
5. The gradual rise in the average temperature of the earth is known as
 - A. humidity
 - B. ozone formation
 - C. condensation
 - D. global warming
6. Which of these gases is a greenhouse gas?
 - A. Nitrogen
 - B. Argon
 - C. Carbon dioxide
 - D. Helium
7. The greenhouse effect is important to life on earth because without it the earth would be
 - A. without an atmosphere at all
 - B. much wetter than it is now
 - C. much colder than it is now
 - D. unable to turn on its axis
8. Which of the following is a NATURAL factor that influences climate change?
 - A. Volcanic eruptions that throw ash and gases into the atmosphere
 - B. The cutting down of forests for timber
 - C. The burning of petrol in motor vehicles
 - D. The burning of refuse in a town
9. A change in the amount of energy given out by the sun can affect the earth's climate because the sun
 - A. supplies the water found in the oceans
 - B. is the main source of the heat that warms the earth

- C. controls the rotation of the earth on its axis
- D. produces the carbon dioxide of the atmosphere

- 11.** Large quantities of ash and gas thrown out by a volcano may cool the earth for some years because the ash
- increases the speed of the earth's rotation
 - absorbs all the carbon dioxide of the air
 - blocks part of the sunlight from reaching the earth's surface
 - destroys the water vapour in the clouds
- 13.** Which of the following is a natural event that can start a wildfire?
- A lightning strike during a dry storm
 - A hunter smoking out grasscutters
 - A farmer clearing land by fire
 - A cigarette thrown into dry grass
- 15.** The burning of fossil fuels such as petrol, diesel and coal adds to climate change chiefly by releasing
- water into the oceans
 - nitrogen into the soil
 - oxygen into the atmosphere
 - carbon dioxide into the atmosphere
- 17.** Which human activity in Ghana contributes MOST directly to the loss of forest cover?
- Logging and the clearing of land for farming and settlement
 - Weaving kente cloth
 - Sweeping the school compound
 - Fishing with hook and line
- 19.** Bush burning before planting contributes to climate change because it
- cools the soil for the next crop
 - adds nitrogen gas to the atmosphere
 - releases carbon dioxide and destroys the plants that take it in
 - increases the humus of the soil permanently
- 10.** Ocean currents influence the climate of coastal lands by
- changing the length of the day
 - producing the winds that blow in space
 - carrying warm or cold water that heats or cools the air above it
 - adding salt to the air over the land
- 12.** Methane released naturally from swamps and wetlands contributes to climate change because methane is
- an acid that burns plant leaves
 - a greenhouse gas that traps heat in the atmosphere
 - a gas that destroys rain clouds
 - a gas that cools the atmosphere
- 14.** Which statement about natural factors of climate change is correct?
- They began only after factories were built
 - They affect only the climate of the oceans
 - They can be completely stopped by governments
 - They have acted on the earth's climate long before human beings appeared
- 16.** Deforestation increases the carbon dioxide of the atmosphere because trees
- store oxygen inside their trunks
 - give out carbon dioxide while they grow
 - take in carbon dioxide for photosynthesis when they are alive
 - prevent the wind from blowing the gas away
- 18.** Galamsey (illegal small-scale mining) affects the environment mainly by
- destroying forest and farmland and polluting rivers with mud and chemicals
 - increasing the rainfall over the mining area
 - cooling the rivers so that fish grow faster
 - adding humus to the soil of the area

- 21.** Keeping large herds of cattle adds to climate change because the animals
- A. drink the water of the rivers
 - B. take in all the oxygen of the air
 - C. trample the soil into dust
 - D. release methane as they digest their food
- 23.** Which of the following would help MOST to reduce the carbon dioxide released in a town?
- A. Burning refuse in every compound
 - B. Using buses and bicycles instead of many private cars
 - C. Cutting the trees along the streets
 - D. Leaving lights on throughout the night
- 25.** A school replaces its filament bulbs with LED bulbs. This action helps to fight climate change because the LED bulbs
- A. use less electricity for the same amount of light
 - B. need no electricity at all to work
 - C. give out more heat than the old bulbs
 - D. produce oxygen while they are on
- 27.** The use of a biogas digester in a school reduces climate change because it
- A. produces carbon dioxide from the air
 - B. prevents waste from decaying at all
 - C. captures the methane from waste and burns it as fuel instead of letting it escape
 - D. sends the methane straight into the air
- 20.** Which of these would release the LEAST greenhouse gas for a journey of ten kilometres?
- A. Travelling in a taxi alone
 - B. Riding on a motorcycle
 - C. Driving alone in a car
 - D. Riding a bicycle
- 22.** Compared with natural factors, human factors in climate change are important chiefly because they
- A. can be controlled or reduced by the actions people take
 - B. have been acting for a much longer time
 - C. cannot be measured by any instrument
 - D. affect only the countries that are cold
- 24.** Afforestation helps to fight climate change because the new trees
- A. prevent the sun from shining on the earth
 - B. absorb carbon dioxide from the atmosphere as they grow
 - C. release carbon dioxide as they grow
 - D. stop the wind from blowing over the land
- 26.** Which farming practice would reduce the release of greenhouse gases?
- A. Burning the farm before every planting season
 - B. Clearing weeds by hoeing and composting them instead of burning them
 - C. Cutting down the shade trees on the farm
 - D. Using more diesel pumps for irrigation
- 28.** Which of the following best describes the idea of 'reduce, reuse and recycle' in the fight against climate change?
- A. It is a way of increasing the waste produced
 - B. It is a method of burning waste more completely
 - C. Using less, using again and remaking materials saves the energy and fuel needed to make

new ones

- D. It has no connection with energy use

- 29.** A community wants to reduce its contribution to climate change at the least cost. The BEST first step would be to
- A. buy electric cars for every household
 - B. import solar panels for every house at once

- C. build a wind farm on the outskirts of the town
- D. plant and protect trees and stop bush burning in the community
- 31.** Which of the following is a renewable source of energy?
- Diesel
 - Petrol
 - Sunlight
 - Coal
- 33.** A hotel installs solar water heaters and plants trees on its grounds. These actions show that the hotel is
- trying to increase its use of fossil fuels
 - practising a green economy by using clean energy and protecting the environment
 - avoiding the payment of electricity bills only
 - reducing the number of its workers
- 35.** A green economy is said to be sustainable mainly because it
- uses up natural resources as quickly as possible
 - makes the largest profit in the shortest time
 - depends entirely on imported technology
 - meets the needs of people today without spoiling the environment for those to come
- 37.** Ghana's programme to replace old street lights with solar and LED lights is an example of
- a method of producing more waste
 - an attempt to stop the use of electricity
 - a plan to increase the use of diesel generators
 - a national action to use energy more efficiently and cut emissions
- 30.** A green economy is one that
- improves the well-being of people while reducing damage to the environment
 - stops all industries from working
 - grows only green vegetables for export
 - uses only money printed on green paper
- 32.** Which of these is an example of a 'green job'?
- Mining gold with mercury in a river
 - Driving a lorry that carries firewood
 - Selling charcoal from cut forest trees
 - Installing and maintaining solar panels
- 34.** Which of the following pairs gives a green economy action and its correct benefit?
- Using solar lamps - less kerosene burnt and less carbon dioxide released
 - Draining a wetland for building - more fish in the river
 - Cutting forest for charcoal - more rainfall in the area
 - Burning tyres for fuel - cleaner air in the town
- 36.** The Green Ghana Day, on which millions of tree seedlings are planted, is an initiative aimed MAINLY at
- providing firewood for export
 - restoring the forest cover of the country
 - building more roads in the forest zone
 - increasing the number of mining sites
- 38.** Which of the following would be the BEST measure of how well a tree-planting initiative has worked?
- The number of seedlings that are still alive and growing after two years

- B. The number of people who attended the launching ceremony
- C. The number of seedlings handed out on the day
- D. The amount of money spent on the ceremony

39. The table below shows the share of electricity from renewable sources in three countries.

- A. All three countries use the same

- amount of renewable energy
- B. Country Z produces more electricity than Country Y
 - C. Country X has the cleanest electricity supply of the three
 - D. Country Y depends least on fuels that release greenhouse gases for its electricity

- 40.** A country reports that it has cut its greenhouse gas emissions, yet its forests continue to disappear. The BEST judgement on the report is that
- A. forests have no connection with greenhouse gases
 - B. the claim needs to be checked against measured data, since losing forest usually raises emissions
 - C. the country should simply be praised for its effort
 - D. the report must be correct, because governments do not make mistakes

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures and the table carefully before you answer the questions on them.

Question 6

- (a) Show all the steps of your working in the questions that involve calculations, and give the unit with every answer.

Question 7

- (a) Use a pencil and a ruler where a diagram, a graph or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 5: Compounds Acids Ba

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. An element is best described as a substance that
 - A. cannot be broken down into simpler substances by a chemical change
 - B. can be separated into its parts by filtering
 - C. is always a shiny solid at room temperature
 - D. is formed when two liquids are stirred together
2. A chemical compound is formed when
 - A. a liquid is heated until it boils
 - B. two solids are poured into the same container
 - C. a solid is ground into a fine powder
 - D. two or more elements combine chemically in a fixed proportion
3. Which of the following is a compound?
 - A. Copper
 - B. Water
 - C. Oxygen
 - D. Air
4. The chemical symbol for sodium is
 - A. Nd
 - B. Na
 - C. Sd
 - D. So
5. Kofi wrote the symbols of four elements as Fe, Cu, K and Mg. The element represented by K is
 - A. calcium
 - B. carbon
 - C. potassium
 - D. krypton
6. The MAIN difference between a mixture of iron filings and sulphur and the compound iron(II) sulphide is that in the compound the
 - A. two substances are present in any proportion the pupil chooses
 - B. iron and the sulphur keep their own properties
 - C. iron filings can be picked out with a magnet
 - D. iron and the sulphur are chemically joined and cannot be separated with a magnet
7. Table salt, which is found in every Ghanaian kitchen, is the compound
 - A. potassium chloride
 - B. calcium carbonate
 - C. sodium chloride
 - D. sodium carbonate
8. A pupil is given a white solid and asked to show that it is a compound and not a mixture. The BEST evidence would be that the solid
 - A. has a fixed composition and properties different from those of the elements in it
 - B. is white in colour
 - C. dissolves in water to give a clear solution
 - D. can be ground into a powder
9. A binary compound is a compound made up of
 - A. two molecules of the same element
 - B. two different elements
 - C. any number of elements and a mixture

D. two different mixtures

- 11.** The common name of the binary compound hydrogen oxide, which is used for drinking, cooking and washing, is
- bleach
 - water
 - vinegar
 - kerosene
- 13.** Iron(II) sulphide is formed when
- sulphur is dissolved in water and poured on iron
 - iron filings are stirred into powdered sulphur in a beaker
 - iron and sulphur are heated together until they glow
 - iron is left in water until it rusts
- 15.** Which pair of compounds commonly found in the home are BOTH binary compounds?
- Vinegar and baking soda
 - Water and table salt
 - Sugar and marble
 - Fertiliser and detergent
- 17.** When a sodium atom loses one electron it becomes
- a sodium ion with a single positive charge
 - a sodium ion with a single negative charge
 - a sodium molecule
 - a chlorine atom
- 19.** The molecular formula of magnesium oxide is
- MgO₂
 - Mg₂O₃
 - Mg₂O
 - MgO
- 21.** In the formula CaCl₂, the figure 2 shows that
- the chlorine carries two positive charges
 - two calcium atoms are present in the compound
 - the compound has two molecules in all
 - two chlorine atoms are combined with one calcium atom
- 10.** Which of the following is a binary compound?
- Calcium trioxocarbonate(IV)
 - Sodium hydrogen carbonate
 - Sugar solution
 - Magnesium oxide
- 12.** A trader at Makola market sells a white powder used to make soap harder and to treat water. The compound calcium oxide is commonly called
- quicklime
 - common salt
 - washing powder
 - baking soda
- 14.** Hydrogen chloride gas dissolves in water to give a solution that is used in industry for cleaning metals. The solution is
- sodium hydroxide solution
 - lime water
 - hydrochloric acid
 - salt solution
- 16.** An atom that has lost or gained one or more electrons is called
- a mixture
 - a compound
 - an ion
 - a molecule
- 18.** A molecule is best described as
- the smallest part of a mixture
 - two or more atoms chemically combined together
 - a charged particle formed from an atom
 - a particle that carries no mass
- 20.** The formula of the compound formed between sodium (valency 1) and oxygen (valency 2) is
- NaO₂
 - NaO
 - Na₂O
 - Na₂O₂

- 23.** Ama models the compound water with balls and sticks. The model she builds should show
- two hydrogen atoms joined to one oxygen atom
 - a single ball labelled water
 - two hydrogen atoms and two oxygen atoms standing apart
 - one hydrogen atom joined to two oxygen atoms
- 25.** Which of the following is a base?
- Carbon(IV) oxide
 - Sodium hydroxide
 - Hydrochloric acid
 - Sodium chloride
- 27.** The acid found in the lime and the orange sold at the roadside is
- citric acid
 - nitric acid
 - ethanoic acid
 - sulphuric acid
- 29.** A solution turns red litmus paper blue and feels soapy between the fingers. The solution is
- alkaline
 - a strong acid
 - acidic
 - neutral
- 31.** A solution with a pH of 2 is
- a strong alkali
 - a weak alkali
 - a strong acid
 - a weak acid
- 22.** Which of the following statements is TRUE about carbon(IV) oxide and iron(II) sulphide?
- Each is a binary compound, but one is a gas of non-metals and the other a solid containing a metal
 - Both are gases at room temperature
 - Both are made of two metals joined together
 - Neither of them has a fixed formula
- 24.** A characteristic property of acids is that they
- feel soapy and have no effect on litmus
 - are always solids at room temperature
 - taste bitter and turn red litmus paper blue
 - taste sour and turn blue litmus paper red
- 26.** Bases that dissolve in water are called
- indicators
 - solvents
 - alkalis
 - salts
- 28.** A salt is formed when
- two acids are mixed together
 - the hydrogen of an acid is replaced by a metal
 - an acid is poured into water
 - a base is heated strongly
- 30.** On the pH scale, a neutral solution such as pure water has a pH of
- 0
 - 10
 - 14
 - 7
- 32.** Universal indicator is preferred to litmus paper when testing a solution because it
- can be used only once
 - shows how strong or weak the acid or alkali is, not only which one it is
 - measures the volume of the solution

- D. changes to a single colour for all solutions

34. A chemical bond is best described as

- A. the space between two atoms in a molecule
- B. the heat given out when a compound is formed
- C. the force of attraction that holds atoms together in a compound
- D. the charge carried by the nucleus of an atom

33. A farmer finds that his soil is too acidic for his crop. The MOST suitable material to add to the soil is

- A. common salt, because it is neutral
- B. vinegar, because it is a liquid
- C. more acid, because crops need acid
- D. lime, because it is a base and will neutralise the acid

35. An ionic bond is formed when

- A. electrons are transferred from a metal atom to a non-metal atom
- B. two atoms are pressed closely together
- C. protons move from one atom to

another

- D. electrons are shared equally between two non-metal atoms

37. Which of the following substances contains an ionic bond?

- A. Water
- B. Carbon(IV) oxide
- C. Sodium chloride
- D. Copper

39. Sodium chloride has a much higher melting point than water chiefly because

- A. its molecules are much heavier than those of water
- B. it is a white solid while water is a liquid
- C. it dissolves easily in water
- D. the strong attraction between its oppositely charged ions needs a lot of heat to overcome

36. A covalent bond is formed by

- A. the complete transfer of electrons
- B. the loss of protons by one atom
- C. the attraction between two ions
- D. the sharing of electrons between atoms

38. The bonding found in copper wire used for house wiring is

- A. metallic bonding
- B. no bonding at all
- C. ionic bonding
- D. covalent bonding

40. Solid sodium chloride does not conduct electricity, but its solution in water does. This is because in the solution the

- A. molecules break down into atoms
- B. ions are free to move and carry the charge
- C. electrons leave the chlorine and move freely
- D. water itself is a very good conductor

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Write all chemical symbols and formulae correctly, using a capital letter for the first letter of a symbol and a small letter for the second.

Question 7

- (a) Use a pencil and a ruler where a table or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 6: Conversion and Conservation

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Energy is best described as
 - A. the force that moves a body
 - B. the ability to do work
 - C. the speed at which work is done
 - D. the mass contained in a body
2. The energy stored in a stone held at the top of a wall is
 - A. kinetic energy
 - B. sound energy
 - C. potential energy
 - D. chemical energy
3. The energy possessed by a moving lorry because of its motion is called
 - A. nuclear energy
 - B. potential energy
 - C. kinetic energy
 - D. light energy
4. The energy stored in food, in a dry cell and in petrol is
 - A. heat energy
 - B. chemical energy
 - C. elastic energy
 - D. electrical energy
5. Which of the following is the SI unit of energy?
 - A. Newton
 - B. Joule
 - C. Watt
 - D. Ampere
6. A stretched catapult rubber possesses
 - A. sound energy
 - B. kinetic energy
 - C. elastic potential energy
 - D. chemical energy
7. Energy conversion means
 - A. the storing of energy inside a body for ever
 - B. the change of energy from one form into another form
 - C. the destruction of energy that is not needed
 - D. the careful use of energy so that none is wasted
8. In a solar panel, energy is converted from
 - A. light energy to electrical energy
 - B. chemical energy to electrical energy
 - C. heat energy to light energy
 - D. electrical energy to light energy
9. A green leaf making food in sunlight converts
 - A. light energy into chemical energy
 - B. electrical energy into chemical energy
 - C. heat energy into kinetic energy
 - D. chemical energy into light energy
10. In a dry-cell torch that is switched on, the chain of energy changes is
 - A. chemical → light → electrical
 - B. chemical → electrical → light
 - C. electrical → chemical → light
 - D. light → electrical → chemical
11. In a motor car engine, the energy stored in the fuel is converted mainly into

- A. electrical energy and then chemical energy
- B. light energy and then sound energy
- C. sound energy and then potential energy
- D. heat energy and then kinetic energy

12. A generator at a filling station converts

- A. sound energy into electrical energy
- B. electrical energy into chemical energy
- C. chemical energy in the fuel into

- electrical energy
D. light energy into electrical energy
- 13.** When a coconut falls from a tree, the energy change that takes place is from
- A. sound energy to kinetic energy
 - B. chemical energy to potential energy
 - C. kinetic energy to potential energy
 - D. potential energy to kinetic energy
- 15.** A rechargeable battery being charged from the mains converts
- A. heat energy into electrical energy
 - B. electrical energy into chemical energy
 - C. chemical energy into electrical energy
 - D. electrical energy into light energy
- 17.** The law of conservation of energy states that energy can
- A. be destroyed completely when a device is switched off
 - B. be stored only in the form of heat
 - C. be created in a power station when needed
 - D. neither be created nor destroyed, but only changed from one form to another
- 19.** A lamp is supplied with 100 J of electrical energy and produces 10 J of light. According to the law of conservation of energy, the other 90 J
- A. has been changed into heat energy
 - B. has been destroyed inside the lamp
 - C. was never supplied at all
 - D. is stored in the wires for later use
- 14.** A loudspeaker in a radio set converts electrical energy into
- A. sound energy
 - B. nuclear energy
 - C. chemical energy
 - D. light energy
- 16.** Which device converts mechanical (kinetic) energy into electrical energy?
- A. An electric motor
 - B. An electric iron
 - C. A dry cell
 - D. A dynamo
- 18.** A machine is supplied with 500 J of energy and gives out 350 J of useful energy. The energy that has been wasted is
- A. 850 J
 - B. 500 J
 - C. 150 J
 - D. 350 J
- 20.** A ball thrown upwards slows down as it rises. During the rise, its kinetic energy is
- A. destroyed by the air
 - B. changed into chemical energy
 - C. changed into potential energy
 - D. changed into sound energy only

- 21.** A pendulum bob swinging to and fro finally comes to rest. The energy of the bob has been
- A. changed into chemical energy in the bob
 - B. changed mainly into heat and sound by friction and air resistance
 - C. stored in the string for later use
 - D. completely destroyed by gravity
- 23.** A motor is supplied with 1 200 J of energy. It gives out 700 J of kinetic energy, 60 J of sound energy and the rest as heat. The heat produced is
- A. 440 J
 - B. 640 J
 - C. 1 960 J
 - D. 500 J
- 22.** Energy is said to be wasted in a machine when it is
- A. changed into kinetic energy
 - B. stored in the moving parts
 - C. changed into a form that is not useful for the job being done
 - D. destroyed inside the machine
- 24.** The chief difference between energy conversion and energy conservation is that conversion is
- A. a change of energy from one form to another, while conservation is the careful use of energy to avoid waste
 - B. the storing of energy, while conservation is the sale of energy

- C. possible only in machines, while conservation is possible only in living things
D. the careful use of energy, while conservation is a change of form
- 26.** A pupil winds a torch that lights a lamp. The change from the turning of the handle to the light of the lamp is an example of energy
A. storage only
B. creation
C. conservation
D. conversion
- 28.** Both energy conversion and energy conservation are important to a country because together they
A. make the energy that is available do more useful work
B. remove the need for fuels of every kind
C. destroy the energy that cannot be used
D. create new energy for the power stations
- 30.** At the Akosombo hydro-electric station, the energy of the stored water is converted into electrical energy. The energy chain is
A. electrical → mechanical → kinetic → potential
B. chemical → kinetic → potential → electrical
C. kinetic → potential → chemical → electrical
D. potential → kinetic → mechanical → electrical
- 25.** Switching off the lights in an empty classroom is an example of energy
A. conversion
B. conservation
C. creation
D. destruction
- 27.** Which pair of actions shows conservation of energy and conversion of energy, in that order?
A. A fan turning when switched on; ironing all the clothes at once
B. Ironing all the clothes at once; a fan turning when switched on
C. Burning a heap of refuse; leaving a bulb on all night
D. Boiling a full kettle for one cup; a stone falling from a wall
- 29.** A pupil says, 'Since energy is never destroyed, there is no need to conserve it.' The best reply is that
A. energy can be created whenever it is needed
B. the law of conservation of energy is not true for machines
C. energy is in fact destroyed whenever it is used
D. the energy wasted is changed into forms such as heat that cannot easily be used again
- 31.** A biogas plant on a farm converts

- A. light energy into chemical energy
- B. electrical energy into chemical energy
- C. chemical energy in animal and plant waste into a fuel gas that burns to give heat
- D. heat energy in the soil into chemical energy

33. Which of these is a renewable source of energy used in Ghana?

- A. Diesel
- B. Petrol
- C. Sunlight
- D. Liquefied petroleum gas

32. In a solar home system, the battery is included mainly to

- A. store energy so that the lamps can work at night
- B. increase the voltage of the panel
- C. cool the solar panel during the day
- D. change the current into sunlight

34. Cooking with a well-made clay stove instead of an open fire saves fuel mainly because

- A. the firewood contains more energy in a stove

- B. the food needs less energy when a stove is used
- C. the stove creates extra heat energy
- D. more of the heat produced reaches the pot instead of escaping into the air
- 35.** A family that dries its clothes in the sun instead of using an electric drier is practising
- A. energy destruction by avoiding electricity
 - B. energy conversion from heat to chemical energy
 - C. energy conservation by using a free, renewable source
 - D. energy creation by using the sun
- 37.** In carrying out an energy audit, the FIRST step should be to
- A. buy new appliances for every room
 - B. switch off the main supply to the school
 - C. ask the electricity company to change the meter
 - D. list all the appliances and note their power ratings and the hours they are used
- 39.** Which measurement would be the MOST useful evidence in a report on energy conservation in a school?
- A. The number of pupils who read the report
 - B. The number of lamps in the school
 - C. The cost of the paper used for the report
 - D. The number of units (kWh) recorded on the meter before and after the changes
- 36.** An energy audit of a school means
- A. a careful survey of where and how energy is used and wasted in the school
 - B. a test of the strength of the school's wiring
 - C. a list of all the money paid for electricity only
 - D. a record of the number of pupils in each class
- 38.** A group found that the computer laboratory air conditioners run all night, that two security lamps burn all day and that a tap of hot water drips. The BEST single recommendation is to
- A. close the computer laboratory to pupils
 - B. fit timers and switches, and repair the faults, so that energy is used only when it is needed
 - C. replace the whole wiring of the school
 - D. remove the air conditioners from the laboratory
- 40.** A school replaces 20 filament lamps of 100 W each with LED lamps of 10 W each, used for 5 hours a day. The energy saved each day is
- A. 9 kWh
 - B. 10 kWh
 - C. 1 kWh
 - D. 90 kWh

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps of your working in the questions that involve calculations, and give the unit with every answer.

Question 7

- (a) Use a pencil and a ruler where a table, a flow chart or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 7: Crop Maturation

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The maturity of a crop is best described as
 - A. the time the farmer clears the land
 - B. the day on which the seed is sown
 - C. the stage at which the crop or its part is fully developed and ready for harvesting
 - D. the height reached by the crop at flowering
2. The correct order of the growth stages of a crop such as maize is
 - A. seedling, maturity, flowering, fruiting
 - B. germination, flowering, seedling, vegetative growth
 - C. flowering, germination, seedling, maturity
 - D. germination, seedling, vegetative growth, flowering, fruiting, maturity
3. The stage at which a crop produces leaves and stems but no flowers is the
 - A. harvesting stage
 - B. vegetative stage
 - C. dormant stage
 - D. reproductive stage
4. A crop that completes its life cycle and is harvested within one season is described as
 - A. a permanent crop
 - B. a late maturing crop
 - C. a perennial crop
 - D. an early maturing (short duration) crop
5. Which of the following crops matures earliest after sowing?
 - A. Oil palm
 - B. Okro
 - C. Cassava
 - D. Cocoa
6. Maize in Ghana is usually ready for harvesting as dry grain about
 - A. three to four months after sowing
 - B. three years after sowing
 - C. one year after sowing
 - D. two weeks after sowing
7. Which of the following is a sign that maize has reached full maturity?
 - A. The leaves are still dark green and soft
 - B. The tassel has just appeared at the top
 - C. The husks and silk are dry and brown and the grains are hard
 - D. The plant is about knee high
8. Two crops sown on the same day matured at different times. The MOST likely reason is that
 - A. the two crops are different in kind and have different lengths of growing period
 - B. the farmer used the same seed for both
 - C. the sun shone on only one of them
 - D. crops always mature on the same day
9. A seed bed is best described as
 - A. the store in which seeds are kept
 - B. the place where harvested crops are dried
 - C. the piece of soil prepared so that seeds or seedlings can be

- planted in it
- D. the bag in which seeds are carried

- 11.** Ridges and mounds are prepared mainly to
- make the farm look beautiful
 - stop the crop from flowering
 - prevent the soil from holding any water
 - give the roots deeper loose soil and allow excess water to drain away
- 13.** Crops grown on a loamy soil usually mature earlier than the same crops grown on a sandy soil because loam
- contains no humus at all
 - holds water so tightly that roots cannot use it
 - holds enough water and nutrients while still allowing air to reach the roots
 - has larger particles than sand
- 15.** Tomato seedlings raised in a waterlogged seed bed grow poorly chiefly because
- the seedlings receive too much sunlight
 - the soil becomes too rich in nitrogen
 - the roots take in too much humus
 - the water fills the air spaces, so the roots cannot get enough oxygen
- 17.** Adding well-rotted manure to a seed bed before sowing helps the crop to mature earlier because the manure
- makes the soil very hard
 - supplies nutrients and improves the water-holding and structure of the soil
 - prevents the seeds from germinating
 - kills all the organisms in the soil
- 19.** When comparing the growth of a crop on two different soils, a class must
- weed only one of the two plots
 - use a different crop on each plot
 - water one plot and leave the other dry
 - treat both plots in exactly the same way apart from the soil
- 10.** Which of the following is NOT a type of seed bed?
- A flat bed
 - A drying floor
 - A mound
 - A ridge
- 12.** A nursery bed is used to
- keep farm tools during the rains
 - store crops after harvesting
 - dry grains before milling
 - raise seedlings which are later transplanted to the field
- 14.** Maize grown on a poor sandy soil is likely to
- produce more grain than maize on loam
 - grow faster than maize on loam
 - mature on the same day as maize on loam
 - grow slowly, be short and mature late or give a poor yield
- 16.** Which crop is usually grown on mounds or ridges in Ghana?
- Oil palm
 - Yam
 - Cocoa
 - Rice in a valley bottom
- 18.** The BEST way for a class to find out how long a crop takes to mature is to
- ask one farmer and write down his answer only
 - count the number of leaves on one plant
 - sow the crop and record the date of sowing and the date of harvesting
 - guess from the height of the plants
- 20.** Which of the following records would be MOST useful in a study of crop maturity?
- The date of sowing, the height of the plants each week and the date of flowering
 - The names of all the pupils in the class
 - The colour of the farmer's clothes
 - The price of seed in the market

- 21.** In an experiment, the plot that is treated in the usual way and used for comparison is called the
- A. control
 - B. sample
 - C. variable
 - D. average
- 23.** A class recorded that okro on plot 1 flowered in 6 weeks and okro on plot 2 flowered in 9 weeks. The BEST conclusion is that
- A. the conditions on plot 1 suited the crop better than those on plot 2
 - B. the two plots were treated in exactly the same way
 - C. the okro on plot 2 was a different crop
 - D. okro always flowers in six weeks
- 25.** Dry mature maize grain is used mainly for
- A. feeding to fish in a pond only
 - B. milling into flour for kenkey, banku or porridge
 - C. roasting on the cob at the roadside
 - D. making palm oil
- 27.** Dry cowpea seeds are valued in the Ghanaian diet mainly as a source of
- A. water
 - B. protein
 - C. fat and oil
 - D. vitamin C
- 29.** Ripe plantain is preferred for kelewele and 'red red' because it is
- A. harder than the green one
 - B. sweeter, since much of its starch has changed into sugar
 - C. richer in protein than the green one
 - D. free from all food value
- 31.** Pepper is picked while still green for some dishes but allowed to turn red for others. This shows that
- A. the stage of maturity determines the use to which a crop is put
 - B. green pepper is a different crop from red pepper
- 22.** Ten plants were measured on each plot instead of one plant because
- A. the teacher wanted more work for the pupils
 - B. ten plants grow faster than one plant
 - C. an average of several plants gives a more reliable result
 - D. one plant is too difficult to measure
- 24.** Fresh green maize on the cob is commonly
- A. ground into flour for banku
 - B. roasted or boiled and eaten as food
 - C. used only as seed for planting
 - D. stored for two years before use
- 26.** Cowpea harvested while the pods are still green and tender is used as
- A. seed for planting next season
 - B. material for making rope
 - C. a fresh vegetable, cooked with the pods
 - D. dry seed for storing in sacks
- 28.** Green (unripe) plantain is commonly used for
- A. kelewele and other sweet dishes only
 - B. roofing houses
 - C. making soap
 - D. ampesi, fufu or plantain chips
- 30.** Cassava is best harvested for gari when the tubers are
- A. mature, with enough starch stored in them
 - B. left in the ground for ten years
 - C. only two weeks old
 - D. still very thin and watery

- C. pepper has only one use
- D. red pepper is never eaten in Ghana

33. Knowledge of the maturity stages of crops helps a farmer MAINLY to

- A. choose the clothes he wears to the farm
- B. decide when to harvest each crop for the use he has in mind
- C. know how many children to send to school
- D. decide the colour to paint his store

32. Rice harvested too early, while the grains are still at the milk stage, will

- A. produce more grains per plant
- B. store better than fully mature rice
- C. give a poor yield of grain that shrivels and breaks during milling
- D. give the heaviest yield of all

- 35.** Harvesting a crop too late may lead to
- A. loss of quality, shattering of grain or rotting of tubers in the ground
 - B. a much higher market price always
 - C. faster drying of the grain in the store
 - D. an increase in the number of seeds formed
- 37.** Knowing that cowpea matures in about 70 days while cassava needs about 12 months helps a farmer to
- A. grow only one crop on the whole farm
 - B. stop weeding the farm completely
 - C. harvest both crops on the same day
 - D. plan which crop to plant where and when, so that food is available at different times
- 39.** A farmer in a district with a short rainy season would be BEST advised to grow
- A. a late maturing variety that needs a year of rain
 - B. a crop that cannot grow in that district
 - C. an early maturing variety that will ripen before the rains end
 - D. no crop at all during the rains
- 34.** A farmer who wants to sell green maize at a festival in December should
- A. sow any crop and wait for December
 - B. harvest the maize as soon as it germinates
 - C. sow the maize in December itself
 - D. count back the number of weeks the crop needs and sow at the right time
- 36.** A market trader who buys okro for sale as a fresh vegetable should choose fruits that are
- A. young and tender
 - B. still inside the flower
 - C. already rotting
 - D. old, dry and fibrous
- 38.** Intercropping an early maturing crop with a late maturing one is useful because the farmer
- A. needs no seed for the second crop
 - B. destroys the late crop at harvest
 - C. harvests food early while the late crop is still growing, and uses the land better
 - D. can avoid weeding the farm
- 40.** Which of the following shows the BEST use of knowledge of maturity stages on a family farm?
- A. Planting crops of different maturity periods so that the family has food all year round
 - B. Leaving all crops in the field until the next rainy season
 - C. Planting all the crops on the same day and harvesting them together
 - D. Harvesting every crop as soon as it flowers

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Read all values from a graph carefully and give the unit with every measurement.

Question 7

- (a) Use a pencil and a ruler where a table or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 8: Ecosystem

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. An ecosystem is best described as
 - A. all the plants growing in one place only
 - B. the climate of a particular country
 - C. a community of living things together with the non-living surroundings in which they live
 - D. a collection of animals kept in a zoo
2. The particular place where an organism lives is called its
 - A. population
 - B. community
 - C. habitat
 - D. niche of the soil
3. Which of the following best shows the difference between a habitat and an ecosystem?
 - A. A habitat contains only animals; an ecosystem contains only plants
 - B. A habitat is the place where an organism lives; an ecosystem includes all the organisms of an area and their non-living surroundings
 - C. A habitat has no living things, while an ecosystem has living things
 - D. A habitat is always larger than an ecosystem
4. All the different populations of organisms living together in one area form a
 - A. species
 - B. habitat
 - C. community
 - D. biome of soil
5. Which of the following is a NON-LIVING component of a pond ecosystem?
 - A. Water lily plants
 - B. Dissolved oxygen in the water
 - C. Frogs on the bank
 - D. Tilapia in the water
6. All the tilapia living in one pond make up a
 - A. ecosystem
 - B. community
 - C. food web
 - D. population
7. Which of the following is an example of an ecosystem in Ghana?
 - A. A bag of maize in a store
 - B. The Kakum forest
 - C. A photograph of an elephant
 - D. A list of animals in a notebook
8. The non-living parts of an ecosystem include
 - A. plants, animals and bacteria
 - B. only the soil and the rocks
 - C. the farmers who work in the area
 - D. sunlight, water, air, soil and temperature

9. Green plants are described as producers in an ecosystem because they

- A. make their own food by photosynthesis
- B. break down dead bodies in the soil
- C. produce young ones every season
- D. eat other organisms for food

11. A grasshopper that feeds on grass is a

- A. decomposer
- B. producer
- C. secondary consumer
- D. primary consumer

10. Organisms that feed on other organisms are called

- A. decomposers
- B. consumers
- C. producers
- D. autotrophs

- 13.** Animals that feed on both plants and animals are described as
- omnivores
 - herbivores
 - carnivores
 - producers
- 15.** Which of the following shows interdependence between a plant and an animal?
- A rock lies in the shade of a tree
 - Wind blows the leaves of a tree
 - A bee feeds on nectar and at the same time pollinates the flower
 - Rain falls on a grassland
- 17.** Two kinds of bird feeding on the same insects in one area are said to be in
- pollination
 - cooperation
 - decomposition
 - competition
- 19.** In a food chain, the arrow between two organisms means
- runs away from
 - lives near
 - is eaten by
 - is bigger than
- 21.** Every food chain begins with
- a predator
 - a decomposer
 - a secondary consumer
 - a producer
- 12.** Bacteria and fungi are important in an ecosystem because they
- make food for green plants by photosynthesis
 - give out oxygen during the night only
 - break down dead plants and animals and release nutrients into the soil
 - hunt and kill small animals for food
- 14.** The ultimate source of energy for almost every ecosystem on the Earth is
- the soil
 - water in rivers
 - decomposers in the ground
 - the Sun
- 16.** Oxygen given out by green plants is used by animals, and carbon dioxide breathed out by animals is used by plants. This relationship shows that
- neither plants nor animals need gases
 - plants can live without animals of any kind
 - plants and animals depend on each other for survival
 - animals are more important than plants
- 18.** A food chain shows
- the way water moves through the soil
 - the number of animals in a habitat
 - the way food energy passes from one organism to another in an ecosystem
 - the order in which crops are planted
- 20.** Which of the following is a correctly written food chain?
- Grass -> grasshopper -> lizard -> hawk
 - Lizard -> grass -> hawk -> grasshopper
 - Hawk -> lizard -> grasshopper -> grass
 - Grasshopper -> grass -> hawk -> lizard

- 23.** A food web is best described as
- a list of all the plants in a habitat
 - a net used to catch fish in a river
 - a single food chain with many animals
 - a number of food chains that are joined together in an ecosystem
- 25.** Only a small part of the energy taken in at one level of a food chain passes to the next level because most of it is
- stored for ever in the bones
 - destroyed by the Sun
 - used for life processes and lost as heat
 - changed into new soil
- 27.** Which of the following human activities destroys an aquatic ecosystem MOST quickly?
- Illegal small-scale mining (galamsey) in and along a river
 - Fishing with a line and hook
 - Walking along the river bank
 - Washing hands in the river once
- 29.** Bush burning before farming is harmful to the ecosystem because it
- destroys the humus, the small organisms in the soil and the homes of many animals
 - increases the number of earthworms
 - adds large amounts of water to the soil
 - makes the soil hold more nutrients
- 31.** Overhunting of an animal species in a forest is likely to
- improve the balance of the ecosystem
 - have no effect on any other organism
 - reduce its numbers so much that the species may disappear from the area
 - increase the number of that species
- 22.** In the food chain maize → weaver bird → hawk, the hawk is the
- secondary consumer
 - primary consumer
 - producer
 - decomposer
- 24.** A food web gives a better picture of feeding in an ecosystem than a single food chain because it shows that
- most organisms feed on more than one kind of food
 - animals never compete for food
 - only one animal eats each plant
 - energy is created in the ecosystem
- 26.** In a food web, the snakes are removed by hunters. The MOST likely immediate result is that the number of
- mice falls to zero
 - mice and lizards increases
 - hawks increases rapidly
 - grass plants increases at once
- 28.** Galamsey harms a river chiefly because it
- makes the river flow faster to the sea
 - fills the water with mud and poisonous chemicals such as mercury
 - adds useful nutrients to the water
 - increases the number of fish in the water
- 30.** Spraying a large quantity of pesticide on a farm may harm the ecosystem because the pesticide
- turns into harmless water in one day
 - kills useful organisms as well as pests and may build up in the bodies of animals along a food chain
 - attracts more bees to the flowers
 - is used up at once by the crop

- 33.** Cutting down a large area of forest for farming usually leads to
- A. cooler weather and richer soil every year
 - B. loss of habitats, soil erosion and reduced rainfall in the area
 - C. an increase in the number of forest animals
 - D. deeper topsoil on the cleared land
- 35.** Which of the following practices helps to protect an ecosystem?
- A. Burning the bush every dry season
 - B. Dumping refuse into the nearest stream
 - C. Hunting animals in the breeding season
 - D. Replanting trees in areas where they have been cut down
- 37.** A community wishes to protect the fish in its river. The BEST measure is to
- A. agree on a closed season and stop the use of poisons and small-mesh nets
 - B. pour chemicals in to kill the weeds
 - C. catch all the big fish before they breed
 - D. allow mining in the river during the rains
- 39.** A class finds that the number of frogs in a stream has fallen sharply since a farm began using a strong pesticide nearby. The BEST conclusion is that
- A. frogs always disappear in the dry season
 - B. the frogs were eaten by the farmers
 - C. pesticides cannot reach a stream
 - D. the pesticide washed into the stream may be responsible, and further investigation is needed
- 32.** Which of the following is a NATURAL event that can upset the balance of an ecosystem?
- A. Illegal mining
 - B. Bush burning by farmers
 - C. The use of pesticides
 - D. A volcanic eruption
- 34.** Waste water from a factory containing poisonous chemicals is discharged into a lagoon. The FIRST organisms to be affected are likely to be
- A. the plants and small organisms living in the water
 - B. the birds flying over the lagoon
 - C. the goats grazing on a distant hill
 - D. the people living in the next region
- 36.** Areas such as Mole National Park are set aside MAINLY to
- A. store the refuse of the nearby towns
 - B. protect wildlife and their habitats
 - C. provide land for building factories
 - D. allow free hunting throughout the year
- 38.** The planting of trees along a river bank helps the ecosystem by
- A. drying up the river quickly
 - B. holding the soil, shading the water and providing homes for organisms
 - C. increasing the mud washed into the river
 - D. preventing fish from breeding
- 40.** Which of the following statements BEST explains why the balance of an ecosystem must be protected?
- A. Every organism depends on others, so damage to one part affects the whole system, including people
 - B. Human beings live outside the ecosystem
 - C. Ecosystems repair themselves fully within a week

D. Only large animals matter in an ecosystem

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) When you write a food chain, use arrows and make sure each arrow points from the organism that is eaten to the organism that eats it.

Question 7

- (a) Use a pencil and a ruler where a table or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 9: Electricity and E

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. An electric circuit is best described as
 - A. a wire that stores electricity for later use
 - B. a device that changes current into voltage
 - C. any arrangement of switches in a building
 - D. a complete path along which an electric current can flow
2. For a current to flow in a circuit, the circuit must be
 - A. open at one point
 - B. made only of insulators
 - C. closed and contain a source of electrical energy
 - D. connected to an ammeter at both ends
3. Which of the following is used to measure the current in a circuit?
 - A. A voltmeter
 - B. A rheostat
 - C. An ammeter
 - D. A galvanometer scale in volts
4. A voltmeter is always connected
 - A. between the two terminals of the ammeter
 - B. after the switch has been removed
 - C. in series with the component being measured
 - D. across (in parallel with) the component being measured
5. The symbol consisting of a long thin line and a short thick line side by side represents
 - A. a switch
 - B. a cell
 - C. a lamp
 - D. a resistor
6. Which material is the BEST conductor of electricity?
 - A. Dry wood
 - B. Rubber
 - C. Copper
 - D. Plastic
7. A switch is placed in a circuit in order to
 - A. increase the voltage of the cells
 - B. measure the current in the circuit
 - C. change alternating current into direct current
 - D. open or close the circuit so that current can be stopped or allowed
8. In a series circuit, the components are connected
 - A. in separate circuits with separate cells
 - B. with each one having its own switch only
 - C. side by side across the same two points
 - D. one after another in a single path
9. Three identical lamps are connected in series to a battery. If one lamp burns out,
 - A. the other two lamps burn more brightly

- B. all the lamps go off
- C. only the burnt lamp goes off
- D. the other two lamps continue to burn normally

- 10.** Two identical lamps are connected in parallel across a battery. If one lamp is removed,
- A. the battery is short-circuited
 - B. the current in the circuit becomes zero

- C. the other lamp continues to burn
D. the other lamp also goes off
- 11.** The current is the SAME at every point in
A. a series circuit
B. any circuit with more than one lamp
C. every branch of a parallel circuit
D. a circuit with no cell
- 13.** Lamps in a house are connected in parallel mainly because
A. less wire is needed for the work
B. each lamp can be switched on or off without affecting the others
C. the lamps then use no electrical energy
D. the voltage across each lamp is reduced
- 15.** Two identical lamps joined in series to a battery glow dimly. When the same lamps are joined in parallel to the same battery, they
A. do not glow at all
B. glow more brightly, because each receives the full voltage of the battery
C. glow more dimly, because the current is shared
D. glow exactly as before
- 17.** In an electric lamp, electrical energy is changed mainly into
A. sound energy and light energy
B. potential energy and heat energy
C. light energy and heat energy
D. chemical energy and light energy
- 19.** An electric iron changes electrical energy mainly into
A. heat energy
B. light energy
C. sound energy
D. chemical energy
- 21.** When a cell drives a current round a circuit, the energy in the cell is
A. light energy changed into
- C. the other lamp continues to burn
D. the other lamp also goes off
- 12.** In a parallel circuit the quantity that is the same across each branch is the
A. current in each branch
B. resistance of each branch
C. length of wire in each branch
D. potential difference
- 14.** Two resistors of $3\ \Omega$ and $5\ \Omega$ are joined in series. Their combined resistance is
A. $15\ \Omega$
B. $2\ \Omega$
C. $8\ \Omega$
D. $1.9\ \Omega$
- 16.** A battery supplies a current of $0.9\ \text{A}$ to two lamps connected in parallel. If one lamp takes $0.5\ \text{A}$, the current in the other lamp is
A. $0.9\ \text{A}$
B. $0.4\ \text{A}$
C. $1.4\ \text{A}$
D. $0.45\ \text{A}$
- 18.** The energy change that takes place in an electric fan is from electrical energy to
A. chemical energy
B. light energy
C. nuclear energy
D. kinetic (mechanical) energy
- 20.** In a radio, electrical energy is changed mainly into
A. heat energy only
B. light energy
C. sound energy
D. chemical energy

- electrical energy
- B. electrical energy changed into chemical energy
 - C. heat energy changed into chemical energy
 - D. chemical energy changed into electrical energy
- 23.** The relationship between potential difference, current and resistance is given by
- A. $V = IR$
 - B. $V = R/I$
 - C. $V = I + R$
 - D. $V = I/R$
- 22.** A filament lamp is said to be inefficient because
- A. it changes all the energy supplied into light
 - B. most of the electrical energy supplied to it is changed into heat rather than light
 - C. it destroys some of the energy supplied to it
 - D. it uses a very small current

- 25.** A current of 2 A flows through a resistor of $6\ \Omega$. The potential difference across the resistor is
- 0.33 V
 - 12 V
 - 3 V
 - 4 V
- 27.** A lamp of resistance $24\ \Omega$ is connected to a 12 V battery. The current in the lamp is
- 288 A
 - 0.5 A
 - 12 A
 - 2 A
- 29.** Three identical resistors are joined in series across a 12 V battery. The potential difference across each resistor is
- 3 V
 - 12 V
 - 36 V
 - 4 V
- 31.** A diode is a component that
- allows current to pass equally in both directions
 - allows current to pass in one direction only
 - changes heat energy into electrical energy
 - stores electric charge for later use
- 33.** A diode is said to be forward biased when its
- positive terminal is joined to the negative terminal of the supply
 - two terminals are joined together
 - terminals are both joined to the negative of the supply
 - positive terminal (anode) is joined to the positive terminal of the supply
- 35.** An LED in a circuit fails to light when the cells are connected one way but lights when the cells are turned round. This shows that the LED
- needs a larger capacitor to work
 - conducts equally well in both directions
- 24.** The unit of resistance is the
- ohm
 - watt
 - ampere
 - volt
- 26.** A potential difference of 9 V drives a current of 3 A through a conductor. The resistance of the conductor is
- $3\ \Omega$
 - $0.33\ \Omega$
 - $12\ \Omega$
 - $27\ \Omega$
- 28.** Two resistors, each of $4\ \Omega$, are connected in series across a 16 V supply. The current in the circuit is
- 8 A
 - 0.5 A
 - 4 A
 - 2 A
- 30.** A pupil doubles the resistance in a circuit while keeping the voltage the same. The current in the circuit
- falls to one quarter of its former value
 - stays exactly the same
 - falls to half its former value
 - doubles
- 32.** The letters LED stand for
- light-emitting diode
 - low energy device
 - linked electrode diode
 - light energy detector
- 34.** When a diode is reverse biased,
- practically no current flows through it
 - a very large current flows through it
 - it stores the current until the supply is reversed
 - it behaves exactly like a piece of copper wire

- C. conducts only when it is forward biased
- D. has been damaged by the cells

37. A resistor is joined in series with an LED mainly to

- A. store charge for the LED
- B. change the direction of the current
- C. limit the current so that the LED is not damaged
- D. increase the brightness of the LED

36. In an LED, electrical energy is changed mainly into

- A. light energy
- B. kinetic energy
- C. chemical energy
- D. sound energy

- 39.** A pupil connects an LED directly to a 9 V battery with no other component. The most likely result is that the LED
- A. lights brightly for a moment and is then damaged by the large current
 - B. does not light at all under any condition
 - C. works normally for a very long time
 - D. stores the charge from the battery

- 38.** A capacitor in an electronic circuit is used to
- A. change electrical energy into sound
 - B. store electric charge and release it when needed
 - C. measure the current in the circuit
 - D. allow current to pass in one direction only
- 40.** In a circuit containing a capacitor and a resistor joined to an LED, the LED is seen to stay on for a short time after the supply is switched off. The best explanation is that the
- A. capacitor discharges through the resistor and the LED
 - B. resistor produces current of its own
 - C. switch has failed to open the circuit
 - D. LED stores light energy inside itself

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the circuit diagrams carefully before you answer the questions on them.

Question 6

- (a) Show all the steps of your working in the questions that involve calculations, and give the unit with every answer.

Question 7

- (a) Use a pencil and a ruler when you are asked to draw a circuit diagram, and use the accepted circuit symbols. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 10: Energy

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Conservation of energy, as used in everyday life, means
 - A. making new energy in a power station
 - B. destroying the energy that is not needed
 - C. using energy carefully so that none of it is wasted
 - D. storing electricity inside a wire
2. Which of the following is a way of conserving energy in the home?
 - A. Opening the refrigerator again and again
 - B. Switching off appliances when they are not in use
 - C. Boiling a full kettle for one cup of tea
 - D. Leaving the television on all night
3. Ironing all the week's clothes at one time instead of ironing daily conserves energy because the iron
 - A. uses a smaller current when it is full
 - B. is heated up once instead of many times
 - C. stores the heat it makes for the next day
 - D. works without any current at all
4. An energy-efficient lamp is one that
 - A. uses more energy but lasts longer
 - B. gives a dim light so that it saves money
 - C. works only in the daytime
 - D. gives the same brightness while using less electrical energy
5. Which lamp converts the LARGEST fraction of the electrical energy supplied to it into waste heat?
 - A. A fluorescent tube
 - B. A filament bulb
 - C. A light-emitting diode (LED) lamp
 - D. A compact fluorescent lamp
6. Opening the windows and curtains during the day instead of switching on the lights is an example of
 - A. increasing the power of the lamps
 - B. storing energy for the night
 - C. using natural daylight to conserve energy
 - D. converting sunlight into electricity
7. Which of these practices WASTES electrical energy?
 - A. Using a pressure cooker to cook beans
 - B. Fitting a lid on a cooking pot
 - C. Leaving a phone charger plugged in at the socket after the phone is removed
 - D. Drying clothes on a line in the sun
8. A school wishes to cut its electricity bill. The BEST first step is to
 - A. ask the pupils to close the school earlier every day
 - B. carry out an energy audit to find where energy is being wasted
 - C. buy a bigger meter
 - D. change all the sockets in the school

- 9.** One important benefit of conserving energy in the home is that
- the meter stops recording
 - the voltage of the mains supply increases
 - the family pays a smaller electricity bill
 - the appliances work faster
- 11.** Which of the following is a non-renewable source of energy?
- Sunlight
 - Running water
 - Crude oil
 - Wind
- 13.** During a period of load shedding, the step by consumers that would help MOST is
- using candles in every room at all times
 - running generators in every house
 - switching off air conditioners and other heavy appliances at peak hours
 - disconnecting the meter from the supply
- 15.** The electrical energy used by an appliance is measured by the electricity company in
- kilowatt-hours
 - newton metres per ampere
 - amperes
 - volts per second
- 17.** A 100 W lamp is left on for 10 hours. The energy it uses is
- 10 kWh
 - 1 kWh
 - 100 kWh
 - 1 000 kWh
- 19.** An appliance of power 2 kW is used for 3 hours. The energy consumed is
- 5 kWh
 - 6 kWh
 - 0.67 kWh
 - 60 kWh
- 10.** Conserving energy helps to protect the environment mainly because it reduces
- the burning of fuels and the release of carbon dioxide and other pollutants
 - the number of rivers in the country
 - the rate at which plants make food
 - the amount of sunlight reaching the earth
- 12.** Energy conservation is important to a country because it
- increases the number of appliances in the shops
 - makes the limited supply of electricity serve more people and industries
 - removes the need for power stations
 - makes electricity free for everybody
- 14.** A community decides to replace all its filament street lamps with LED lamps. The most important long-term gain is that
- the electricity supply will become free
 - the streets become brighter than daylight
 - the lamps will never need any maintenance
 - the same lighting is obtained for far less energy, saving fuel and money each year
- 16.** Power, current and voltage are related by the equation
- $P = IV$
 - $P = I + V$
 - $P = V / I$
 - $P = I / V$
- 18.** An electric iron is marked 1 500 W. Expressed in kilowatts, its power is
- 0.15 kW
 - 150 kW
 - 15 kW
 - 1.5 kW

- 21.** A refrigerator takes a current of 2 A from a 240 V supply. Its power is
- 120 W
 - 242 W
 - 4 800 W
 - 480 W
- 23.** Two lamps, one of 40 W and one of 60 W, are both switched on for 8 hours. The energy used altogether is
- 8 kWh
 - 0.1 kWh
 - 800 kWh
 - 0.8 kWh
- 25.** The image formed on the screen of a pinhole camera is
- virtual and upright
 - virtual and inverted
 - real and inverted
 - real and upright
- 27.** In a lens camera, the part that controls the amount of light entering the camera is the
- view finder
 - aperture
 - film or sensor
 - shutter release button
- 29.** A shadow is formed because light
- bends round every object it meets
 - is absorbed by the air around the object
 - is destroyed when it strikes an object
 - travels in straight lines and cannot pass through an opaque object
- 31.** The partially dark region at the edge of a shadow is the
- penumbra
 - image
 - umbra
 - focus
- 33.** An eclipse of the sun occurs when
- the sun comes between the earth and the moon
 - the moon moves behind the sun
 - the earth comes between the sun and the moon
 - the moon comes between the
- 20.** If one kilowatt-hour costs GH¢1.50, the cost of using a 2 kW heater for 4 hours is
- GH¢12.00
 - GH¢48.00
 - GH¢6.00
 - GH¢3.00
- 22.** A 60 W fan is used for 5 hours each day. The energy it uses in 30 days is
- 900 kWh
 - 9 kWh
 - 0.3 kWh
 - 90 kWh
- 24.** In a simple pinhole camera, light enters the box through
- a very small hole in the front
 - a lamp placed inside the box
 - a mirror fixed at the back
 - a wide opening in the side
- 26.** The inside of a camera box is painted black in order to
- absorb stray light so that the image is clear
 - make the image larger
 - reflect light on to the screen
 - keep the box cool during use
- 28.** An object 12 cm tall stands 60 cm in front of a pinhole camera whose box is 15 cm long. The height of the image on the screen is
- 48 cm
 - 12 cm
 - 6 cm
 - 3 cm
- 30.** The region of a shadow that receives no light at all is called the
- penumbra
 - umbra
 - aperture
 - eclipse
- 32.** A penumbra is formed only when the source of light is
- very small, like a pinhole
 - a source of coloured light
 - broad, so that different parts of it light different parts of the screen
 - placed behind the screen

sun and the earth

- 35.** A total eclipse of the sun is seen only by observers standing in the
- penumbra of the moon's shadow
 - bright region outside the shadow
 - umbra of the moon's shadow on the earth
 - umbra of the earth's shadow
- 37.** Light changes its path on passing from one medium to another because its
- colour changes in the new medium
 - energy is destroyed at the surface
 - speed changes in the new medium
 - wavelength is turned into sound
- 39.** A straight stick held partly in a bucket of water appears bent at the water surface. This is because light from the part under water
- changes direction as it leaves the water and enters the air
 - is completely absorbed by the water
 - travels in a curved path inside the water
 - is reflected by the bottom of the bucket
- 34.** During an eclipse of the moon,
- the moon casts its shadow on the earth
 - the sun passes behind the earth
 - the moon stops reflecting light for ever
 - the earth casts its shadow on the moon
- 36.** A ray of light travelling from air into a glass block changes direction. This change of direction is called
- refraction
 - dispersion
 - reflection
 - diffusion
- 38.** A ray of light passing from air into water is bent
- towards the normal
 - away from the normal
 - back along its own path
 - along the water surface
- 40.** A fisherman at Elmina sees a fish in clear water and aims his spear straight at where he sees it, but misses. The best explanation is that the fish
- swims away each time the spear is thrown
 - is a reflection in the surface of the water
 - appears deeper than it really is, so he aims too low
 - is deeper than it appears, because light from it is refracted as it leaves the water

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps of your working in the questions that involve calculations, and give the unit with every answer.

Question 7

- (a) Use a pencil and a ruler where a table or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 11: Force and Motion

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Pressure is best described as
 - A. the total force acting on a surface
 - B. the force acting normally on each unit area of a surface
 - C. the weight of a body divided by its volume
 - D. the force needed to lift a body
2. The formula for pressure is
 - A. $\text{pressure} = \text{force} + \text{area}$
 - B. $\text{pressure} = \text{force} \div \text{area}$
 - C. $\text{pressure} = \text{force} \times \text{area}$
 - D. $\text{pressure} = \text{area} \div \text{force}$
3. The SI unit of pressure is the
 - A. pascal
 - B. joule
 - C. newton
 - D. newton metre
4. One pascal is the same as
 - A. one newton per square metre
 - B. one joule per square metre
 - C. one newton square metre
 - D. one newton per metre
5. For a given force, the pressure exerted on a surface increases when the area of contact is
 - A. turned upside down
 - B. kept exactly the same
 - C. made larger
 - D. made smaller
6. Two boxes of the same weight rest on the floor. The box that exerts the greater pressure is the one with the
 - A. greater height above the floor
 - B. rougher surface in contact with the floor
 - C. larger area of contact with the floor
 - D. smaller area of contact with the floor
7. Which of the following is measured in newtons?
 - A. Area
 - B. Force
 - C. Pressure
 - D. Mass
8. A force acting on a very large area produces
 - A. a small pressure
 - B. a large pressure
 - C. a large area
 - D. no force at all
9. A knife is sharpened before use because a sharp edge
 - A. increases the weight of the knife
 - B. has a larger area, so it exerts a larger force
 - C. reduces the force the hand must apply to zero
 - D. has a very small area, so it exerts a large pressure on the material
10. A tractor that works on soft farmland is fitted with wide tyres mainly to
 - A. increase the pressure on the soil so that it grips
 - B. spread its weight over a large area and so reduce the pressure on the soil
 - C. reduce the weight of the tractor
 - D. make the tractor heavier

- 11.** The foundation of a building is made wider than the wall above it in order to
- increase the pressure on the soil
 - allow water to pass under the wall
 - reduce the weight of the building
 - spread the load over a greater area and reduce the pressure on the soil
- 12.** A person drinks through a straw. The drink rises in the straw because
- the straw pulls the drink upwards with a force
 - the drink becomes lighter inside the straw
 - the air inside the straw pushes the drink up
 - the pressure inside the straw is lowered and the atmosphere pushes the drink up
- 13.** Air is pumped into a car tyre until it is hard. The air inside the tyre
- presses equally on all parts of the inside of the tyre
 - presses only on the valve
 - presses only on the bottom of the tyre
 - loses its pressure as soon as the pump is removed
- 14.** At a washing bay, water from a hose cleans better when a narrow nozzle is fitted because the
- water becomes heavier in the narrow nozzle
 - water leaves through a smaller area and so strikes the car with greater pressure
 - nozzle increases the volume of water used
 - water is cooled by the narrow nozzle
- 15.** Walking on soft sand is easier in flat sandals than in shoes with narrow heels because flat sandals
- make the person lighter on the sand
 - press the sand with a greater force
 - spread the weight over a larger area, lowering the pressure on the sand
 - increase the pressure under the foot
- 16.** Which of these is an application of pressure in everyday life?
- Using a pulley to raise a load
 - Using a syringe to draw up a liquid
 - Using a lever to lift a stone
 - Using a thermometer to measure temperature
- 17.** A force of 200 N acts on an area of 4 m². The pressure exerted is
- 204 Pa
 - 0.02 Pa
 - 50 Pa
 - 800 Pa
- 18.** A box exerts a pressure of 300 Pa on a floor through an area of 2 m². The force exerted by the box is
- 302 N
 - 600 N
 - 150 N
 - 0.0067 N
- 19.** A block of weight 480 N rests on a face of area 0.24 m². The pressure on the ground is
- 20 000 Pa
 - 115.2 Pa
 - 500 Pa
 - 2 000 Pa
- 20.** A girl of weight 500 N stands on one foot of area 0.025 m². The pressure she exerts on the ground is
- 2 000 Pa
 - 12.5 Pa
 - 20 000 Pa
 - 500 Pa

- 21.** A pressure of 4 000 Pa is produced by a force of 1 200 N. The area over which the force acts is
- A. 4 800 000 m²
 - B. 0.3 m²
 - C. 3.3 m²
 - D. 3 000 m²
- 23.** Newton's third law of motion states that
- A. for every action there is an equal and opposite reaction
 - B. action is always greater than reaction
 - C. force is equal to mass times acceleration
 - D. a body continues at rest unless a force acts on it
- 25.** A boy pushes a wall with a force of 50 N. The force with which the wall pushes back on the boy is
- A. 25 N
 - B. zero
 - C. 50 N
 - D. 100 N
- 27.** A gun recoils when it is fired. The recoil is explained by the fact that
- A. the air pushes the gun backwards
 - B. the burning powder pushes only the bullet
 - C. the gun is heavier than the bullet
 - D. the bullet is pushed forward and the gun is pushed back with an equal force
- 29.** A man standing in a small canoe steps forward on to the bank. The canoe moves backwards because
- A. the water pushes the canoe backwards
 - B. the canoe is lighter than the man
 - C. the force he exerts backwards on the canoe is matched by the force the canoe exerts forwards on him
 - D. his weight increases as he steps
- 22.** A crate exerts a pressure of 1 500 Pa when it stands on a face of 0.4 m². When the same crate stands on a face of 0.2 m², the pressure becomes
- A. 3 000 Pa
 - B. 1 500 Pa
 - C. 750 Pa
 - D. 6 000 Pa
- 24.** The action and the reaction described by Newton's third law
- A. act on the same body
 - B. are always unequal in size
 - C. act on two different bodies
 - D. act in the same direction
- 26.** A swimmer moves forward in water by pushing the water
- A. sideways, so that the water lifts her up
 - B. backwards, so that the water pushes her forwards
 - C. forwards, so that the water pushes her forwards
 - D. downwards, so that the water pushes her backwards
- 28.** When a balloon full of air is released and the air rushes out backwards, the balloon
- A. moves backwards with the escaping air
 - B. stays still because the two forces cancel
 - C. rises straight up because air is light
 - D. moves forward because the escaping air pushes the balloon the other way
- 30.** A pupil says that action and reaction always cancel each other, so nothing can ever move. The mistake in this statement is that the two forces
- A. are never really equal in size
 - B. act at different times, so one comes first
 - C. act on different bodies, so they cannot cancel each other
 - D. are both pushes and never pulls

- 31.** Newton's first law of motion states that a body remains at rest, or moves with uniform velocity in a straight line, unless
- it is placed on a rough surface
 - its weight is removed
 - its mass is increased
 - an external force acts on it
- 33.** Passengers in a bus are thrown forward when the driver brakes suddenly. This is explained by
- Newton's first law, since the passengers tend to continue moving forward
 - Newton's third law, since the seat pushes them
 - the law of conservation of energy
 - the increase in the weight of the passengers
- 35.** Wearing a seat belt in a vehicle is important mainly because the belt
- removes the inertia of the passenger
 - reduces the speed of the vehicle
 - provides the force needed to stop the passenger moving forward in a sudden stop
 - increases the mass of the passenger
- 37.** A body of mass 50 kg is given an acceleration of 2 m/s^2 . The force acting on it is
- 100 N
 - 0.04 N
 - 52 N
 - 25 N
- 39.** A loaded lorry and an empty lorry move at the same speed. The loaded lorry is harder to stop because it has
- a smaller mass but a greater speed
 - less momentum than the empty lorry
 - greater momentum, so a larger force is needed to stop it in the same time
 - no inertia at all
- 32.** The tendency of a body to remain in its state of rest or of uniform motion is called
- inertia
 - pressure
 - momentum
 - acceleration
- 34.** Newton's second law of motion leads to the relation
- $F = m + a$
 - $F = a/m$
 - $F = m/a$
 - $F = ma$
- 36.** A force of 20 N acts on a mass of 4 kg. The acceleration produced is
- 5 m/s^2
 - 24 m/s^2
 - 80 m/s^2
 - 0.2 m/s^2
- 38.** The momentum of a moving body is given by
- force $\times$ time
 - mass $\times$ acceleration
 - mass $\times$ velocity
 - mass $\div$ velocity
- 40.** A trolley of mass 2 kg is pushed with a force of 6 N on a smooth bench. If the force is doubled and the mass is kept the same, the acceleration
- stays at 3 m/s^2
 - also doubles, from 3 m/s^2 to 6 m/s^2
 - becomes 12 m/s^2
 - is halved, from 3 m/s^2 to 1.5 m/s^2

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps of your working in the questions that involve calculations, and give the unit with every answer.

Question 7

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 12: Human Health

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. According to the World Health Organisation (WHO), health is
 - A. a state of complete physical, mental and social well-being and not merely the absence of disease
 - B. a condition in which a person has never visited a hospital
 - C. the ability to work from morning until evening
 - D. the absence of pain in any part of the body
2. A disease may be described as
 - A. any pain felt after hard work
 - B. any feeling of hunger in the morning
 - C. any condition that upsets the normal working of the body or mind
 - D. any change in the weight of a person
3. Which of the following statements BEST shows the relationship between health and disease?
 - A. Health and disease have no connection with each other
 - B. A person in good health can never fall ill
 - C. Disease lowers the level of health of a person
 - D. Disease improves the health of a person who recovers
4. A girl has no illness but is unhappy, lonely and unable to mix with others in her class. Using the WHO definition, she is
 - A. not completely healthy, because her mental and social well-being is poor
 - B. completely healthy, because she has no disease
 - C. suffering from a communicable disease
 - D. healthy, since only the body matters in health
5. A disease that can be passed from one person to another is said to be
 - A. non-communicable
 - B. communicable
 - C. hereditary only
 - D. deficiency-related
6. Which of the following is a communicable disease?
 - A. Diabetes
 - B. Cholera
 - C. High blood pressure
 - D. Asthma
7. Non-communicable diseases are diseases that
 - A. are not passed from one person to another
 - B. are caused only by bacteria
 - C. affect only old people
 - D. are spread by mosquitoes
8. Which of the following is a non-communicable disease?
 - A. Ringworm
 - B. Tuberculosis
 - C. Measles
 - D. Diabetes

- 9.** Kwashiorkor and marasmus in young children are caused mainly by
- the bite of a tsetse fly
 - drinking boiled water
 - playing in the sun
 - malnutrition
- 11.** Frequent passing of urine, great thirst and loss of weight are symptoms of
- measles
 - ringworm
 - diabetes mellitus
 - a common cold
- 13.** Which of these habits is a risk factor for several non-communicable diseases?
- Smoking tobacco and taking alcohol in large amounts
 - Sleeping for about eight hours a night
 - Taking part in sports twice a week
 - Eating a balanced diet daily
- 15.** Cancer is a disease in which
- the blood loses all its red cells
 - certain body cells divide in an uncontrolled way and form a growth
 - a virus destroys the bones of the body
 - the body produces too much urine
- 17.** A sore that does not heal, a lump under the skin and unexplained loss of weight are
- signs of good health
 - symptoms of ringworm
 - warning signs of cancer
 - symptoms of malaria
- 19.** Women are advised to examine their breasts regularly mainly in order to
- prevent the breasts from paining
 - find any lump early, when treatment is more likely to succeed
 - avoid catching an infection from others
 - make the breasts grow larger
- 10.** Which of the following is a risk factor for high blood pressure?
- Eating too much salt and fatty food
 - Walking to school every morning
 - Eating fruits and vegetables
 - Drinking clean water daily
- 12.** A worker in a quarry who breathes in dust for many years without a nose mask is MOST likely to develop a disease of the
- kidneys
 - lungs
 - ears
 - teeth
- 14.** The BEST general way of preventing many non-communicable diseases is to
- boil all drinking water before use
 - take antibiotics every month
 - sleep under a treated mosquito net
 - eat a balanced diet, exercise regularly and avoid tobacco and excess alcohol
- 16.** Which of the following is a known cause or risk factor of cancer?
- Eating a diet rich in vegetables
 - Drinking clean water
 - Taking regular exercise
 - Long exposure to tobacco smoke
- 18.** Which cancer is MOST commonly linked with smoking?
- Bone cancer
 - Skin cancer
 - Lung cancer
 - Mouth ulcer
- 20.** Which of the following lifestyle changes would do MOST to reduce a person's risk of cancer?
- Taking painkillers at the first sign of pain
 - Stopping smoking and eating more fruits and vegetables
 - Avoiding all contact with sick people
 - Sleeping in the afternoon every day

- 21.** A common disease in a community is one that
- A. has no known treatment anywhere
 - B. appears only once in many years
 - C. occurs in a large number of people in that community
 - D. is found only in children under five years
- 23.** Malaria is spread in a community by the bite of
- A. a tsetse fly
 - B. a housefly
 - C. a female *Anopheles* mosquito
 - D. a cockroach
- 25.** Which of the following pairs gives a disease and its correct method of prevention?
- A. Malaria - boiling all drinking water
 - B. Diabetes - sleeping under a treated net
 - C. Cholera - using a toilet and washing the hands with soap
 - D. Ringworm - drinking plenty of water
- 27.** A survey in a town showed 120 cases of malaria out of 300 cases reported in a month. The percentage of the cases that were malaria is
- A. 60%
 - B. 40%
 - C. 12%
 - D. 30%
- 29.** Fungal diseases of humans are caused by
- A. fungi, which grow on the skin or in moist parts of the body
 - B. viruses carried in the air
 - C. worms living in the intestine
 - D. a shortage of vitamins in the diet
- 31.** Ringworm is spread from one pupil to another mainly by
- A. eating from the same bowl of rice
 - B. being bitten by a mosquito
 - C. sharing combs, towels, caps and clothing with an infected person
- 22.** The BEST way for a class to find out the common diseases of its community is to
- A. count the number of chemical shops in the town
 - B. copy a list of diseases from a textbook
 - C. ask one elder in the town to name them
 - D. carry out a survey and record the cases reported at the clinic and in the homes
- 24.** Cholera spreads rapidly in a crowded community mainly through
- A. shaking hands with a patient
 - B. drinking water and food contaminated with faeces
 - C. breathing the air near a patient
 - D. sharing a seat with a patient
- 26.** Prolonged coughing with blood-stained sputum, chest pain, night sweating and loss of weight suggest
- A. diabetes
 - B. tuberculosis
 - C. candidiasis
 - D. a sprained chest muscle
- 28.** A community records many cases of diarrhoea every rainy season. The MOST useful long-term action for the community is to
- A. close the clinic during the rainy season
 - B. give every household painkillers
 - C. ask people to stay indoors when it rains
 - D. provide safe drinking water and proper toilets and keep the gutters clean
- 30.** Ringworm of the scalp is BEST recognised by
- A. round, scaly patches with broken hair and loss of hair
 - B. yellow colouring of the eyes
 - C. swelling of the legs and feet
 - D. severe pain in the joints

D. breathing the same air in class

- 33.** Fungal infection of the toe nails and the skin between the toes is encouraged by
- A. cutting the nails short and keeping them clean
 - B. washing the feet with soap every day
 - C. wearing tight, damp shoes for long periods
 - D. walking barefoot on dry, clean sand
- 35.** The BEST way for a boarding school to prevent fungal skin infections among its pupils is to
- A. provide enough clean water, insist on personal towels and combs, and treat cases early
 - B. send away every pupil who falls ill
 - C. ask pupils to bathe only once a week
 - D. spray the dormitory with insecticide every night
- 37.** Which of the following is a role of the government in managing common diseases?
- A. Washing the clothes of every patient
 - B. Providing clinics, medicines and trained health workers for the people
 - C. Deciding what each family eats at home
 - D. Cutting the hair of pupils in schools
- 39.** A class is asked to design a poster on ringworm for the community. The MOST useful poster would
- A. describe the fungus in long scientific sentences
 - B. show only a picture of a sick child with no words
 - C. list the names of all the pupils in the class
 - D. show the signs of the disease and give short, clear steps on how to avoid and treat it
- 32.** Candidiasis, also known as thrush, is MOST commonly seen as
- A. white patches in the mouth or an itching infection of moist body folds
 - B. a deep wound on the sole of the foot
 - C. a hard swelling on the shin
 - D. a yellow discolouring of the teeth
- 34.** A pupil with ringworm should
- A. cover the patch with a plaster and tell nobody
 - B. stay away from water until it heals
 - C. rub the patch with sand to remove it
 - D. report to the clinic for treatment and stop sharing combs, caps and towels
- 36.** The role of an individual in controlling fungal diseases in the home is MAINLY to
- A. keep the illness secret from the family
 - B. buy drugs from the market without advice
 - C. wait for the health workers to visit the house
 - D. keep the body, clothing and bedding clean and dry, and avoid sharing personal articles
- 38.** A community can BEST help to control a disease outbreak by
- A. blaming the families in which the cases occurred
 - B. hiding the number of cases from health workers
 - C. waiting until the rains stop before acting
 - D. joining clean-up exercises and reporting cases early to the health post
- 40.** A town has a clinic, clean water and health education, yet records many cases of high blood pressure and diabetes. The MOST likely explanation is that
- A. the people's way of living, such

as diet, exercise and alcohol use,
has not changed

- B. the water supply is causing both diseases
- C. the clinic is spreading the diseases to the people
- D. the two diseases are passed from person to person

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps of your working in the questions that involve calculations.

Question 7

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 13: Life Cycle Grasshopper

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The grasshopper belongs to the group of animals called
 - A. worms
 - B. crabs
 - C. insects
 - D. spiders
2. The body of an adult grasshopper is divided into
 - A. thorax and abdomen only
 - B. head and trunk only
 - C. head, neck and tail
 - D. head, thorax and abdomen
3. The number of legs on an adult grasshopper is
 - A. six
 - B. four
 - C. eight
 - D. ten
4. The legs and wings of a grasshopper are attached to its
 - A. antennae
 - B. thorax
 - C. abdomen
 - D. head
5. The hind legs of the grasshopper are long and thick because they are used for
 - A. digging the soil
 - B. holding food
 - C. jumping
 - D. sensing the surroundings
6. The antennae of a grasshopper are used mainly for
 - A. breathing in air
 - B. chewing grass
 - C. laying eggs
 - D. feeling and smelling the surroundings
7. A grasshopper is MOST likely to be found in
 - A. grassland, a farm or a weedy field
 - B. a dark cupboard indoors
 - C. the roof of a house only
 - D. the bottom of a deep river
8. The life cycle of the grasshopper has
 - A. five stages
 - B. three stages
 - C. four stages
 - D. two stages
9. The correct order of the stages in the life cycle of the grasshopper is
 - A. egg, nymph, adult
 - B. nymph, egg, adult
 - C. egg, pupa, nymph
 - D. egg, larva, pupa, adult
10. The female grasshopper lays her eggs
 - A. on the surface of still water
 - B. in a pod in moist soil
 - C. under the leaves of a tree only
 - D. inside the body of another insect
11. The eggs of the grasshopper are laid in a pod chiefly to
 - A. help them float on water
 - B. protect them from drying up and from enemies

- C. make them hatch into pupae
- D. allow the wind to scatter them

13. The nymph of the grasshopper is best described as

- A. a worm-like creature quite unlike the adult
- B. a swimming stage that lives in water
- C. a small wingless insect that looks very much like the adult
- D. a resting stage covered by a hard case

12. The young one that hatches from the egg of a grasshopper is called

- A. a larva
- B. a pupa
- C. a nymph
- D. a maggot

- 15.** The nymph has to moult as it grows because
- it needs to change into a pupa
 - its colour must change to green each week
 - its hard outer covering cannot stretch as the body grows
 - it must lose its wings before flying
- 17.** Which stage of the grasshopper is able to fly and to reproduce?
- The young nymph
 - The egg
 - The adult
 - The pupa
- 19.** The life cycle of the grasshopper is described as incomplete metamorphosis because it
- has no pupal stage
 - has no adult stage
 - has no egg stage
 - takes place entirely in water
- 21.** The stage found in the life cycle of the mosquito but NOT in that of the grasshopper is the
- nymph
 - pupa
 - adult
 - egg
- 23.** One important difference between a nymph and a larva is that the nymph
- resembles the adult and feeds on the same kind of food
 - never moults as it grows
 - is quite unlike the adult in shape and food
 - lives only inside water
- 25.** A pupil watching an insect found that the young stage ate the same leaves as the adult and simply grew bigger after each moult. The insect MOST likely undergoes
- incomplete metamorphosis
 - metamorphosis with two pupal stages
 - no change at all
 - complete metamorphosis
- 14.** As the nymph grows, it casts off its hard outer covering from time to time. This casting off is called
- germination
 - moulting
 - fertilisation
 - pupation
- 16.** Wings appear on the grasshopper
- suddenly on the day the egg hatches
 - only when the insect enters a pupal case
 - only after the insect has laid eggs
 - gradually as wing buds on the later nymphs, becoming full wings in the adult
- 18.** The series of changes through which an insect passes from egg to adult is called
- pollination
 - photosynthesis
 - respiration
 - metamorphosis
- 20.** Which insect undergoes complete metamorphosis?
- The housefly
 - The grasshopper
 - The cockroach
 - The termite
- 22.** In complete metamorphosis, the stage that hatches from the egg is the
- larva
 - adult
 - pupa
 - nymph
- 24.** Which of the following pairs of insects BOTH undergo incomplete metamorphosis?
- Butterfly and housefly
 - Housefly and mosquito
 - Grasshopper and cockroach
 - Mosquito and butterfly
- 26.** Which statement about the pupa of the housefly is CORRECT?
- It feeds actively on rubbish
 - It is a resting stage during which the body is rebuilt into the adult

form

- C. It is the stage that flies about
- D. It is the stage that lays the eggs

27. Grasshoppers feed mainly on

- A. small fish in ponds
 - B. dead wood only
 - C. grasses, weeds and the leaves of crops
 - D. the blood of animals
- 29.** The MAIN harm done by grasshoppers to humans is that they
- A. bite people and spread malaria
 - B. carry the germ of cholera on their feet
 - C. spoil food stored in tins
 - D. destroy crops in the field by eating the leaves and young shoots
- 31.** Which of the following is a benefit of grasshoppers to humans?
- A. They kill the mosquitoes on a farm
 - B. They pollinate all the cereal crops
 - C. They add nitrogen to the soil
 - D. They are eaten as food rich in protein in some communities
- 33.** A maize farmer found that the leaves of his young plants had ragged holes with the veins left behind. The damage was MOST likely done by
- A. nitrogen shortage in the soil
 - B. too much rain falling on the leaves
 - C. grasshoppers feeding on the leaves
 - D. bacteria in the soil water
- 35.** Which of the following is a cultural method of controlling grasshoppers on a farm?
- A. Ploughing the land to expose and destroy the egg pods in the soil
 - B. Burning the whole farm after planting
 - C. Releasing birds that have been bought from the market
 - D. Spraying the crop with an insecticide
- 37.** The biological method of controlling grasshoppers makes use of
- A. their natural enemies, such as birds and lizards
 - B. fire set to the bush each dry
- 28.** The mouth parts of the grasshopper are of the
- A. sponging type
 - B. piercing and sucking type
 - C. siphoning type
 - D. biting and chewing type
- 30.** A swarm of migratory grasshoppers that destroys crops over a wide area is commonly called
- A. a colony of termites
 - B. a swarm of locusts
 - C. a hive of bees
 - D. a flock of weaver birds
- 32.** Grasshoppers are also useful in the environment because they
- A. carry seeds from one farm to another
 - B. break stones into soil
 - C. serve as food for birds, lizards and other animals in the food chain
 - D. produce honey for human use
- 34.** Grasshoppers do the greatest damage to crops when they
- A. appear singly during the dry season
 - B. fly away from the farm to the forest
 - C. remain as eggs in the soil
 - D. appear in very large numbers in the growing season
- 36.** Keeping a farm and its surroundings free of tall weeds helps to control grasshoppers because it
- A. attracts more grasshoppers to the weeds
 - B. raises the temperature of the soil
 - C. makes the soil richer in nitrogen
 - D. removes the places where they feed, hide and lay their eggs

season

- C. nets spread over the whole farm
- D. strong chemicals sprayed from aeroplanes

- 39.** A farmer who sees a few grasshoppers on his young maize is BEST advised to
- A. burn the maize and plant again
 - B. watch the farm closely, remove weeds and hand-pick the insects before they multiply
 - C. do nothing at all until harvest
 - D. spray the whole farm heavily with insecticide at once
- 38.** A disadvantage of using chemical insecticides to control grasshoppers is that they
- A. make the crop grow too quickly
 - B. cannot be bought in Ghana
 - C. work only on the eggs in the soil
 - D. may kill useful insects and pollute the soil and water
- 40.** Which combination of measures would control grasshoppers on a farm with the LEAST harm to the environment?
- A. Heavy spraying of insecticide every week
 - B. Early ploughing, clean weeding, protecting birds and careful hand-picking
 - C. Burning the bush around the farm each year
 - D. Flooding the farm with water for several weeks

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Where a diagram is asked for, draw it large and in pencil, and label it with a ruler. Do not shade or colour your diagram.

Question 7

- (a) Use a pencil and a ruler where a table or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 14: Manure Prep

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Manure is best described as
 - A. the water used to irrigate a vegetable bed
 - B. material of plant or animal origin added to the soil to supply nutrients and improve it
 - C. a chemical made in a factory for killing weeds
 - D. the seed kept for planting the next season
2. Which of the following is an ORGANIC manure?
 - A. Sulphate of ammonia
 - B. Well-rotted cow dung
 - C. Superphosphate
 - D. Muriate of potash
3. The chief difference between manure and an inorganic fertiliser is that manure
 - A. supplies nutrients far more quickly than fertiliser
 - B. is always made in a factory
 - C. contains no plant nutrients at all
 - D. comes from plant and animal waste and also improves the structure of the soil
4. A mixture of animal dung and the bedding material from the animal house is known as
 - A. farmyard manure
 - B. green manure
 - C. liquid fertiliser
 - D. mulch of stones
5. Compost is best described as
 - A. dried cow dung burnt to ash
 - B. decayed plant and animal waste that has been rotted down together
 - C. fresh grass spread on the soil surface
 - D. a chemical dissolved in water
6. A crop such as mucuna grown and then ploughed into the soil while still fresh and green provides
 - A. bone meal
 - B. farmyard manure
 - C. green manure
 - D. compost
7. Which of the following is the MAIN advantage of using manure rather than fertiliser alone?
 - A. Manure contains no organisms at all
 - B. Manure cannot be made on the farm
 - C. Manure improves the structure of the soil and its ability to hold water as well as supplying nutrients
 - D. Manure works on the very day it

is applied

- 9.** Which of the following is a PLANT waste used in making manure?
- A. Cow dung
 - B. Bone meal
 - C. Poultry droppings
 - D. Cocoa pod husks
- 11.** Poultry droppings are valued in manure making chiefly because they are rich in
- A. water only
 - B. nitrogen
 - C. plastic material
 - D. sand

- 8.** Which of the following best explains why manure is described as a way of recycling?
- A. It is carried from one village to another
 - B. Waste materials from plants and animals are returned to the soil to grow new crops
 - C. It is burnt and the ash thrown away
 - D. It is bought again every season
- 10.** Which of the following is an ANIMAL waste used in making manure?
- A. Rice husk
 - B. Saw dust
 - C. Pig dung
 - D. Cassava peels

- 13.** Wood ash from the kitchen fire is useful in the garden because it supplies
- A. protein
 - B. humus in large amounts
 - C. potassium (potash)
 - D. nitrogen gas
- 15.** Human excreta (night soil) can be used as manure, but it must first be
- A. mixed with drinking water
 - B. composted or treated thoroughly to destroy disease-causing organisms
 - C. spread fresh on vegetables ready for harvest
 - D. left uncovered beside the classroom
- 17.** A farmer at Techiman has plenty of cocoa pod husks, cassava peels and poultry droppings. These materials are useful because they
- A. contain no nutrients for crops
 - B. are available free on the farm and can be rotted down into good manure
 - C. can be fed directly to the growing crop
 - D. must be burnt before they are of any value
- 19.** On a heavy clay soil, well-rotted manure helps by
- A. opening up the soil so that air and water can move through it
 - B. sealing the surface against rain
 - C. making the soil stickier when wet
 - D. removing the humus from the soil
- 21.** In an area of heavy rainfall, manure is best worked into the soil rather than left on the surface because, on the surface, it
- A. turns into fertiliser
 - B. is eaten by the crop at once
 - C. becomes too dry to rot
 - D. is easily washed away by the rain
- 12.** Bone meal added to the soil supplies mainly
- A. phosphorus and calcium
 - B. carbon dioxide
 - C. nitrogen only
 - D. potassium only
- 14.** Which of the following materials must NOT be put into a compost heap?
- A. Dry leaves from the school compound
 - B. Pieces of plastic and broken glass
 - C. Vegetable peels from the kitchen
 - D. Weeds pulled from the garden before they seed
- 16.** Saw dust and wood shavings rot slowly in a compost heap because they
- A. contain no carbon at all
 - B. are too rich in nitrogen
 - C. contain much woody material and little nitrogen
 - D. are always wet when added
- 18.** Manure is especially useful on a sandy soil because it
- A. removes all the air from the soil
 - B. helps the soil to hold water and nutrients
 - C. makes the soil drain even faster
 - D. turns the sand into clay
- 20.** Fresh poultry droppings should be rotted before use because, when fresh, they
- A. are so strong that they can scorch and kill young plants
 - B. turn the soil into stone
 - C. contain no nitrogen at all
 - D. are too weak to be of any use
- 22.** A vegetable gardener wants a manure that acts quickly on his lettuce. He should choose
- A. fresh saw dust
 - B. dry maize stalks
 - C. well-rotted poultry manure
 - D. coarse wood shavings

- 23.** Manure is best applied to a field
- A. on the day the crop is harvested
 - B. only when the crop begins to flower
 - C. two to three weeks before planting, so that it mixes with the soil
 - D. once in five years, in large heaps
- 25.** The first step in preparing compost from household and garden waste is to
- A. sort the waste and remove materials that will not rot
 - B. burn the waste to ashes
 - C. cover the waste with cement
 - D. pour chemicals over the waste
- 27.** Which of the following is added in thin layers to a compost heap to introduce the organisms that cause decay?
- A. Good topsoil or old compost
 - B. Clean river sand
 - C. Dry cement powder
 - D. Broken bottles
- 29.** The compost heap is turned over every two or three weeks MAINLY to
- A. cool the heap down to the temperature of the air
 - B. make the heap smaller for storage
 - C. let air into the heap and mix the materials so that they rot evenly
 - D. remove the small organisms from the heap
- 31.** The heat of a well-made compost heap is useful because it
- A. prevents the materials from rotting
 - B. destroys many weed seeds and disease-causing organisms
 - C. attracts insects to the heap
 - D. dries the heap completely
- 33.** A heap that is packed too tightly and kept very wet begins to smell badly because
- A. the materials are too dry to decay
 - B. too many weed seeds are
- 24.** A farmer who uses manure year after year will find that his soil
- A. becomes darker, richer in humus and easier to work
 - B. loses all its nutrients
 - C. can no longer hold any water
 - D. becomes hard and stony
- 26.** A compost heap is built in layers so that
- A. animals can feed on each layer
 - B. rain cannot reach the heap at all
 - C. the different materials are well mixed and rot evenly
 - D. the heap looks attractive to visitors
- 28.** A compost heap is watered from time to time because the organisms that cause decay
- A. use water as their only food
 - B. are killed by any water
 - C. turn the water into nutrients
 - D. need moisture in order to work
- 30.** The inside of a well-made compost heap becomes hot. This heat is produced by
- A. the direct rays of the Sun on the heap
 - B. the water poured on the heap
 - C. the activity of micro-organisms breaking down the waste
 - D. a chemical added by the farmer
- 32.** Compost is ready for use when it is
- A. dry, hard and dusty like cement
 - B. wet, slimy and foul smelling
 - C. still green and sharp smelling
 - D. dark brown, crumbly and smells like fresh earth

- present
- C. the heap is too small to rot
 - D. air cannot enter, so the waste rots without oxygen
- 35.** Before building the heap, large materials such as maize stalks and pod husks should be
- A. tied into bundles and left whole
 - B. soaked in kerosene
 - C. dried and burnt to ash
 - D. cut or chopped into smaller pieces
- 34.** A compost pit or heap on a school compound should be sited
- A. away from classrooms and from any well or stream, in a shady place
 - B. on the bank of the nearest stream
 - C. inside the school kitchen
 - D. in front of the headteacher's office

- 37.** A pupil pushes a stick into the heap and pulls it out after a few minutes. A warm, moist stick shows that
- A. the heap is working well
 - B. the heap must be removed at once
 - C. the heap is too dry
 - D. the heap has finished rotting
- 39.** A class prepared compost in two pits. Pit 1 was turned every two weeks and kept moist; pit 2 was never turned. After three months, the compost in pit 1 was ready while pit 2 was not. The BEST conclusion is that
- A. turning and moistening the heap speed up the rotting of the waste
 - B. compost is always ready in three months
 - C. pit 2 contained the wrong materials
 - D. compost can only be made in pits
- 36.** Covering a compost heap with palm fronds or grass helps to
- A. keep the heap cold at all times
 - B. stop the materials from rotting
 - C. prevent air from ever entering
 - D. keep the heap moist and warm and protect it from heavy rain and the Sun
- 38.** Which of the following is the BEST safety practice when handling manure?
- A. Handle the manure with bare hands and eat soon after
 - B. Store the manure in the classroom
 - C. Smell the manure closely to test it
 - D. Wear gloves or use a tool, and wash the hands with soap afterwards
- 40.** Which of the following shows the BEST use of waste on a school farm?
- A. Burning all the waste at the edge of the field
 - B. Throwing all the waste into the nearby gutter
 - C. Leaving the waste in heaps around the classrooms
 - D. Sorting the waste, composting what will rot and using the compost on the garden beds

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Where you are asked to describe a practical task, write the steps in the order in which they are carried out.

Question 7

- (a) Use a pencil and a ruler where a table or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 15: Nitrogen

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The percentage of nitrogen in the air is about
 - A. 21%
 - B. 50%
 - C. 78%
 - D. 0.03%
2. Nitrogen gas in the air cannot be used directly by green plants because it is
 - A. too heavy to rise from the ground
 - B. unreactive, so plants cannot take it in through their roots in that form
 - C. present in the air in very small amounts
 - D. poisonous to the roots of plants
3. A natural cycle is best described as
 - A. a change that takes place only once in nature
 - B. the movement of the earth round the sun
 - C. a series of changes that is repeated over and over again in nature
 - D. a chain of animals feeding on one another
4. The nitrogen cycle is described as a repeated pattern in nature because
 - A. nitrogen passes from the air to the soil, into living things and back to the air again, over and over
 - B. the amount of nitrogen in the air keeps rising every year
 - C. it takes place only in the rainy season
 - D. nitrogen is used up completely each year and replaced by rain
5. In which form do green plants take up nitrogen from the soil?
 - A. As nitrogen gas trapped in the soil air
 - B. As protein washed from dead leaves
 - C. As ammonia gas rising from the soil
 - D. As nitrates dissolved in the soil water
6. Nitrogen is needed by living things chiefly for the making of
 - A. vitamin C
 - B. fats and oils
 - C. starch and sugar
 - D. proteins
7. Which of the following shows that nitrogen does not disappear from nature?
 - A. Nitrogen gas is made fresh in factories every year
 - B. The air gains new nitrogen from the sun
 - C. The nitrogen in a dead plant returns to the soil and is used again by other plants
 - D. Nitrogen is destroyed when a plant dies
8. The process by which nitrogen gas from the air is changed into compounds that plants can use is called
 - A. nitrogen fixation
 - B. denitrification
 - C. transpiration
 - D. ammonification

- 9.** Nitrogen-fixing bacteria are found in the
- water of streams and rivers only
 - stomach of cattle only
 - root nodules of leguminous plants
 - leaves of all green plants
- 11.** Besides bacteria, nitrogen in the air is also fixed by
- lightning during a thunderstorm
 - the moon at full moon
 - the heat of the sun at noon
 - the wind blowing over the sea
- 13.** The change of ammonium compounds in the soil into nitrates is called
- nitrogen fixation
 - assimilation
 - denitrification
 - nitrification
- 15.** The relationship between a leguminous plant and the bacteria in its root nodules is best described as one in which
- both the plant and the bacteria gain
 - only the plant gains and the bacteria are killed
 - neither the plant nor the bacteria gains anything
 - only the bacteria gain and the plant is harmed
- 17.** The taking in of nitrates by plants and their use in making plant protein is called
- denitrification
 - evaporation
 - nitrification
 - assimilation
- 19.** The breaking down of the protein in dead plants and animals into ammonium compounds is called
- nitrogen fixation
 - ammonification
 - nitrification
 - assimilation
- 10.** Which of the following crops grown in Ghana bears root nodules?
- Rice
 - Cowpea
 - Cassava
 - Maize
- 12.** During a thunderstorm, lightning causes nitrogen and oxygen in the air to combine to form oxides of nitrogen, which
- rise and form a layer above the clouds
 - dissolve in rain water and reach the soil as nitrates
 - are breathed in directly by animals
 - turn back into nitrogen gas at once
- 14.** Nitrification is carried out by
- nitrifying bacteria in the soil
 - earthworms in the soil
 - the roots of grasses
 - the heat of the sun on the soil
- 16.** A farmer at Ejura plants cowpea on a plot that has carried maize for three seasons. The MAIN scientific reason for his choice is that cowpea
- has deeper roots that break up rocks
 - is harvested earlier than maize
 - has bacteria in its root nodules that add nitrogen compounds to the soil
 - uses less water than maize
- 18.** Animals obtain the nitrogen they need for growth by
- feeding on plants or on other animals and digesting the protein in them
 - drinking water that contains nitrogen gas
 - absorbing nitrates through their skin
 - breathing in nitrogen gas from the air
- 20.** Ammonification in the soil is brought about mainly by
- nitrates dissolved in water
 - sunlight falling on the soil

- C. green plants
- D. decomposers such as bacteria and fungi

21. Besides dead bodies, nitrogen compounds are returned to the soil by

- A. the carbon dioxide breathed out by animals
 - B. the urine and droppings of animals
 - C. the water vapour lost from leaves
 - D. the heat given out by the soil at night
- 23.** Denitrifying bacteria are MOST active in a soil that is
- A. well drained and loose
 - B. dry and sandy
 - C. waterlogged and poorly aerated
 - D. covered with growing legumes
- 25.** A farmer who drains a waterlogged field helps his crops chiefly because draining
- A. reduces the activity of denitrifying bacteria and keeps more nitrates in the soil
 - B. turns the soil into clay
 - C. adds nitrogen gas directly to the soil
 - D. kills all the bacteria in the soil
- 27.** A plant short of nitrogen usually shows
- A. very dark green leaves and rapid growth
 - B. yellowing of the leaves and stunted growth
 - C. roots growing above the soil
 - D. thick woody stems and no leaves
- 29.** Which farming practice adds nitrogen to the soil at the least cost to the farmer?
- A. Growing a leguminous crop such as groundnut in rotation with other crops
 - B. Buying and spreading more inorganic fertiliser each season
 - C. Ploughing the land many times before planting
 - D. Burning the grass on the land before planting
- 31.** Decomposers are important in the nitrogen cycle because they
- A. make protein for the use of animals
 - B. carry nitrates from the soil to the leaves
- 22.** The process by which nitrates in the soil are changed back into nitrogen gas is called
- A. ammonification
 - B. fixation
 - C. nitrification
 - D. denitrification
- 24.** Which process returns nitrogen to the air and so completes the nitrogen cycle?
- A. Assimilation
 - B. Denitrification
 - C. Nitrification
 - D. Ammonification
- 26.** Nitrogen is important to the environment chiefly because it
- A. keeps the earth warm at night
 - B. is needed by all living things for making the proteins used in growth and repair
 - C. is used by plants for respiration
 - D. gives plants their green colour directly
- 28.** The nitrogen cycle is important to farmers because it
- A. raises the temperature of the soil
 - B. prevents weeds from growing
 - C. keeps returning nitrogen compounds to the soil for the next crop
 - D. removes all the water from the soil
- 30.** Applying well-rotted manure to a farm adds nitrogen to the soil because the manure
- A. contains nitrogen gas trapped between its particles
 - B. kills the bacteria that use up nitrates
 - C. changes the sand in the soil into nitrates
 - D. is decayed organic matter that releases ammonium compounds and then nitrates

- C. break down dead bodies and waste, returning nitrogen compounds to the soil
- D. take in nitrogen gas from the air

- 33.** Too much nitrate fertiliser washed into a river is harmful because it
- A. makes water plants and algae grow too fast, so the water loses oxygen and fish die
 - B. makes the river water too cold for fish
 - C. turns the river water into acid at once
 - D. removes all the nitrogen from the air above the river
- 35.** Leaching reduces the fertility of a soil because it
- A. kills the roots of the crop at once
 - B. adds more nitrogen gas to the soil
 - C. turns the nitrates into protein
 - D. carries dissolved nitrates down beyond the reach of the roots
- 37.** Which of the following practices would MOST likely reduce the amount of nitrogen returned to a soil?
- A. Spreading compost on the land
 - B. Clearing and uprooting all the leguminous plants on the land
 - C. Planting groundnut after maize
 - D. Ploughing crop remains back into the soil
- 39.** A field was left bare after harvest and was beaten by heavy rains for two months. The nitrate content of its topsoil would MOST likely
- A. fall, because the nitrates would be leached away
 - B. stay exactly the same, because nitrates do not dissolve
 - C. rise, because rain stops denitrification
 - D. rise, because rain water contains protein
- 32.** A pupil said, 'Without the nitrogen cycle, life on earth would be difficult.' The BEST reason for this statement is that
- A. the air would contain no oxygen
 - B. the earth would become too cold for plants
 - C. rain would no longer fall on the land
 - D. the supply of usable nitrogen compounds for making protein would run out
- 34.** The washing away of nitrates from the topsoil by heavy rain is called
- A. transpiration
 - B. denitrification
 - C. erosion of the air
 - D. leaching
- 36.** Bush burning interferes with the nitrogen cycle mainly because it
- A. increases the number of nitrifying bacteria
 - B. makes the soil hold too much water
 - C. destroys the organic matter and the soil organisms that return nitrogen to the soil
 - D. adds too much nitrate to the soil
- 38.** A farmer burns his field every year before planting. Over several seasons his yields fall. The BEST explanation is that burning has
- A. removed all the sand from the soil
 - B. destroyed the humus and the organisms that recycle nitrogen in the soil
 - C. made the soil too cold for the crop
 - D. increased the number of decomposers
- 40.** Which of the following would be the BEST advice to a farmer who wants to keep the nitrogen cycle working on his land?
- A. Flood the land so that it stays waterlogged
 - B. Remove all crop remains from the land after harvest
 - C. Practise crop rotation with

legumes, add compost and avoid
burning the bush

- D. Burn the land each year and
leave it bare after harvest

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Where a process is asked for, give the scientific name of the process.

Question 7

- (a) Use a pencil and a ruler where a table or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 16: Non Planeta

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The solar system is made up of
 - A. the Earth and its moon only
 - B. the eight planets without the Sun
 - C. the Sun and all the bodies that move round it
 - D. all the stars seen in the night sky
2. The Sun is correctly described as
 - A. a comet
 - B. a natural satellite
 - C. a planet
 - D. a star
3. The force that keeps the planets and other bodies moving round the Sun is
 - A. the magnetic pull of the Earth
 - B. the push of the solar wind
 - C. the force of friction in space
 - D. the gravitational pull of the Sun
4. Which of the following is NOT a planet of the solar system?
 - A. Ceres
 - B. Neptune
 - C. Mars
 - D. Jupiter
5. Bodies of the solar system that are not planets, such as comets and asteroids, are described as
 - A. constellations
 - B. fixed stars
 - C. galaxies
 - D. non-planetary bodies
6. A natural satellite is best described as a body that
 - A. moves round the Sun only
 - B. burns up in the atmosphere
 - C. gives out its own light
 - D. moves round a planet
7. The Moon shines at night because it
 - A. reflects light from the Sun
 - B. is heated by the Earth
 - C. produces its own light like a star
 - D. burns gases on its surface
8. The path followed by a body as it moves round the Sun is called its
 - A. orbit
 - B. axis
 - C. equator
 - D. latitude
9. Asteroids are best described as
 - A. stars that have stopped shining
 - B. balls of ice and dust with long tails
 - C. small rocky bodies that move round the Sun
 - D. pieces of rock that have landed on the Earth
10. Most of the asteroids in the solar system are found in a belt between the orbits of
 - A. Neptune and the Sun
 - B. Venus and the Earth
 - C. Mars and Jupiter
 - D. the Earth and Mars
11. Which of the following best describes the shape of a typical asteroid?
 - A. Irregular, like a lump of rock

- B. A flat disc
- C. A long cylinder
- D. A perfect sphere

- 12.** Compared with the orbit of a comet, the orbit of an asteroid in the belt is
- A. a straight line through the solar system
 - B. very long and narrow
 - C. nearly circular and much shorter

- D. a path round the Earth
- 13.** The largest body in the asteroid belt, now classed as a dwarf planet, is
- A. Ceres
 - B. Eris
 - C. Titan
 - D. Halley
- 15.** Scientists study asteroids that pass close to the Earth MAINLY because such a body
- A. could be used as a new planet for farming
 - B. controls the seasons on the Earth
 - C. could one day collide with the Earth and cause great damage
 - D. gives out light that helps ships at sea
- 17.** A comet is best described as a body made of
- A. solid iron only
 - B. burning gases like the Sun
 - C. soil and water like the Earth
 - D. ice, dust and frozen gases
- 19.** The cloud of gas and dust that surrounds the solid centre of a comet as it nears the Sun is the
- A. nucleus
 - B. coma
 - C. corona of the Earth
 - D. orbit
- 21.** A comet develops a bright tail as it comes near the Sun because
- A. it catches fire on entering the atmosphere
 - B. it collides with a planet on the way
 - C. it begins to produce its own light like a star
 - D. heat from the Sun turns its ice into gas, and dust and gas are pushed away from it
- 23.** A comet moves fastest along its orbit when it is
- A. nearest to the Sun
 - B. passing behind a planet
 - C. at the same distance from the
- 14.** Asteroids and the planets move round the Sun in the same direction and in nearly the same plane. This suggests that they
- A. were thrown out of the Earth
 - B. are all made of ice
 - C. are moved by the same wind
 - D. were formed from the same cloud of material at about the same time
- 16.** A small body that leaves the asteroid belt and moves towards the inner solar system is called a
- A. meteorite
 - B. meteoroid
 - C. nebula
 - D. satellite
- 18.** The solid central part of a comet is called the
- A. tail
 - B. crater
 - C. nucleus
 - D. coma
- 20.** The orbit of a comet round the Sun is usually
- A. a long, narrow ellipse
 - B. a straight line
 - C. a square path
 - D. a perfect circle
- 22.** The tail of a comet always points
- A. away from the Sun
 - B. towards the Earth
 - C. in the direction in which the comet is travelling
 - D. towards the Sun

- Sun as the Earth
D. farthest from the Sun

25. Why is a comet seen from the Earth only for a short period and then not seen again for many years?

- A. It is destroyed each time it passes the Sun
- B. The Earth stops moving while it passes
- C. Its orbit is very long, so it spends most of the time far from the Sun where it cannot be seen
- D. It changes into an asteroid after being seen

24. Halley's comet returns near the Earth about once every

- A. 76 days
- B. 760 years
- C. 7 years
- D. 76 years

26. A meteoroid is

- A. a cloud of gas round the Sun
- B. a small piece of rock or metal moving through space
- C. a star that is about to die

- D. a large planet beyond Neptune
- 28.** A meteor is commonly called a
- A. northern light
 - B. morning star
 - C. shooting star
 - D. falling planet
- 30.** A body from space that is not completely burnt up and lands on the surface of the Earth is called a
- A. meteor
 - B. meteorite
 - C. meteoroid
 - D. comet
- 27.** The streak of light seen in the night sky when a small body burns in the Earth's atmosphere is called a
- A. comet
 - B. meteor
 - C. satellite
 - D. meteorite
- 29.** A meteoroid becomes very hot and glows as it enters the Earth's atmosphere because of
- A. the heat of the Moon
 - B. friction between the body and the air
 - C. the burning of water vapour in the air
 - D. the magnetic field of the Sun
- 31.** Which of the following is the correct order as a body from space approaches and reaches the ground?
- A. Meteoroid, meteorite, meteor
 - B. Meteorite, meteor, meteoroid
 - C. Meteor, meteoroid, meteorite
 - D. Meteoroid, meteor, meteorite
- 33.** The atmosphere protects living things on the Earth from most meteoroids because it
- A. turns them into harmless comets
 - B. causes them to burn up before they reach the ground
 - C. pushes them back into space with wind
 - D. attracts them to the north pole
- 35.** One way in which a comet differs from an asteroid is that a comet
- 32.** A shower of meteors is often seen when the Earth
- A. passes between Mars and Jupiter
 - B. passes through the trail of dust left behind by a comet
 - C. moves closer to the Moon
 - D. turns on its axis at night
- 34.** Which of the following bodies does NOT move round the Sun?
- A. An asteroid in the belt
 - B. The Earth
 - C. A comet
 - D. The Moon

- A. contains a large amount of ice and develops a tail near the Sun
 - B. is made of rock and metal only
 - C. always stays in the belt between Mars and Jupiter
 - D. moves round a planet instead of the Sun
- 37.** Which of the following bodies is classed as a dwarf planet?
- A. Mercury
 - B. The Moon
 - C. Pluto
 - D. Halley's comet
- 39.** A large meteorite striking the Earth is dangerous chiefly because it
- A. makes the Earth spin faster on its axis
 - B. releases a great deal of energy, forming a crater and throwing dust into the air
 - C. cools the Earth by blocking the Moon
 - D. adds useful water to the oceans
- 36.** Which of the following is TRUE of both comets and asteroids?
- A. They both have long tails of gas and dust
 - B. They both move round the Earth
 - C. They both move round the Sun under its gravitational pull
 - D. They are both larger than the Earth
- 38.** A pupil at Dodowa reads that a comet will be visible next month. The BEST way for the class to prepare is to
- A. find out the date and the part of the sky from a reliable source and observe it on a clear night
 - B. look at the Sun directly with the naked eye at noon
 - C. wait for the comet to land on the school field
 - D. do nothing, since comets cannot be seen from Ghana

- 40.** The MOST important reason for studying non-planetary bodies such as comets and asteroids is that they
- A. shine brightly enough to light the Earth at night
 - B. carry material left from the formation of the solar system and help us to understand how it began
 - C. can be used at once as sources of food
 - D. control the climate of the Earth each year

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) The figures are not drawn to scale; do not measure distances on them.

Question 7

- (a) Use a pencil and a ruler where a table or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 17: Science and

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. An industry is BEST described as
 - A. a single building in which people work
 - B. a group of workers who belong to the same family
 - C. a group of firms that produce the same kind of goods or services
 - D. any place where goods are sold to buyers
2. Which of the following is an example of an industry?
 - A. All the firms in the country that make soap
 - B. A lorry that carries soap to the north
 - C. A family that uses soap at home
 - D. A shop that sells soap in a market
3. Industries that take raw materials directly from nature, such as mining and fishing, are described as
 - A. secondary industries
 - B. tertiary industries
 - C. service industries
 - D. primary industries
4. A factory that changes cocoa beans into chocolate belongs to a
 - A. service industry
 - B. secondary (manufacturing) industry
 - C. primary industry
 - D. mining industry
5. Banking, transport and teaching belong to the group of industries known as
 - A. manufacturing industries
 - B. extractive industries
 - C. service (tertiary) industries
 - D. construction industries
6. Which of the following gives the correct order of production?
 - A. Finished product, processing, raw material, marketing
 - B. Raw material, processing, finished product, marketing
 - C. Processing, marketing, raw material, finished product
 - D. Marketing, raw material, processing, finished product
7. The chief raw material used by a chocolate factory at Tema is
 - A. bauxite
 - B. cassava tubers
 - C. palm fruits
 - D. cocoa beans
8. Bauxite mined at Awaso is used mainly in the production of
 - A. common salt
 - B. glass
 - C. cement
 - D. aluminium
9. Which industry uses limestone and clay as its main raw materials?
 - A. The soap industry
 - B. The cement industry
 - C. The fishing industry
 - D. The textile industry
10. A gari processing factory in a village depends chiefly on
 - A. groundnuts
 - B. maize
 - C. cocoyam
 - D. cassava

- 11.** Which of the following products is obtained from palm fruits?
- Palm oil
 - Groundnut paste
 - Shea butter
 - Coconut fibre
- 13.** The MAIN reason why many industries in Ghana are built near their source of raw material is to
- keep the workers away from towns
 - obtain more rain for the factory
 - make the factory look attractive
 - reduce the cost of transporting the raw material
- 15.** Common salt is obtained from sea water in a salt pan by
- filtration of the sea water through sand
 - distillation of the salt out of the water
 - evaporation of the water, followed by crystallisation of the salt
 - boiling the sea water until the salt burns
- 17.** Bread rises before it is baked because the yeast in the dough produces
- oxygen, which burns in the oven
 - water, which swells the flour
 - nitrogen, which pushes the dough up
 - carbon dioxide, which forms bubbles in the dough
- 19.** In a juice factory the juice is heated for a short time and cooled quickly. This process is called
- saponification
 - sedimentation
 - pasteurisation
 - chlorination
- 21.** A factory passes its waste water through settling tanks before letting it out. The scientific principle used in the tanks is
- fermentation, in which sugar is changed to alcohol
 - evaporation, in which the water turns to vapour
- 12.** Crude oil brought ashore at Takoradi is treated in a refinery mainly in order to
- make it heavier so that it can be stored
 - remove all the carbon from it
 - change it into natural gas only
 - separate it into useful products such as petrol, kerosene and diesel
- 14.** The chemical process in which oil or fat is boiled with caustic soda to produce soap is called
- saponification
 - distillation
 - fermentation
 - evaporation
- 16.** In a brewery and in a bakery, the organism that brings about fermentation is
- a green alga
 - yeast
 - an earthworm
 - a virus
- 18.** Water meant for bottling is passed through sand and charcoal filters mainly to
- make the water taste salty
 - change the water into steam
 - remove suspended solid particles and substances that give taste and smell
 - add minerals to the water
- 20.** Which scientific principle is applied when iron ore is heated strongly with coke in a furnace?
- A chemical reaction in which the oxygen is removed from the ore to leave the metal
 - The ore is beaten until the metal separates
 - The ore is cooled until the metal crystallises out
 - The ore is dissolved in water and filtered

- C. sedimentation, in which heavy solid particles settle to the bottom
- D. saponification, in which fat is changed to soap

- 23.** Which of the following is an indigenous industry of Ghana?
- Oil refining
 - Motor vehicle assembly
 - Aluminium smelting
 - Kente weaving
- 25.** The indigenous industry that produces cutlasses, hoes and knives from scrap metal is
- weaving
 - pottery
 - blacksmithing
 - basketry
- 27.** Alata samina (local black soap) is made mainly from
- sea water and clay
 - charcoal and groundnut shells
 - palm kernel oil and the ash of plantain peels or cocoa pods
 - cassava starch and sand
- 29.** Which pair correctly matches an indigenous industry with its product?
- Pottery - cutlasses
 - Salt winning - common salt
 - Weaving - palm oil
 - Blacksmithing - cooking pots
- 31.** The roasting of the sieved cassava meal over a hot pan serves to
- add salt to the gari
 - wet the meal before it is bagged
 - drive off the remaining water and cook the starch so that the gari keeps long
 - dissolve the fibre in the meal
- 22.** An indigenous industry is one that
- uses local materials and skills handed down within the community
 - is owned by a foreign company
 - produces goods only for export
 - uses only machines imported from abroad
- 24.** Pottery is an indigenous industry in which the chief raw material is
- clay
 - glass
 - limestone
 - bauxite
- 26.** Shea butter is produced in northern Ghana from the
- leaves of the dawadawa tree
 - bark of the baobab tree
 - roots of the mango tree
 - nuts of the shea tree
- 28.** Kente weaving, adinkra printing and basketry are classified together as
- indigenous craft industries
 - service industries
 - heavy chemical industries
 - extractive industries
- 30.** In the making of gari, the grated cassava is packed into bags and pressed for about three days mainly to
- make the mash heavier before roasting
 - squeeze out the liquid and allow fermentation to reduce the harmful substance in the cassava
 - allow the starch to change into fat
 - add water to the mash so that it softens
- 32.** Palm oil is obtained from boiled palm fruits by pounding and then adding water, so that the oil
- floats on the water and is skimmed off, because oil is less dense than water
 - dissolves in the water and is filtered off
 - evaporates with the steam and is condensed

D. sinks below the water and is drawn off

34. Local soap makers boil palm kernel oil with the ash of plantain peels. The ash is used because it

- A. contains an alkali that reacts with the oil to form soap
- B. prevents the oil from boiling over
- C. is a fuel that keeps the pot hot
- D. gives the soap its black colour only

33. In the firing of clay pots, strong heating is necessary because it

- A. makes the pot able to dissolve in water
- B. drives off the water and changes the clay permanently into a hard material
- C. turns the clay into a metal
- D. adds water to the clay to soften it

35. A dye from plant leaves is used to colour cloth and the cloth is then rinsed in salt solution. Which scientific idea BEST explains this

practice?

- A. The salt bleaches the cloth to a lighter colour
- B. The salt helps to fix the dye in the cloth so that it does not wash out easily
- C. The salt changes the dye into a gas
- D. The salt makes the cloth heavier for sale

37. One clear advantage of indigenous industries to a community is that they

- A. depend wholly on imported machines
- B. need large sums of foreign money to begin
- C. use local raw materials and provide work for local people at little cost
- D. require every worker to hold a degree

39. Which of the following would BEST improve a village gari industry without destroying its local character?

- A. Introducing a motorised grater and a screw press, and training the women in hygiene
- B. Asking the women to work longer hours by hand
- C. Replacing cassava with imported wheat flour
- D. Closing the village factory and importing gari

36. Compared with a modern factory, an indigenous industry usually

- A. uses heavy machines and produces on a large scale
- B. uses simple tools and hand labour and produces on a small scale
- C. employs only trained engineers
- D. produces goods of one fixed size and quality

38. A weakness of many indigenous industries is that

- A. their products are never bought in the community
- B. they always pollute the air more than factories
- C. they can never use local raw materials
- D. output is small and the quality of the product varies from one batch to another

40. A group of pupils argued that indigenous industries have no science in them. The BEST reply is that

- A. indigenous industries work only by chance
- B. scientific principles such as fermentation, evaporation and the action of heat are used at almost every stage
- C. only modern factories make use of chemical change
- D. science begins only when machines are used

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 18: Specialised

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The cell is best described as
 - A. a group of tissues working together
 - B. the smallest unit of a living thing that can carry out the processes of life
 - C. the outer covering of a living thing
 - D. the smallest particle of any substance
2. A specialised cell is a cell that has
 - A. the same shape as every other cell in the body
 - B. no cell membrane around it
 - C. lost its nucleus and cannot divide
 - D. a particular shape and structure that suit it to one special job
3. The correct order of organisation in a living thing, from the simplest, is
 - A. organ, tissue, cell, organ system
 - B. tissue, cell, organ, organ system
 - C. cell, tissue, organ, organ system
 - D. cell, organ, tissue, organ system
4. A group of similar cells that work together to do the same job is called
 - A. an organ
 - B. an organism
 - C. an organ system
 - D. a tissue
5. Which of the following is a specialised cell?
 - A. An amoeba
 - B. A bacterium in the soil
 - C. A grain of pollen dust
 - D. A red blood cell
6. Specialisation of cells is an advantage to a plant or an animal because
 - A. it stops the cells from dividing
 - B. each kind of cell carries out its own job very well, so the whole body works efficiently
 - C. it allows every cell to do every job
 - D. it makes the body of the organism larger
7. A pupil observed cells from two parts of the same plant and found that they differed in shape. The BEST explanation is that the cells
 - A. are specialised for different jobs in the plant
 - B. were damaged while the slide was being prepared
 - C. came from two different plants
 - D. had different numbers of nuclei
8. New cells are formed in a living thing by
 - A. cell feeding
 - B. evaporation
 - C. cell division
 - D. osmosis
9. The process by which a newly formed cell changes to take on a special shape and structure for a special job is called
 - A. respiration
 - B. transpiration

- C. differentiation
- D. pollination

- 10.** In a dicotyledonous plant, the regions where cells divide most actively to produce new cells are the
- A. mature leaves and flowers
 - B. root hairs on the older parts of the root
 - C. dead xylem vessels in the stem
 - D. meristems, found at the tips of roots and shoots and in the

cambium

- 11.** In humans, specialised cells are formed from
- unspecialised (stem) cells that divide and then differentiate
 - nerve cells that divide throughout life
 - red blood cells that change back into bone
 - cells taken in with food
- 13.** Which statement about specialised cells is TRUE?
- Once a cell is fully specialised it usually loses the ability to divide and change into another kind of cell
 - A fully specialised cell can change into any other kind of cell at any time
 - Specialised cells are formed without any cell division
 - All the specialised cells of an organism have the same shape
- 15.** The epidermal cells that cover the leaf of a plant are important because they
- form a protective covering that reduces the loss of water from the leaf
 - transport water up the stem
 - divide to make the stem thicker
 - carry out most of the photosynthesis of the leaf
- 17.** Unlike the other cells of the leaf epidermis, guard cells contain
- no cytoplasm
 - chloroplasts
 - no cell wall
 - no nucleus
- 19.** The MAIN function of the stomata of a leaf is to
- support the leaf blade
 - store starch made in the leaf
 - allow gases to pass into and out of the leaf
 - absorb water from the soil
- 21.** The xylem vessel is well suited to its
- 12.** Before a cell divides, the material in the nucleus is
- shared out so that each new cell gets half of it
 - passed out of the cell through the membrane
 - copied, so that each new cell receives a complete set
 - destroyed, so that the new cells start afresh
- 14.** Which of the following is a specialised cell of a dicotyledonous plant?
- A sperm cell
 - A nerve cell
 - A guard cell
 - A white blood cell
- 16.** The two cells that surround a stoma and control its opening and closing are the
- cambium cells
 - epidermal hair cells
 - guard cells
 - xylem cells
- 18.** A stoma opens when the guard cells
- lose water and become flaccid
 - take in water, become turgid and curve apart
 - divide into two new cells
 - lose their chloroplasts
- 20.** Water and dissolved mineral salts are carried from the root to the leaves of a plant through the
- xylem
 - cambium
 - phloem
 - epidermis

work because it is

- A. a short living cell packed with cytoplasm
- B. a thin-walled cell that divides every day
- C. a cell with a large nucleus and many chloroplasts
- D. a long hollow tube of dead cells with no cross walls, strengthened with lignin

23. The cambium of a dicotyledonous stem is important because it

- A. controls water loss from the stem
- B. divides to form new xylem and phloem, so the stem grows thicker
- C. protects the stem from insects
- D. carries manufactured food to the roots

22. Besides carrying water, the xylem also helps the plant by

- A. giving support to the stem because of its strong, thickened walls
- B. absorbing carbon dioxide from the air
- C. controlling the opening of the stomata
- D. making food for the plant

- 25.** Which of the following is a specialised animal cell?
- A. A guard cell
 - B. A nerve cell
 - C. A cambium cell
 - D. A xylem vessel
- 27.** The red colour of the red blood cell is due to
- A. cytoplasm
 - B. haemoglobin
 - C. the cell wall
 - D. chlorophyll
- 29.** The MAIN function of the white blood cells is to
- A. carry oxygen to the tissues
 - B. store food for the body
 - C. carry messages from the brain
 - D. defend the body against disease-causing organisms
- 31.** A nerve cell is suited to its work because it
- A. is round and has no nucleus
 - B. is very long and has fine branches at its ends for passing on messages
 - C. has a thick wall of lignin
 - D. is filled with haemoglobin
- 33.** The male gamete of a human being is the
- A. nerve cell
 - B. sperm cell
 - C. egg-white cell
 - D. red blood cell
- 35.** The biconcave disc shape of the red blood cell is important because it
- A. allows the cell to make its own food
 - B. makes the cell heavier than the plasma
 - C. prevents the cell from entering narrow capillaries
 - D. gives the cell a large surface area for taking in and giving out oxygen
- 24.** A farmer cut a ring of bark, including the cambium, right round the trunk of a young orange tree. The tree later died. The BEST explanation is that
- A. water could no longer enter the roots from the soil
 - B. the tree lost all its leaves at once
 - C. the tissues that carry food and form new xylem and phloem were destroyed
 - D. the roots were cut off from the soil
- 26.** The cell that carries oxygen round the human body is the
- A. white blood cell
 - B. nerve cell
 - C. muscle cell
 - D. red blood cell
- 28.** A mature human red blood cell is unusual because it has
- A. no nucleus, which leaves more room for haemoglobin
 - B. a thick cell wall of cellulose
 - C. many chloroplasts for making food
 - D. two nuclei, so that it can divide quickly
- 30.** Messages are carried from the brain to the muscles of the leg by
- A. nerve cells
 - B. muscle cells
 - C. red blood cells
 - D. sperm cells
- 32.** The cells that bring about movement of the parts of the body by shortening are the
- A. red blood cells
 - B. sperm cells
 - C. muscle cells
 - D. nerve cells
- 34.** The tail of the sperm cell is important because it
- A. carries oxygen to the egg
 - B. enables the sperm to swim towards the egg
 - C. digests the food taken in by the sperm
 - D. holds the sperm firmly in one place

- 37.** Which of the following pairs of a cell and its function is CORRECT?
- A. Guard cell - controls the opening of the stoma
 - B. Red blood cell - fights disease-causing organisms
 - C. Nerve cell - carries water up the stem
 - D. Xylem vessel - makes food for the plant
- 39.** A pupil sees under the microscope a cell that is long and narrow, has no nucleus and has thick walls with no cross walls. The cell is MOST likely
- A. a sperm cell of a human being
 - B. a nerve cell of an animal
 - C. a guard cell of a leaf
 - D. a xylem vessel of a plant
- 36.** Muscle cells contain many mitochondria because they
- A. store the food eaten by the person
 - B. have to carry oxygen round the body
 - C. need a large supply of energy from respiration in order to contract
 - D. must divide many times each day
- 38.** A person with too few red blood cells is likely to feel weak and tired chiefly because
- A. the muscles grow larger than normal
 - B. the blood can no longer clot
 - C. the nerve cells stop carrying messages
 - D. less oxygen reaches the body cells, so less energy is released in respiration
- 40.** Which statement BEST shows why specialised cells matter to the whole organism?
- A. Each kind of cell carries out one job so well that together they keep the whole organism alive
 - B. Specialised cells allow the organism to live without food
 - C. Specialised cells make the organism grow taller than other organisms
 - D. Specialised cells remove the need for organ systems

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Where a diagram is asked for, draw it large and in pencil, and label it with a ruler. Do not shade or colour your diagram.

Question 7

- (a) Use a pencil and a ruler where a table or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 19: Understanding the

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The part of a plant that absorbs water and mineral salts from the soil is the
 - A. fruit
 - B. flower
 - C. root
 - D. leaf
2. The chief work of the stem of a plant is to
 - A. support the leaves and flowers and carry water and food through the plant
 - B. produce the seeds of the plant
 - C. attract insects to the plant
 - D. absorb water from the soil
3. Food is made in the green plant mainly in the
 - A. roots
 - B. fruits
 - C. leaves
 - D. flowers
4. The part of a plant that develops into a fruit after fertilisation is the
 - A. leaf
 - B. root
 - C. flower
 - D. stem
5. Which part of a plant protects the seeds and helps in their dispersal?
 - A. The fruit
 - B. The root
 - C. The stem
 - D. The leaf
6. Which of the following pairs gives a plant part and a structure that grows from it?
 - A. Root - petal
 - B. Flower - tap root
 - C. Leaf - root hair
 - D. Stem - branch
7. Cassava is planted mainly by using
 - A. its leaves
 - B. its roots
 - C. its flowers
 - D. pieces of its mature stem
8. Which plant part is used to plant yam?
 - A. Setts cut from the tuber
 - B. The leaves
 - C. The tendrils
 - D. The flowers
9. Plantain and banana are usually propagated by means of
 - A. suckers
 - B. flower buds
 - C. leaf cuttings
 - D. seeds from the fruit
10. Sweet potato is commonly planted using
 - A. dry seeds
 - B. the flowers
 - C. pieces of the leaf
 - D. vine cuttings
11. Maize, cowpea and okro are planted by means of
 - A. seeds
 - B. suckers
 - C. tubers
 - D. stem cuttings

- 13.** A farmer selects cassava stems from the middle of a healthy, mature plant for planting because such stems
- carry the flowers of the plant
 - contain no water at all
 - are the thinnest part of the plant
 - have well-formed buds and enough stored food to sprout quickly
- 15.** Mulching the soil with dry leaves helps the crop MAINLY because the mulch
- increases the temperature of the soil greatly
 - reduces loss of water from the soil and checks the growth of weeds
 - prevents air from entering the soil
 - makes the soil more acidic at once
- 17.** Which plant part is used as the handle of a hoe or a cutlass?
- A straight piece of a hardwood stem
 - A broad green leaf
 - A fresh flower stalk
 - A soft root hair
- 19.** Which of the following plant materials is used as feed for livestock?
- Dry sand and gravel
 - Groundnut haulms and maize stover
 - Burnt plastic bags
 - Palm kernel shells only
- 21.** A farmer crushes neem leaves, soaks them in water and sprays the liquid on his vegetables. The liquid is used as a
- natural pesticide
 - hormone for ripening
 - liquid fertiliser
 - weedicide for grasses
- 23.** The bark, leaves and roots of some plants are used in the community for
- refining crude oil
 - making cement blocks
 - preparing herbal medicines
- 12.** An advantage of planting with stem cuttings rather than with seeds is that the new plants
- are always taller than the parent plant
 - never suffer from any disease
 - do not need water to grow
 - are exactly like the parent plant and mature earlier
- 14.** Dry grass and leaves spread on the soil around a crop are called
- humus
 - compost
 - mulch
 - manure
- 16.** Leaves, peels and plant stalks heaped and allowed to rot are used to make
- charcoal
 - inorganic fertiliser
 - compost manure
 - herbicide
- 18.** Palm fronds and bamboo are used on a poultry farm chiefly for
- treating sick birds
 - feeding the birds daily
 - hatching the eggs
 - building cages, nests and shelter for the birds
- 20.** Leguminous plants such as mucuna are grown and ploughed into the soil in order to
- stop water from entering the soil
 - add nitrogen and organic matter to the soil
 - make the soil hard and compact
 - take nitrogen out of the soil
- 22.** Which of the following is an advantage of using a plant extract rather than a factory-made pesticide on a vegetable garden?
- It also supplies all the nutrients the crop needs
 - It is cheaper, is made from local materials and leaves less harmful residue on the crop
 - It kills every insect in the garden at once
 - It never needs to be applied a second time

D. producing electricity

- 25.** Timber from the trunks of trees is used mainly for
- building and making furniture
 - treating malaria
 - feeding cattle
 - making common salt
- 27.** Which of these is a NON-agricultural use of a plant part?
- Using cassava stems for planting
 - Carving a stool from the trunk of a tree
 - Ploughing mucuna into the soil
 - Feeding maize stover to cattle
- 29.** Charcoal used in many Ghanaian homes is produced from
- the flowers of the shea tree
 - wood obtained from tree stems and branches
 - dry maize leaves only
 - the roots of grasses
- 31.** A Basic 9 class wishes to make a rope for tying up a goat. The MOST suitable plant material is
- the soft fruit of a pawpaw
 - fibre stripped from a plant stem or a palm frond
 - fresh green leaves of a vegetable
 - the petals of a flower
- 33.** Before a neem leaf extract is sprayed on a crop, it should be
- left in the sun for a month
 - strained through a cloth so that the leaf pieces do not block the sprayer
 - boiled until no liquid remains
 - mixed with petrol to make it stronger
- 24.** Which plant part is used in the roofing of a traditional hut?
- The flowers of the palm
 - The tap root of the grass
 - The fruits of the grass
 - Thatch made from grass and palm fronds
- 26.** Sponges, baskets and mats produced in Ghanaian communities are made from
- clay from the river
 - the roots of grasses only
 - the fibres, stems and leaves of plants
 - animal bones
- 28.** The use of palm fronds and leaves at funerals and festivals in Ghana is an example of the use of plant parts for
- pest control
 - animal feeding
 - soil improvement
 - ceremonial and cultural purposes
- 30.** Dyes used in colouring cloth in some Ghanaian communities are obtained from
- the leaves, bark and fruits of certain plants
 - cattle bones
 - sea sand
 - iron filings
- 32.** Which plant material would a class use to make seedling pots for a nursery?
- Sheets of metal from a drum
 - Stones gathered from the roadside
 - Pieces of broken glass
 - Banana or plantain leaves rolled and tied into cups
- 34.** A class making compost from plant parts should
- use only dry sand and water
 - cut the material into small pieces, keep the heap moist and turn it regularly
 - cover the heap with a sheet of plastic and never open it
 - pour kerosene on the heap to

speed up decay

36. Which test would BEST show a class that the rope they have made from plant fibre is fit for use?

- A. Measuring the length of the rope only
- B. Hanging a known load on the rope and checking that it does not break or stretch too much
- C. Smelling the rope after it has dried
- D. Looking at the colour of the rope

35. A scarecrow made from maize stalks and dry palm fronds is set in a field in order to

- A. attract insects for pollination
- B. provide shade for the crop
- C. supply nutrients to the soil
- D. frighten birds away from the ripening crop

37. A group wants to make a mat for the school library from plant materials. The MOST suitable material is

- A. green leaves cut that morning
- B. fresh flowers from the garden
- C. dried strips of raffia, straw or palm leaf
- D. roots dug from the school farm

39. A class strips the bark completely round the trunk of a young tree to obtain fibre. The BEST judgement on this action is that it is

- A. right, because the tree has many other branches
- B. wrong, because the fibre of bark is of no use
- C. wrong, because removing a complete ring of bark kills the tree
- D. right, because the bark grows back within a week

38. Plant material to be used for weaving is first dried in the sun mainly because drying

- A. makes the material tough and prevents it from rotting or shrinking after weaving
- B. removes all the fibre from the material
- C. adds colour to the material
- D. makes the material heavier to handle

40. Which practice shows the MOST sustainable use of plant parts by a community?

- A. Burning the bush to make the plants sprout faster
- B. Uprooting plants before they have produced seed
- C. Harvesting only mature stems and leaves and planting new seedlings to replace what is taken
- D. Cutting whole trees for every basket that is needed

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 20: Waste Management

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Waste is best described as
 - A. any material that can be sold at a market
 - B. any material that is harmful to living things
 - C. material that is no longer wanted by the person or place that produced it
 - D. material that has been buried in the ground
3. Waste that cannot be broken down by micro-organisms is described as
 - A. organic
 - B. non-biodegradable
 - C. liquid
 - D. compostable
5. Used syringes, expired drugs and soiled bandages from a clinic are classified as
 - A. construction waste
 - B. clinical (hospital) waste
 - C. agricultural waste
 - D. domestic waste
7. Empty water sachets, plastic bottles and rubber bags found in a Ghanaian town are together described as
 - A. organic waste
 - B. plastic waste
 - C. clinical waste
 - D. electronic waste
9. Choked gutters filled with rubbish in a town are MOST likely to cause
 - A. an increase in the oxygen of the air
 - B. a rise in the yield of nearby farms
 - C. a fall in the temperature of the town
 - D. flooding when it rains
2. Which of the following is an example of biodegradable waste?
 - A. A broken bottle
 - B. A rusted tin
 - C. An empty water sachet
 - D. Cassava peels
4. Which of these is a LIQUID waste produced in the home?
 - A. Water used in washing clothes
 - B. Empty tins of milk
 - C. Yam peels
 - D. Ash from the kitchen fire
6. Which of the following is an example of industrial waste?
 - A. Peels from a kitchen
 - B. Leaves from a school compound
 - C. Oil and chemicals discharged from a factory
 - D. Sweepings from a classroom
8. Old mobile phones, dead batteries and broken television sets are called
 - A. building waste
 - B. agricultural waste
 - C. food waste
 - D. electronic waste (e-waste)
10. Water collecting in old tins and tyres left lying about is dangerous because it
 - A. provides a breeding place for mosquitoes
 - B. makes the soil more fertile
 - C. attracts useful bees to the area
 - D. increases the amount of drinking water

- 11.** A heap of rotting refuse close to a house is a health risk chiefly because it
- A. uses up the carbon dioxide of the air
 - B. makes the soil around the house dry
 - C. attracts flies and rats that carry disease organisms to food
 - D. lowers the humidity of the air in the house
- 13.** Liquid that drains from a heap of waste and enters a stream is harmful because it
- A. adds oxygen to the water
 - B. carries dissolved chemicals and germs that poison the water
 - C. removes all the mud from the water
 - D. makes the water cooler for fish
- 15.** Animals grazing near a rubbish dump sometimes die after feeding there because they
- A. eat too much grass at the dump
 - B. swallow plastics and poisonous materials with the food
 - C. drink too much clean water there
 - D. are frightened by the smell
- 17.** The process by which waste materials are treated and used again to make new articles is called
- A. evaporation
 - B. littering
 - C. recycling
 - D. dumping
- 19.** Burning waste in a closed furnace at a very high temperature is known as
- A. composting
 - B. incineration
 - C. sorting
 - D. sedimentation
- 21.** The FIRST step in managing waste scientifically in a school is to
- A. bury all the waste in one deep hole
 - B. sweep all the waste into the gutter
- 12.** Burning rubbish in the open in a town pollutes the air mainly by releasing
- A. pure water droplets
 - B. nitrogen only
 - C. smoke and poisonous gases
 - D. oxygen and water vapour
- 14.** Plastic waste left in farm soil for many years will MOST likely
- A. remain in the soil and prevent water and roots from passing freely
 - B. dissolve in rain water and disappear
 - C. rot within a few weeks and enrich the soil
 - D. turn into humus that feeds the crops
- 16.** Which of the following is the MOST serious effect of dumping waste into the sea at a fishing community?
- A. Fish and other sea animals are poisoned or choked, reducing the catch
 - B. More sand is washed on to the beach
 - C. The waves become smaller than before
 - D. The sea water becomes cooler for swimming
- 18.** Composting is a method of waste management in which
- A. plant and food waste are allowed to decay to form manure
 - B. waste is sorted and stored in bags for sale
 - C. waste is burnt at a very high temperature
 - D. waste is poured into a flowing stream
- 20.** A sanitary landfill differs from an ordinary rubbish dump because in a sanitary landfill the waste is
- A. poured into a river to be carried away
 - B. burnt in the open every evening
 - C. placed in a lined pit, spread in layers and covered with soil
 - D. left in the open for animals to feed on

- C. burn all the waste behind the classrooms
- D. separate the waste into different types at the point where it is produced

- 23.** The three Rs of waste management are
- A. rake, roast and return
 - B. remove, rinse and rest
 - C. reduce, reuse and recycle
 - D. repeat, report and refuse
- 25.** The decay of material in a compost heap is brought about mainly by
- A. sunlight alone
 - B. bacteria and fungi
 - C. the wind blowing over the heap
 - D. rainwater only
- 27.** A compost heap that is too wet and tightly packed begins to smell badly because
- A. the material has become too dry to decay
 - B. the bacteria in the heap have all died of heat
 - C. the heap has lost all its nitrogen to the air
 - D. air cannot enter, so decay takes place without oxygen and foul gases are formed
- 29.** Metals such as aluminium can be recycled because when heated strongly they
- A. dissolve in water and can be poured
 - B. change into a completely different metal
 - C. decay quickly into humus
 - D. melt and can be cast into new shapes without losing their properties
- 31.** Which of the following is an INNOVATIVE way of managing waste for sustainable development?
- A. Sweeping waste into the nearest gutter at night
 - B. Carrying waste to a neighbouring village to dump
 - C. Burning waste in the open behind the classroom block
 - D. Using biogas from decaying waste to cook in the school kitchen
- 22.** Which of the following wastes is MOST suitable for composting?
- A. Vegetable peels and dry leaves
 - B. Broken louver blades
 - C. Empty metal tins
 - D. Used batteries
- 24.** A pupil washes an empty jam bottle and uses it to store groundnuts. This practice is an example of
- A. composting
 - B. incineration
 - C. recycling
 - D. reuse
- 26.** A compost heap is turned from time to time in order to
- A. kill the earthworms in the heap
 - B. cool the heap so that decay stops
 - C. let air reach the material so that the decomposers work faster
 - D. remove all the water from the heap
- 28.** The rise in temperature inside a working compost heap is evidence that
- A. energy is being released by the micro-organisms as they break down the waste
 - B. the sun has heated the whole heap through
 - C. water is condensing inside the heap
 - D. the heap is about to catch fire
- 30.** Which scientific principle explains why a sanitary landfill is lined at the bottom before waste is placed in it?
- A. The lining makes the waste heavier so that it settles
 - B. The lining keeps the waste warm so that it decays faster
 - C. The lining stops the liquid from the waste from soaking through to the underground water
 - D. The lining allows air to reach the lowest layer of waste

- 33.** The gas formed in a biogas digester from animal dung and food waste is mainly
- methane
 - chlorine
 - oxygen
 - nitrogen
- 35.** Of the following actions, the one that follows the 'reduce' step BEST is
- carrying a reusable bottle to school instead of buying sachet water daily
 - burning the sachets collected at the end of the week
 - separating sachets from tins before dumping them
 - burying the sachets in a pit behind the school
- 37.** In a week a school collected 60 kg of waste, of which 24 kg was food and plant waste. The percentage of the waste that could be composted is
- 60%
 - 40%
 - 36%
 - 24%
- 39.** A criticism that can fairly be made of waste management in many Ghanaian towns is that
- there are no waste bins anywhere in the country
 - all waste is now recycled and nothing is dumped
 - households are forbidden to sort their waste
 - waste is often collected without being separated, so materials that could be recycled are buried or burnt
- 32.** A company collects plastic water sachets and uses them to make paving blocks. This practice is valuable because it
- removes the need to sort waste at all
 - makes the sachets decay faster in the soil
 - turns a waste that does not rot into a useful product and creates jobs
 - increases the amount of plastic thrown away
- 34.** Sustainable waste management is BEST described as managing waste in a way that
- removes waste from the town as cheaply as possible
 - meets the needs of today without damaging the environment for future generations
 - produces the largest possible quantity of waste
 - keeps all waste hidden in one place for ever
- 36.** A waste audit carried out in a school means
- checking the money spent on cleaning the school
 - collecting, sorting, weighing and recording the waste the school produces
 - paying a company to carry the school waste away
 - writing the names of pupils who litter the compound
- 38.** The MOST useful first action for a school whose audit shows that most of its waste is paper and food waste is to
- place separate labelled bins for paper and for food waste in every block
 - ask pupils to take their waste home
 - buy one large bin for all the waste
 - burn the paper and bury the food waste together

- 40.** A town sorts its waste but the collecting truck tips all the sorted bins into one load. The BEST judgement on this practice is that it
- A. makes no difference, because sorting is of no scientific value
 - B. is an improvement, because mixing helps the waste to decay
 - C. is acceptable, because the waste is still removed from the town
 - D. wastes the effort of the households, since the materials are mixed again and cannot easily be recycled

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps of your working in the questions that involve calculations, and give the unit with every answer.

Question 7

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

ANSWER KEY

and Marking Schemes

This section contains correct answers and detailed marking schemes.

Agricultural Tool

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	15.1 Simple Agricultural Tools and Their Uses	Remember
2	D	15.1 Simple Agricultural Tools and Their Uses	Remember
3	D	15.1 Simple Agricultural Tools and Their Uses	Remember
4	A	15.1 Simple Agricultural Tools and Their Uses	Remember
5	C	15.1 Simple Agricultural Tools and Their Uses	Understand
6	D	15.1 Simple Agricultural Tools and Their Uses	Understand
7	A	15.1 Simple Agricultural Tools and Their Uses	Remember
8	B	15.1 Simple Agricultural Tools and Their Uses	Apply
9	A	15.2 Materials Used in Making Simple Tools	Understand
10	A	15.2 Materials Used in Making Simple Tools	Understand
11	C	15.2 Materials Used in Making Simple Tools	Analyse
12	C	15.2 Materials Used in Making Simple Tools	Understand
13	A	15.2 Materials Used in Making Simple Tools	Understand
14	B	15.2 Materials Used in Making Simple Tools	Understand
15	A	15.2 Materials Used in Making Simple Tools	Understand
16	B	15.2 Materials Used in Making Simple Tools	Apply
17	D	15.3 Connecting the Parts of a Tool	Remember
18	D	15.3 Connecting the Parts of a Tool	Understand
19	C	15.3 Connecting the Parts of a Tool	Understand
20	A	15.3 Connecting the Parts of a Tool	Understand
21	B	15.3 Connecting the Parts of a Tool	Remember
22	B	15.3 Connecting the Parts of a Tool	Analyse
23	C	15.4 Processes Involved in Making a Tool	Understand
24	B	15.4 Processes Involved in Making a Tool	Understand
25	D	15.4 Processes Involved in Making a Tool	Remember
26	C	15.4 Processes Involved in Making a Tool	Understand
27	D	15.4 Processes Involved in Making a Tool	Understand
28	B	15.4 Processes Involved in Making a Tool	Analyse
29	C	15.4 Processes Involved in Making a Tool	Understand
30	D	15.4 Processes Involved in Making a Tool	Evaluate
31	D	15.5 Constraints and Precautions in Making...	Understand
32	B	15.5 Constraints and Precautions in Making...	Remember
33	A	15.5 Constraints and Precautions in Making...	Understand
34	C	15.5 Constraints and Precautions in Making...	Remember
35	A	15.5 Constraints and Precautions in Making...	Evaluate
36	A	15.6 Making, Testing and Caring for Tools	Understand
37	C	15.6 Making, Testing and Caring for Tools	Understand

Q	Ans	Curriculum Area	Cognitive Level
38	D	15.6 Making, Testing and Caring for Tools	Apply
39	C	15.6 Making, Testing and Caring for Tools	Evaluate
40	B	15.6 Making, Testing and Caring for Tools	Analyse

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 3 (7%) | Evaluate: 3 (7%) | Remember: 10 (25%) | Understand: 20 (50%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [4 marks]

- (i) **Weeding / making mounds, ridges or seedbeds** (1 mark).
- (ii) **Lifting and transplanting seedlings / digging small holes** (1 mark).
- (iii) **Levelling the seedbed / gathering dry leaves and rubbish** (1 mark).
- (iv) **Carrying manure, soil or produce about the garden** (1 mark).
- **Marking note: One correct use is enough in each case.**

(b) [2 marks]

- Tool: **a pair of secateurs** (accept pruning shears or pruning saw) (1 mark).
- Reason: it **cuts the small branches cleanly without tearing the bark**, and it can be worked with one hand (1 mark).
- **Marking note: Accept any sensible reason that fits the tool named.**

(c) [4 marks]

- Cutlass - working part: **steel** (accept hardened iron) (1 mark); handle: **wood** (accept hard plastic or rubber) (1 mark).
- Hoe - working part: **steel or iron** (1 mark); handle: **wood** (accept bamboo) (1 mark).
- **Marking note: The table must be copied; a list of four materials in the right order also earns the marks.**

Question 2 [Total: 12 marks]

(a) [5 marks]

- A - **handle** (1 mark).
- B - **rivet** (accept bolt or pin) (1 mark).
- C - **tang** (the part of the blade that runs inside the handle) (1 mark).
- D - **blade** (1 mark).
- E - **cutting edge** (accept sharpened edge) (1 mark).
- **Marking note: One mark for each correctly named part; no half marks.**

(b) [3 marks]

- (i) Material: **steel** (accept hardened iron) (1 mark); property: **hard and strong, takes and keeps a sharp edge** (1 mark).
- (ii) A steel handle would be **heavy, slippery and hot or cold to hold**, while wood is light, warm and easy to grip (1 mark).
- **Marking note: In (ii) any one clear comparison between the two materials earns the mark.**

(c) [2 marks]

- The **tang** of the blade is fitted into (or between) the handle (1 mark).
- **Rivets** (or bolts) are passed through holes in the handle and the tang and hammered tight to hold the two parts firmly (1 mark).
- *Marking note: Accept a clear labelled sketch in place of words.**

(d) [2 marks]

- Cause: **the rivets have become loose / the wooden handle has shrunk or worn** so that it no longer grips the tang (1 mark).
- Correction: **tighten or replace the rivets, or fit a new well-seasoned handle** (accept adding a ferrule or wedge) (1 mark).
- *Marking note: Do not award the second mark for 'buy a new cutlass'; the question asks for a repair.**

Question 3 [Total: 10 marks]**(a)** [4 marks]

- L - **the head (cross bar) of the rake** (1 mark).
- J - **wood** (accept bamboo) (1 mark).
- K - **wire** (accept binding wire or metal wire) (1 mark).
- M - **iron (nails)** (accept steel) (1 mark).
- *Marking note: The name of the part and the three materials, one mark each.**

(b) [2 marks]

- Work: it **binds (lashes) the handle firmly to the head** so that the joint does not loosen or twist in use (1 mark).
- Other method: **nailing, screwing, bolting, or fitting the handle into a socket and wedging it** (1 mark).
- *Marking note: Any one sensible alternative method earns the second mark.**

(c) [2 marks]

- Test: **pull the rake through loose soil or over dry leaves several times** (accept any reasonable trial on the work the tool is meant to do) (1 mark).
- Result: **the teeth remain straight and firm, the joint stays tight and the leaves or soil are gathered easily** (1 mark).
- *Marking note: The test must be on the work the tool is meant to do.**

(d) [2 marks]

- Any TWO of: **use thicker, harder nails or steel rods**; **drive the teeth deeper into the head**; **set the teeth closer together**; **make the teeth shorter**; **use a harder wood for the head** (1 mark each, maximum 2 marks).
- *Marking note: The change must act on the strength or fixing of the teeth; 'be careful when using it' earns no mark.**

Question 4

(a) [5 marks]

- Order: ****R, Q, T, S, P**** (4 marks; award 2 marks if only one pair of letters is out of place).
- Process in T: ****forging**** (1 mark).
 - ***Marking note: Award the 4 marks only for the full correct order; a single transposition scores 2.***

(b) [3 marks]

- (i) The hot metal is ****soft and can be beaten into shape**** (1 mark).
- (ii) Quenching ****hardens**** the blade and tempering ****removes the brittleness**** so that it does not snap (1 mark).
- (iii) Seasoned wood ****will not shrink, crack or work loose**** (1 mark).
 - ***Marking note: In (ii) either hardening or tempering correctly explained earns the mark.***

(c) [2 marks]

- Any TWO of: ****cheap and easily available locally****; ****made of good steel that hardens well****; ****reduces the cost of the finished tool****; ****recycles scrap and keeps the environment clean**** (1 mark each).
- ***Marking note: Two distinct advantages are required.***

Question 5 [Total: 10 marks]**(a)** [3 marks]

- Any THREE of: ****shortage of money****; ****lack of proper tools and equipment (forge, anvil, vice, files)****; ****lack of skilled hands or trained supervision****; ****limited time on the timetable****; ****poor or uneven quality of the scrap materials****; ****lack of safety equipment**** (1 mark each).
- ***Marking note: The constraint must be one that affects the making of the tool.***

(b) [4 marks]

- Any FOUR of: ****wear an apron, gloves, goggles and stout shoes****; ****handle hot metal with tongs only****; ****clamp the work in a vice and keep the hands behind the edge****; ****keep the workshop tidy, well lit and ventilated****; ****carry sharp tools with the edge downwards and away from the body****; ****work under supervision and report injuries at once****; ****keep a first aid box and water nearby**** (1 mark each).
- ***Marking note: Four distinct precautions; repeated ideas score once.***

(c) [3 marks]

- Reason: aluminium is ****too soft - it bends in hard soil and cannot be hardened to keep an edge**** (1 mark).
- Any TWO ways: ****look for scrap steel (vehicle springs, old hoe blades, drum**

sheet)**; **ask a local blacksmith or technical school for material or help**;
 make fewer trowels with the steel available; **fold and rivet the aluminium
 over a steel strip to stiffen it** (1 mark each).

- *Marking note: Do not accept 'stop the project' as one of the two ways; the
 question asks how the work can be completed.*

Question 6 [Total: 8 marks]

(a) [3 marks]

- Materials: any TWO of **a hardwood branch or broken handle**, **an old tyre
 or rubber strip**, **a metal pipe or cap**, **bamboo** (1 mark each).
- Tools: **a saw or cutlass, a knife or file or rasp, sandpaper, a hammer and
 nails** - any suitable set (1 mark).
- *Marking note: The tools named must match the materials chosen.*

(b) [3 marks]

- Recognisable shape of a dibber: a straight body with a tapered point (1 mark).
- A handle shown clearly at the top (1 mark).
- All three parts correctly labelled: **handle**, **body**, **point** (1 mark).
- *Marking note: Marks are for shape and labels, not for artistic quality; a ruled
 pencil sketch is enough.*

(c) [2 marks]

- Any TWO actions with a reason: **smooth and oil the wood** - it does not
 splinter or take in water (1 mark); **clean, dry and store it under cover after
 use** - the wood does not rot and metal does not rust (1 mark); **cap or harden
 the point** - it does not wear away quickly (1 mark). Maximum 2 marks.
- *Marking note: No mark for the action alone; the reason must be given.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 5: Agricultural Tools |
Class: Basic 9 (JHS 3)

Content Standard: **B9/JHS3.4.5.1** - Demonstrate knowledge and skill

Animal Feed and Feeding

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	6.1 Feed Ingredients and the Preparation o...	Remember
2	D	6.1 Feed Ingredients and the Preparation o...	Remember
3	B	6.1 Feed Ingredients and the Preparation o...	Remember
4	B	6.1 Feed Ingredients and the Preparation o...	Remember
5	D	6.1 Feed Ingredients and the Preparation o...	Understand
6	D	6.1 Feed Ingredients and the Preparation o...	Understand
7	B	6.1 Feed Ingredients and the Preparation o...	Apply
8	C	6.1 Feed Ingredients and the Preparation o...	Apply
9	B	6.1 Feed Ingredients and the Preparation o...	Understand
10	D	6.2 Ruminants, Monogastrics, Poultry and t...	Remember
11	B	6.2 Ruminants, Monogastrics, Poultry and t...	Remember
12	C	6.2 Ruminants, Monogastrics, Poultry and t...	Remember
13	C	6.2 Ruminants, Monogastrics, Poultry and t...	Remember
14	A	6.2 Ruminants, Monogastrics, Poultry and t...	Understand
15	D	6.2 Ruminants, Monogastrics, Poultry and t...	Understand
16	A	6.2 Ruminants, Monogastrics, Poultry and t...	Remember
17	C	6.2 Ruminants, Monogastrics, Poultry and t...	Remember
18	A	6.2 Ruminants, Monogastrics, Poultry and t...	Remember
19	C	6.2 Ruminants, Monogastrics, Poultry and t...	Apply
20	C	6.3 Selecting and Recording Feed on the Farm	Understand
21	A	6.3 Selecting and Recording Feed on the Farm	Apply
22	D	6.3 Selecting and Recording Feed on the Farm	Understand
23	B	6.3 Selecting and Recording Feed on the Farm	Apply
24	C	6.3 Selecting and Recording Feed on the Farm	Understand
25	B	6.3 Selecting and Recording Feed on the Farm	Apply
26	C	6.4 Feed for the Different Stages of Growth	Remember
27	D	6.4 Feed for the Different Stages of Growth	Remember
28	C	6.4 Feed for the Different Stages of Growth	Remember
29	C	6.4 Feed for the Different Stages of Growth	Understand
30	D	6.4 Feed for the Different Stages of Growth	Understand
31	A	6.4 Feed for the Different Stages of Growth	Remember
32	D	6.4 Feed for the Different Stages of Growth	Apply
33	A	6.4 Feed for the Different Stages of Growth	Understand
34	B	6.4 Feed for the Different Stages of Growth	Analyse
35	B	6.5 Feeding Practice on the Farm	Understand
36	A	6.5 Feeding Practice on the Farm	Understand
37	A	6.5 Feeding Practice on the Farm	Understand

Q	Ans	Curriculum Area	Cognitive Level
38	B	6.5 Feeding Practice on the Farm	Analyse
39	D	6.5 Feeding Practice on the Farm	Apply
40	A	6.5 Feeding Practice on the Farm	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 2 (5%) | Apply: 8 (20%) | Evaluate: 1 (2%) | Remember: 15 (37%) | Understand: 14 (35%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [2 marks]

- (i) The **quantity of feed, made up of different ingredients, given to an animal in one day to meet its needs** (1 mark).
- (ii) The animal **grows fast and stays healthy and produces more meat, milk or eggs**; feed is not wasted and disease is reduced (1 mark).
- **Marking note: Accept 'the daily feed allowance of an animal' in (i).**

(b) [4 marks]

- Maize - **carbohydrate (energy)** (1 mark).
- Fish meal - **protein** (1 mark).
- Oyster shell - **minerals, chiefly calcium** (1 mark).
- Premix - **vitamins and trace minerals** (1 mark).
- **Marking note: Accept 'energy food' for maize and 'body-building food' for fish meal.**

(c) [4 marks]

- **Weigh out each ingredient** in the correct proportion for 100 kg (1 mark).
- On a **clean, dry floor spread the ingredient of the largest quantity (the maize meal) first** and the smaller ones on top (1 mark).
- **Mix the premix and the salt with a little maize meal first** so that the small quantity spreads evenly through the heap (1 mark).
- **Turn the heap over with a shovel several times until the mixture is uniform**, then bag it, label it with the date and store it in a cool dry place on pallets (1 mark).
- **Marking note: Marks are for the steps, not for the order alone; a candidat**

Question 2 [Total: 12 marks]

(a) [3 marks]

- (i) **P** (1 mark).
- (ii) W - **rumen** (1 mark).
- (ii) Z - **abomasum** (true stomach) (1 mark).
- **Marking note: Do not accept 'stomach' alone for Z.**

(b) [3 marks]

- (i) The feed is **stored and softened**, and is later **returned to the mouth to be chewed again as cud** (1 mark).
- (i) **Micro-organisms (bacteria) break down the cellulose (fibre)** of the feed by fermentation (1 mark).

- (ii) **Pig** (accept fowl, rabbit or dog) (1 mark).
- **Marking note:** Either of the two points in (i) may be given in any order.*

(c) [3 marks]

- (i) P can feed on **bulky, fibrous roughage** (grass, hay, silage, crop residues), while Q needs feed that is **low in fibre** (1 mark).
- (i) Q must be given a **concentrate rich in nutrients, in small regular feeds**, while P can be allowed to graze over a long period (1 mark).
- (ii) **Elephant grass, guinea grass, hay, silage, maize stover, groundnut haulms or cassava peels** (1 mark).
- **Marking note:** Any named local roughage is accepted in (ii).*

(d) [3 marks]

- The pig has a **simple (one-chambered) stomach and no rumen with micro-organisms**, so it **cannot digest the cellulose** in the grass (1 mark).
- Most of the grass therefore **passes out undigested** and the animal gets too little energy and protein, so it remains thin (1 mark).
- He should feed a **concentrate ration of maize, bran and a protein source such as fish meal or soya bean meal** (with clean water), using greens only as a small part of the feed (1 mark).
- **Marking note:** The first mark requires the idea of cellulose or fibre.*

Question 3 [Total: 10 marks]

(a) [3 marks]

- (i) **2.4 kg** (1 mark).
- (ii) **Ration M** (1 mark).
- (iii) **2.4 kg - 1.2 kg = 1.2 kg** (1 mark).
- **Marking note:** The unit kg is required in (i) and (iii).*

(b) [3 marks]

- (i) **The higher the protein content of the ration, the greater the average weight of the birds** (1 mark).
- (ii) Ration M supplied **too little protein, which is needed to build muscle and other body tissue**, so the birds grew slowly (1 mark).
- (iii) **Broilers should be fed a ration high in protein (about 20%) if they are to grow well** (1 mark).
- **Marking note:** In (i) accept 'weight increases as protein increases'.*

(c) [2 marks]

- Whether the **extra weight gained is worth more than the extra money spent on the feed** - that is, the profit per bird (1 mark).
- The **price the market pays for heavier birds**, the **quantity of feed needed per kilogramme of meat**, or whether he can **mix a ration of the same quality**

more cheaply from local ingredients** (1 mark).

- *Marking note: Two distinct considerations are needed.*

(d) [2 marks]

- The **same breed, age and number of chicks** in each group (1 mark).

- The **same housing, space, water, vaccination and length of time** for all three groups (1 mark).

- *Marking note: Any two distinct factors earn the marks.*

Question 4 [Total: 10 marks]

(a) [3 marks]

- (i) **Young, growing and mature (adult)** animals (1 mark for all three).

- (ii) **Colostrum** (1 mark).

- (ii) It contains **antibodies that protect the young animal against disease** (accept: it is rich in protein and vitamins) (1 mark).

- *Marking note: No mark in (i) unless all three stages are given.*

(b) [4 marks]

- Day-old to four weeks - **chick (starter) mash** (1 mark).

- Nutrient - **protein** (accept vitamins) for rapid growth (1 mark).

- Laying hen - **layer mash** (1 mark).

- Nutrient - **calcium** for the shells of the eggs (accept protein) (1 mark).

- *Marking note: Accept 'chick starter' and 'layers' mash' as written.*

(c) [3 marks]

- (i) The eggs will have **thin, soft shells that break easily**, because the ration is **low in calcium** (1 mark).

- (i) The **number of eggs laid will fall**, and the birds may peck the shells or become weak in the legs (1 mark).

- (ii) He should **change the birds to a layer ration** and may also offer **oyster shell or limestone grit** separately (1 mark).

- *Marking note: Accept 'he is wasting money' as an extra point only, not for the mark.*

Question 5 [Total: 10 marks]

(a) [4 marks]

- **Wash the feeders and drinkers and provide clean, fresh water** first thing in the morning (1 mark).

- **Weigh out the day's ration (a grower or finisher mash) and share it among enough feeders** so that every bird can reach one (1 mark).

- **Observe the birds while they feed and remove any sick or weak bird**;
remove wet, stale or soiled feed (1 mark).

- **Top up feed and water in the evening and record the quantity of feed given** in the farm record book (1 mark).

- **Marking note:** A candidate who only writes 'give them feed and water' earns 1 mark.*

(b) [3 marks]

- **Wash the feeders and drinkers regularly** and keep the area around them clean (1 mark).

- **Give clean, fresh drinking water at all times** (1 mark).

- **Never feed mouldy, stale or soiled feed**, and store feed in a cool, dry, rat-proof store (1 mark).

- **Marking note:** Maximum 3 marks; accept any three correct hygiene rules.*

(c) [3 marks]

- **No, the suggestion should not be accepted** (the position must be clear) (1 mark).

- **Reason 1:** growing broilers must be **fed every day** (feed available for most of the day), otherwise growth stops, the birds lose weight and the flock becomes uneven (1 mark).

- **Reason 2:** feed that has been wet **grows mould**, which produces poisons that can make the birds sick or kill them, so it must be thrown away (1 mark).

- **Marking note:** A candidate who agrees with the suggestion cannot earn the reason marks.*

Question 6 [Total: 8 marks]

(a) [3 marks]

- (i) **$50 \times 0.12 \text{ kg} = 6 \text{ kg}$** per day (1 mark).

- (ii) **$6 \text{ kg} \times 7 = 42 \text{ kg}$** per week (1 mark).

- (iii) **$42 \times \text{GH¢ } 6.50 = \text{GH¢ } 273.00$** (1 mark).

- **Marking note:** Award the later marks on a wrong figure carried forward, provided the working is correct.*

(b) [2 marks]

- To **know how much feed is used**, work out the cost of production and see whether the farm is making a profit (1 mark).

- To **notice at once when the animals eat less** (an early sign of disease), to order feed in good time and to compare one ration or batch with another (1 mark).

- **Marking note:** Two distinct reasons are required.*

(c) [3 marks]

- **Mix their own ration from local ingredients** (maize, bran, palm kernel cake, fish meal) instead of buying ready-made feed (1 mark).

- **Buy ingredients in bulk at harvest time, when prices are low, and store them dry and rat-proof** so that none is lost (1 mark).
- **Reduce wastage: use well-designed feeders, fill them only part full and raise them to the level of the birds' backs**, and control rats and wild birds (1 mark).
 - **Marking note: Accept 'add safe greens or farm by-products while keeping the ration balanced'; do not accept 'feed the birds less'.**

CURRICULUM ASSESSMEN

Circulatory System

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	7.1 Blood: Composition and Functions	Remember
2	C	7.1 Blood: Composition and Functions	Remember
3	D	7.1 Blood: Composition and Functions	Remember
4	A	7.1 Blood: Composition and Functions	Understand
5	A	7.1 Blood: Composition and Functions	Remember
6	D	7.1 Blood: Composition and Functions	Understand
7	D	7.1 Blood: Composition and Functions	Understand
8	A	7.1 Blood: Composition and Functions	Remember
9	D	7.1 Blood: Composition and Functions	Apply
10	C	7.2 The Heart and the Blood Vessels	Remember
11	D	7.2 The Heart and the Blood Vessels	Remember
12	B	7.2 The Heart and the Blood Vessels	Remember
13	C	7.2 The Heart and the Blood Vessels	Understand
14	B	7.2 The Heart and the Blood Vessels	Remember
15	B	7.2 The Heart and the Blood Vessels	Remember
16	C	7.2 The Heart and the Blood Vessels	Remember
17	A	7.2 The Heart and the Blood Vessels	Understand
18	B	7.2 The Heart and the Blood Vessels	Understand
19	B	7.2 The Heart and the Blood Vessels	Remember
20	A	7.3 The Circulation of Blood in the Body	Remember
21	D	7.3 The Circulation of Blood in the Body	Remember
22	A	7.3 The Circulation of Blood in the Body	Remember
23	C	7.3 The Circulation of Blood in the Body	Understand
24	A	7.3 The Circulation of Blood in the Body	Understand
25	B	7.3 The Circulation of Blood in the Body	Apply
26	D	7.4 Health Challenges of the Circulatory S...	Remember
27	B	7.4 Health Challenges of the Circulatory S...	Remember
28	A	7.4 Health Challenges of the Circulatory S...	Remember
29	D	7.4 Health Challenges of the Circulatory S...	Understand
30	B	7.4 Health Challenges of the Circulatory S...	Analyse
31	C	7.4 Health Challenges of the Circulatory S...	Understand
32	C	7.4 Health Challenges of the Circulatory S...	Evaluate
33	A	7.4 Health Challenges of the Circulatory S...	Analyse
34	B	7.5 Respiration and the Work of the Two Sy...	Understand
35	C	7.5 Respiration and the Work of the Two Sy...	Remember
36	C	7.5 Respiration and the Work of the Two Sy...	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	D	7.5 Respiration and the Work of the Two Sy...	Remember
38	D	7.5 Respiration and the Work of the Two Sy...	Understand
39	C	7.5 Respiration and the Work of the Two Sy...	Analyse
40	A	7.5 Respiration and the Work of the Two Sy...	Analyse

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 2 (5%) | Evaluate: 1 (2%) | Remember: 20 (50%) | Understand: 13 (32%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a) [2 marks]**

- **Plasma** and **red blood cells** (1 mark for both).
- **White blood cells** and **platelets** (1 mark for both).
- *Marking note: Award 1 mark where only two or three components are correct.**

(b) [4 marks]

- Red blood cells - **carry oxygen from the lungs to the body cells** by means of haemoglobin (1 mark).
- White blood cells - **defend the body against disease-causing organisms** (1 mark).
- Platelets - **help the blood to clot when a vessel is cut** (1 mark).
- Plasma - **transports digested food, hormones, carbon dioxide and other wastes** (and distributes heat) (1 mark).
- *Marking note: One correct function for each row earns the mark.**

(c) [2 marks]

- (i) **Anaemia** (1 mark).
- (ii) **Kontomire or other dark green leafy vegetable, liver, meat, fish or beans**
- **foods rich in iron** (1 mark).
- *Marking note: Do not accept 'plenty of water' in (ii).**

(d) [2 marks]

- There is **little haemoglobin, so the blood carries too little oxygen** to the cells (1 mark).
- The cells therefore **cannot respire fast enough to release the energy needed**, so the person tires quickly and breathes faster to try to take in more oxygen (1 mark).
- *Marking note: The idea of oxygen shortage must be clear for the first mark.**

Question 2 [Total: 12 marks]**(a) [4 marks]**

- A - **right atrium** (accept right auricle) (1 mark).
- B - **right ventricle** (1 mark).
- E - **vena cava** (accept superior vena cava) (1 mark).
- G - **aorta** (1 mark).
- *Marking note: Accept 'auricle' for atrium throughout the paper.**

(b) [3 marks]

- (i) It is a **valve**: it **allows blood to flow in one direction only (from the atrium into the ventricle) and prevents it from flowing back** (1 mark).
- (ii) **D pumps blood to the whole body, while B pumps blood only to the lungs**, which are close to the heart (1 mark).
- (ii) **D must therefore develop a much greater pressure**, so it has a thicker layer of muscle (1 mark).
- **Marking note: Both reasons in (ii) must refer to the work done by the chamber.**

(c) [3 marks]

- From **E (vena cava) into A, the right atrium** (1 mark).
- Through the **valve into B, the right ventricle** (1 mark).
- Out through **F, the pulmonary artery, to the lungs** (1 mark).
- **Marking note: The order must be correct; a correct list in the wrong order earns 1 mark only.**

(d) [2 marks]

- K is the **septum**; a hole allows **oxygenated and deoxygenated blood to mix** (1 mark).
- The blood sent to the body therefore **carries less oxygen**, the heart has to work harder and the child tires easily and is breathless (1 mark).
- **Marking note: The idea of mixing of the two kinds of blood is essential.**

Question 3 [Total: 10 marks]**(a)** [3 marks]

- (i) **Arteries, veins and capillaries** (1 mark for all three).
- (ii) So that **materials can pass quickly and easily between the blood and the body cells** (1 mark).
- (iii) The **wrist** (accept the neck, the temple or the ankle) (1 mark).
- **Marking note: No mark in (i) unless all three are named.**

(b) [4 marks]

- Direction - artery: **carries blood away from the heart** (1 mark).
- Direction - vein: **carries blood towards the heart** (1 mark).
- Wall - artery: **thick, muscular and elastic**, to stand the high pressure (1 mark).
- Wall - vein: **thinner, with a wider space inside and with valves** (1 mark).
- **Marking note: Marks are for the paired comparison; an answer that describes only one vessel earns half of the available marks.**

(c) [3 marks]

- (i) The blood **passes through the heart twice in one complete round of the body** (1 mark).
- (i) Once on the **short journey to the lungs and back** and once on the **journey to the rest of the body and back** (1 mark).
- (ii) The blood is **pumped again after the lungs, so it reaches the body at high pressure and travels quickly**, and the two kinds of blood are kept separate (1 mark).
- **Marking note: Accept 'pulmonary circulation and systemic circulation' for the second mark in (i).**

Question 4 [Total: 10 marks]**(a)** [3 marks]

- K - the **wall of the air sac (alveolus)** (1 mark).
- L - the **blood capillary** (1 mark).
- M - a **red blood cell** (1 mark).
- **Marking note: Accept 'alveolus wall' for K and 'capillary' alone for L.**

(b) [3 marks]

- (i) Arrow 1 - **oxygen** (from the air sac into the blood) (1 mark).
- (i) Arrow 2 - **carbon dioxide** (from the blood into the air sac) (1 mark).
- (ii) **Diffusion** (1 mark).
- **Marking note: Accept 'gaseous exchange by diffusion' in (ii); 'breathing' earns no mark.**

(c) [2 marks]

- The wall is **very thin (one cell thick) and moist**, so gases cross easily (1 mark).
- The sacs are **very many, giving a large surface area**, and each is **richly supplied with capillaries** carrying blood away all the time (1 mark).
- **Marking note: Any two distinct features earn the marks.**

(d) [2 marks]

- The **oxygenated blood is carried to the heart and pumped through the arteries and capillaries to every cell**, where the oxygen is used in respiration (1 mark).
- The blood then **collects the carbon dioxide made in the cells and carries it back through the veins and the heart to the lungs** to be breathed out (1 mark).
- **Marking note: Both the outward and the return journey are needed for full marks.**

Question 5**(a)** [3 marks]

- (i) The **chemical breakdown of glucose in the cells, using oxygen, to release energy** (1 mark).
- (ii) **glucose + oxygen** on the left of the arrow (1 mark).
- (ii) **carbon dioxide + water + energy** on the right of the arrow (1 mark).
- **Marking note: Accept 'food' for glucose in (i) but not in the equation.**

(b) [3 marks]

- (i) Breathing is the **physical movement of air into and out of the lungs**; respiration is a **chemical reaction that releases energy** (1 mark).
- (i) Breathing takes place in the **breathing organs only**, while respiration takes place in **every living cell** (1 mark).
- (ii) For **movement of the muscles** (accept growth and repair, keeping the body warm, or the working of the organs) (1 mark).
- **Marking note: Two distinct differences are needed in (i).**

(c) [4 marks]

- The **muscles were working hard and needed much more energy, so they respired faster** (1 mark).
- **Breathing became faster and deeper to take in more oxygen** (and to remove carbon dioxide at the lungs) (1 mark).
- The **heart beat faster to carry oxygen and glucose to the muscles and to carry carbon dioxide away** more quickly (1 mark).
- After the race the **muscles needed less energy and the extra carbon dioxide was removed**, so breathing and heart beat slowly returned to normal (1 mark).
- **Marking note: A candidate who writes only 'he was tired' earns no mark.**

Question 6 [Total: 8 marks]**(a)** [3 marks]

- (i) The **force with which the blood presses on the walls of the blood vessels** as the heart pumps (1 mark).
- (ii) A **sphygmomanometer** (1 mark).
- (iii) **Hypertension (high blood pressure)** (1 mark).
- **Marking note: Accept 'BP apparatus' only where the candidate also writes sphygmomanometer.**

(b) [2 marks]

- **Stroke** or bursting of a blood vessel in the brain (1 mark).
- **Heart attack, heart failure, kidney damage or damage to the eyes** (1 mark).
- **Marking note: Two distinct problems are required; 'death' alone earns no mark.**

(c) [3 marks]

- **Use less salt in food** - too much salt **raises the blood pressure** (1 mark).
- **Eat less fried and fatty food and more fruit and vegetables** - fat is **deposited in the arteries and narrows them** (1 mark).
- **Exercise regularly (about thirty minutes on most days), avoid smoking and heavy drinking, and check the blood pressure at the clinic** - this **strengthens the heart and finds trouble early** (1 mark).
- **Marking note: Each mark requires both a piece of advice and a reason; advice without a reason earns no mark.**

CURRICULUM ASSESSMENT BLUEPRINT**Strand:** 3: System

Climate Change and Green

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	19.1 Climate, Climate Change and the Green...	Remember
2	A	19.1 Climate, Climate Change and the Green...	Understand
3	B	19.1 Climate, Climate Change and the Green...	Remember
4	C	19.1 Climate, Climate Change and the Green...	Understand
5	D	19.1 Climate, Climate Change and the Green...	Remember
6	C	19.1 Climate, Climate Change and the Green...	Remember
7	C	19.1 Climate, Climate Change and the Green...	Understand
8	A	19.2 Natural Factors that Influence Climat...	Understand
9	B	19.2 Natural Factors that Influence Climat...	Understand
10	C	19.2 Natural Factors that Influence Climat...	Understand
11	C	19.2 Natural Factors that Influence Climat...	Analyse
12	B	19.2 Natural Factors that Influence Climat...	Understand
13	A	19.2 Natural Factors that Influence Climat...	Understand
14	D	19.2 Natural Factors that Influence Climat...	Analyse
15	D	19.3 Human Factors that Influence Climate ...	Remember
16	C	19.3 Human Factors that Influence Climate ...	Understand
17	A	19.3 Human Factors that Influence Climate ...	Understand
18	A	19.3 Human Factors that Influence Climate ...	Understand
19	C	19.3 Human Factors that Influence Climate ...	Understand
20	D	19.3 Human Factors that Influence Climate ...	Apply
21	D	19.3 Human Factors that Influence Climate ...	Understand
22	A	19.3 Human Factors that Influence Climate ...	Analyse
23	B	19.4 Minimising Human Activities that Caus...	Apply
24	B	19.4 Minimising Human Activities that Caus...	Understand
25	A	19.4 Minimising Human Activities that Caus...	Understand
26	B	19.4 Minimising Human Activities that Caus...	Apply
27	C	19.4 Minimising Human Activities that Caus...	Understand
28	C	19.4 Minimising Human Activities that Caus...	Understand
29	D	19.4 Minimising Human Activities that Caus...	Evaluate
30	A	19.5 The Green Economy	Remember
31	C	19.5 The Green Economy	Remember
32	D	19.5 The Green Economy	Understand
33	B	19.5 The Green Economy	Apply
34	A	19.5 The Green Economy	Analyse
35	D	19.5 The Green Economy	Understand
36	B	19.6 Climate Change and Green Economy Init...	Remember
37	D	19.6 Climate Change and Green Economy Init...	Understand

Q	Ans	Curriculum Area	Cognitive Level
38	A	19.6 Climate Change and Green Economy Init...	Evaluate
39	D	19.6 Climate Change and Green Economy Init...	Analyse
40	B	19.6 Climate Change and Green Economy Init...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 4 (10%) | Evaluate: 3 (7%) | Remember: 8 (20%) | Understand: 20 (50%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [3 marks]

- (i) **The average weather condition of a place over a long period** (1 mark).
- (ii) **Weather changes from day to day; climate is the average over many years** (1 mark).
- (iii) **A lasting change in the average weather pattern of a place or of the earth** (1 mark).
- **Marking note:** In (ii) both halves of the comparison are needed.*

(b) [3 marks]

- Any THREE of: **longer droughts or dry spells**; **heavy rain in a short time causing floods**; **rivers, streams and wells drying up**; **certain plants or animals disappearing from the area**; **crops failing or maturing at unusual times**; **stronger winds and more storms**; **hotter nights** (1 mark each).
- **Marking note:** The sign must be a lasting change, not a single day's weather.*

(c) [4 marks]

- (i) Any TWO of: **volcanic eruptions**; **changes in the sun's energy**; **changes in ocean currents**; **changes in the earth's orbit or tilt**; **methane from natural wetlands**; **wildfires started by lightning** (1 mark each).
- (ii) A correct explanation, for example **volcanic ash blocks sunlight so less heat reaches the earth** (1 mark) **and the earth cools for a period** (1 mark).
- **Marking note:** A human activity named in (i) earns no mark.*

Question 2 [Total: 12 marks]

(a) [4 marks]

- (i) **Radiation (heat and light) from the sun reaching the earth** (1 mark).
- (ii) **Heat from the earth escaping through the layer into space** (1 mark).
- (iii) **Heat from the earth sent back to the surface by the layer** (1 mark).
- (iv) **The layer of greenhouse gases in the atmosphere** (1 mark).
- **Marking note:** In (iii) the idea of the heat returning to the earth is essential.*

(b) [2 marks]

- Any TWO of: **carbon dioxide**, **methane**, **water vapour**, **nitrous oxide** (1 mark each).
- **Marking note:** Oxygen and nitrogen are not greenhouse gases and earn no mark.*

(c) [3 marks]

- (i) **All the heat radiated by the surface would escape into space** (1 mark); **none would be returned, so the earth would lose heat and be very much colder** (1 mark).
- (ii) **The temperature rises (the earth becomes warmer)** (1 mark).
- **Marking note: Accept 'global warming' in (ii).**

(d) [3 marks]

- Any THREE activities each with the correct gas: **burning petrol or diesel - carbon dioxide** (1 mark); **deforestation, charcoal burning or bush burning - carbon dioxide** (1 mark); **cattle rearing, rotting waste in dumps or flooded rice fields - methane** (1 mark).
- **Marking note: No mark for an activity without the gas it adds.**

Question 3 [Total: 10 marks]**(a) [4 marks]**

- (i) **26.0 °C** (1 mark) and **28.0 °C** (1 mark).
- (ii) **28.0 - 26.0 = 2.0 °C** (1 mark).
- (iii) **2.0 ÷ 4 = 0.5 °C per ten years** (1 mark).
- **Marking note: The unit °C is required for the final two marks.**

(b) [2 marks]

- Trend: **the mean temperature rises steadily from 1985 to 2025** (1 mark).
- Meaning: **the district is becoming warmer - its climate is changing** (1 mark).
- **Marking note: 'The line goes up' alone earns the first mark only.**

(c) [4 marks]

- Any FOUR of: **plant and protect trees**; **stop bush burning**; **use improved stoves, gas or solar energy instead of much firewood and charcoal**; **share transport, walk, cycle or use buses**; **switch off lights and use LED lamps**; **compost waste or use a biogas digester instead of burning it** (1 mark each).
- **Marking note: The action must reduce greenhouse gases, not merely tidy the town.**

Question 4 [Total: 10 marks]**(a) [2 marks]**

- (i) **An economy that improves people's well-being while reducing harm to the environment** (1 mark).
- (ii) Any ONE of **solar panel installer**, **tree nursery worker**, **recycling collector**, **biogas technician**, **organic farmer**, **ecotourism guide** (1 mark).

mark).

- *Marking note: Accept any job whose work protects the environment.*

(b) [4 marks]

- Solar panels: **clean electricity from sunlight, so less fuel is burnt and less carbon dioxide released** (1 mark).

- Biogas digester: **the methane from the waste is captured and burnt as fuel instead of escaping, and less firewood is needed** (1 mark).

- Trees: **they absorb carbon dioxide as they grow and store the carbon** (1 mark).

- Recycling plastics: **saves the energy and oil needed to make new plastic and keeps waste from being burnt** (1 mark).

- *Marking note: Each mark needs the link to greenhouse gases or energy saved.*

(c) [2 marks]

- $60 \div 200 \times 100$ (1 mark) $= 30\%$ (1 mark).

- *Marking note: The percentage sign is required for the second mark.*

(d) [2 marks]

- Any TWO of: **units of electricity bought before and after**; **firewood, charcoal or gas used by the kitchen before and after**; **number and height of surviving trees after one or two years**; **mass of plastic collected and sold each term**; **quantity of waste still burnt or dumped** (1 mark each).

- *Marking note: The evidence must be something that can be measured and compared.*

Question 5 [Total: 10 marks]

(a) [3 marks]

- (i) **M** (1 mark).

- (ii) **L** (1 mark).

- (iii) $40\,000 \div 2\,000 = 20$ times (1 mark).

- *Marking note: Only the letter of the country is needed in (i) and (ii).*

(b) [3 marks]

- **M produces more than half its electricity without burning fuel, so little carbon dioxide is released** (1 mark).

- **Its law against felling has kept forest loss lowest, so its trees go on absorbing carbon dioxide** (1 mark).

- **It acts on both the source of the gas and the removal of the gas, while L and N act on one side only** (1 mark).

- *Marking note: The answer must use figures from the table.*

(c) [2 marks]

- Any TWO of: **it still loses 40 000 hectares of forest a year, far more than one day's planting can replace**; **only 15% of its electricity is from renewable sources, so most of its power still releases greenhouse gases**; **no figures are given for how many seedlings survive, so the planting may achieve little** (1 mark each).

- **Marking note:** A reason must be supported by the table or by sound reasoning.*

(d) [2 marks]

- Any TWO actions each with a benefit: **Green Ghana Day tree planting** - restores forest cover and absorbs carbon dioxide (1 mark); **action against illegal mining in forests and rivers** - protects trees and water (1 mark); **solar and LED street lighting** - less electricity used (1 mark); **improved cook stoves and gas for cooking** - less firewood and charcoal burnt (1 mark); **solar power plants feeding the grid** - clean electricity (1 mark). Maximum 2 marks.

- **Marking note:** No mark for an action without its benefit.*

Question 6 [Total: 8 marks]**(a) [3 marks]**

- Example - natural: **volcanic eruption** (accept changes in the sun, ocean currents, earth's orbit) (1 mark); human: **burning of fossil fuels** (accept deforestation, bush burning, cattle rearing) (1 mark).

- Control: **natural factors cannot be controlled by people, while human factors can be reduced or stopped** (1 mark).

- **Marking note:** The table must be copied; both columns of the second row are needed for the third mark.*

(b) [3 marks]

- Any THREE actions each with a benefit: **plant and protect trees** - absorbs carbon dioxide and gives shade and work (1 mark); **a by-law against bush burning and illegal mining** - saves forest, farms and rivers (1 mark); **buses or cycle paths** - less fuel burnt (1 mark); **solar street lights and panels on public buildings** - less electricity bought (1 mark); **a composting or biogas site for market waste** - less methane and new jobs (1 mark). Maximum 3 marks.

- **Marking note:** No mark for an action given without its benefit.*

(c) [2 marks]

- Any TWO of: **many actions cost little or nothing and even save money**; **green actions create jobs and income in the district**; **doing nothing costs more later through floods, failed harvests and disease**; **support is available from national programmes and funding organisations** (1 mark each).

- **Marking note:** A bare 'it is important' earns no mark; an argument is

required.*

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:**** 5: Humans and the Environment | ****Sub-strand:**** 4: Climate Change and Green Economy | ****Class:**** Basic 9 (JHS 3)

****Content Standard:**** ****B9/JHS3.5.4.1**** - D

Compounds Acids Bases

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	1.1 Elements, Compounds and Chemical Symbols	Remember
2	D	1.1 Elements, Compounds and Chemical Symbols	Remember
3	B	1.1 Elements, Compounds and Chemical Symbols	Remember
4	B	1.1 Elements, Compounds and Chemical Symbols	Remember
5	C	1.1 Elements, Compounds and Chemical Symbols	Understand
6	D	1.1 Elements, Compounds and Chemical Symbols	Analyse
7	C	1.1 Elements, Compounds and Chemical Symbols	Remember
8	A	1.1 Elements, Compounds and Chemical Symbols	Analyse
9	B	1.2 Binary Compounds in the Home and Commu...	Remember
10	D	1.2 Binary Compounds in the Home and Commu...	Understand
11	B	1.2 Binary Compounds in the Home and Commu...	Remember
12	A	1.2 Binary Compounds in the Home and Commu...	Remember
13	C	1.2 Binary Compounds in the Home and Commu...	Understand
14	C	1.2 Binary Compounds in the Home and Commu...	Understand
15	B	1.2 Binary Compounds in the Home and Commu...	Analyse
16	C	1.3 Formation and Formulae of Binary Compo...	Remember
17	A	1.3 Formation and Formulae of Binary Compo...	Understand
18	B	1.3 Formation and Formulae of Binary Compo...	Remember
19	D	1.3 Formation and Formulae of Binary Compo...	Remember
20	C	1.3 Formation and Formulae of Binary Compo...	Apply
21	D	1.3 Formation and Formulae of Binary Compo...	Understand
22	A	1.3 Formation and Formulae of Binary Compo...	Analyse
23	A	1.3 Formation and Formulae of Binary Compo...	Apply
24	D	1.4 Acids, Bases and Salts and the pH Scale	Remember
25	B	1.4 Acids, Bases and Salts and the pH Scale	Remember
26	C	1.4 Acids, Bases and Salts and the pH Scale	Remember
27	A	1.4 Acids, Bases and Salts and the pH Scale	Remember
28	B	1.4 Acids, Bases and Salts and the pH Scale	Understand
29	A	1.4 Acids, Bases and Salts and the pH Scale	Apply
30	D	1.4 Acids, Bases and Salts and the pH Scale	Remember
31	C	1.4 Acids, Bases and Salts and the pH Scale	Understand
32	B	1.4 Acids, Bases and Salts and the pH Scale	Understand
33	D	1.4 Acids, Bases and Salts and the pH Scale	Apply
34	C	1.5 Chemical Bonding in the Formation of C...	Remember
35	A	1.5 Chemical Bonding in the Formation of C...	Understand
36	D	1.5 Chemical Bonding in the Formation of C...	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	C	1.5 Chemical Bonding in the Formation of C...	Remember
38	A	1.5 Chemical Bonding in the Formation of C...	Understand
39	D	1.5 Chemical Bonding in the Formation of C...	Analyse
40	B	1.5 Chemical Bonding in the Formation of C...	Analyse

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 4 (10%) | Remember: 18 (45%) | Understand: 12 (30%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a) [2 marks]**

- **Element** - a substance that cannot be split into two or more simpler substances by an ordinary chemical change; it is made of only one kind of atom (1 mark).
- **Compound** - a substance formed when two or more elements combine chemically in a fixed proportion by mass, and whose properties differ from those of the elements in it (1 mark).
- *Marking note: Award the mark for the idea, not for the exact wording. Do not award a mark where a candidate says a compound is 'two things mixed together'.*

(b) [3 marks]

- Iron - **Fe** (1 mark).
- Sodium - **Na** (1 mark).
- Copper - **Cu** (1 mark).
- *Marking note: The second letter must be a small letter. Do not award the mark for FE, NA or CU, nor for the Latin name alone.*

(c) [3 marks]

- The first dish holds a **mixture**; the black solid is a **compound** (1 mark).
- In the mixture the iron and the sulphur **keep their own properties**; in iron(II) sulphide the properties are **new and different** from those of iron and of sulphur (1 mark).
- The mixture can be made in **any proportion** and the parts **separated by physical means** (a magnet); the compound has a **fixed proportion** by mass and can be broken down only by a chemical change (1 mark).
- Any other correct difference, for example that **heat is given out** when the compound is formed but not when the two are mixed.

(d) [2 marks]

- **Test:** pass a magnet, wrapped in a piece of paper or cling film, over the material in the dish (1 mark).
- **Result:** the grey **iron filings are attracted** and are lifted away, leaving the yellow sulphur behind, showing that the two substances are only mixed and not chemically combined (1 mark).
- *Marking note: Accept adding carbon disulphide or warm water as alternative separation tests only where the expected result is correctly stated. The magnet test is the expected answer.*

Question 2

(a) [2 marks]

- A **binary compound** is a compound made up of two different elements chemically combined (1 mark).
- Any correct household example: **water**, **table salt (common salt)**, **quicklime**, **magnesium oxide**, **hydrogen chloride** (1 mark).
- **Marking note:** Do not award the example mark for marble, baking soda or a fertiliser, because each of these contains more than two elements.*

(b) [4 marks]

- Table salt - chemical name **sodium chloride** (1 mark).
- Table salt - one use: **seasoning food**, **preserving fish or meat**, or **making oral rehydration solution** (1 mark).
- Quicklime - chemical name **calcium oxide** (1 mark).
- Quicklime - one use: **making mortar or whitewash for building**, **treating acidic soil on a farm**, or **treating water** (1 mark).
- **Marking note:** The use must be a real use of that material. Do not award a mark for 'used in the laboratory' without saying what it is used for.*

(c) [3 marks]

- **Iron filings and powdered sulphur are mixed** (in the correct proportion by mass) and placed in a hard-glass test tube or on a tin lid (1 mark).
- The mixture is **heated strongly** until the reaction starts (1 mark).
- **Observation:** the mixture **glows red hot** and continues to glow after the heat is removed, and a **black solid, iron(II) sulphide**, is left (1 mark).
- **Marking note:** Accept the word equation iron + sulphur → iron(II) sulphide in place of one of the first two points. A candidate who mentions that the reaction gives out heat has the third mark.*

(d) [3 marks]

- (i) **Na₂O** (1 mark).
- (ii) **MgCl₂** (1 mark).
- (iii) **Al₂O₃** (1 mark).
- **Marking note:** Accept the figures written as subscripts. Award no mark where the symbols are wrongly written, even if the figures are correct.*

Question 3 [Total: 10 marks]**(a)** [2 marks]

- P: **pH 1** (1 mark).
- R: **pH 7** (1 mark).
- **Marking note:** Accept the figure alone; the words 'pH' need not be repeated.*

(b) [3 marks]

- P (pH 1) - **strong acid** (1 mark).
- Q (pH 5) - **weak acid** (1 mark).
- S (pH 13) - **strong alkali** (strong base) (1 mark).
- **Marking note:** Do not award the mark for 'acid' alone where the question asks for the strength as well.*

(c) [3 marks]

- (i) Blue litmus in **Q turns red**, because Q is acidic (1 mark).
- (i) Red litmus in **S turns blue**, because S is alkaline (1 mark).
- (ii) The solution may be a **strong acid or strong alkali that is corrosive or poisonous** and would burn the mouth or make the pupil ill (1 mark).
- **Marking note:** In (ii) accept any answer that names harm to the body. Do not award the mark for 'it is against the rules' alone.*

(d) [2 marks]

- Universal indicator gives a **different colour at each pH**, which is matched against a chart to read a **pH value** (1 mark).
- It therefore shows **how strong or how weak** the acid or alkali is, while litmus shows only **whether** the solution is acidic or alkaline (1 mark).
- **Marking note:** The second mark depends on the comparison with litmus being stated.*

Question 4 [Total: 10 marks]**(a)** [3 marks]

- They have a **bitter taste** (1 mark).
- They **feel soapy or slippery** between the fingers (1 mark).
- They **turn red litmus paper blue** (1 mark).
- They **neutralise acids** to form a salt and water (1 mark).
- The soluble ones (alkalis) give solutions with a **pH greater than 7** (1 mark, maximum 3).
- **Marking note:** Maximum 3 marks, one for each correct property.*

(b) [2 marks]

- **Acid:** ethanoic acid (in vinegar) used in cooking and for preserving food, **or** citric acid (in lime, lemon or orange) used in drinks; the name and the use together earn 1 mark.
- **Salt:** sodium chloride (table salt) used to season food, to preserve fish or meat, or to make oral rehydration solution; the name and the use together earn 1 mark.
- **Marking note:** A name without a use, or a use without a name, scores nothing for that half of the question.*

(c) [3 marks]

- (i) **Neutralisation** (1 mark).
- (ii) **acid + base $\rightarrow$ salt + water** (1 mark).
- (iii) **Shake a sample of the soil with water and test it with universal indicator** (or a soil-testing kit) and find that the pH has risen towards 7 (1 mark).
 - **Marking note:** In (ii) accept 'alkali' for 'base'. In (iii) accept improved crop growth only where the pupil also mentions testing the soil.*

(d) [2 marks]

- The burning feeling is caused by **too much acid (hydrochloric acid) in the stomach** (1 mark).
- The tablet contains a **base** (for example magnesium hydroxide or a carbonate), which neutralises the excess acid*, forming a salt and water and so reducing the acidity (1 mark).
 - **Marking note:** The second mark requires the word neutralise or a clear statement that the base cancels the acid.*

Question 5 [Total: 10 marks]**(a)** [2 marks]

- X - **2, 8, 1** (1 mark).
- Y - **2, 8, 7** (1 mark).
 - **Marking note:** Accept 2.8.1 and 2.8.7. Award no mark where the shells are written in the wrong order.*

(b) [3 marks]

- X has **1 electron** in its outermost shell; Y has **7 electrons** in its outermost shell (1 mark for both).
- **X loses its one outer electron** and is left with a complete outer shell, becoming a **positively charged ion** (1 mark).
- **Y gains one electron** to complete its outer shell of eight, becoming a **negatively charged ion** (1 mark).
 - **Marking note:** Accept the ions written as X⁺ and Y⁻. The idea of a complete or stable outer shell must appear for the second and third marks.*

(c) [3 marks]

- (i) **Ionic bond** (electrovalent bond) (1 mark).
- (ii) **XY** (1 mark).
- (iii) One electron is **transferred from X to Y**, giving X⁺ and Y⁻ ions, and the **opposite charges attract each other strongly**, holding the ions together (1 mark).
 - **Marking note:** In (ii) do not accept X₂Y or XY₂. In (iii) the words transfer and attraction (or their clear equivalents) are both needed.*

(d) [2 marks]

- It is a **crystalline solid at room temperature** (1 mark).
- It has a **high melting point and boiling point** (1 mark).
- It **dissolves in water** (1 mark).
- It **conducts electricity when molten or dissolved in water, but not when solid** (1 mark, maximum 2).
- **Marking note: Maximum 2 marks, one for each correct property.**

Question 6 [Total: 8 marks]**(a)** [3 marks]

- **Ionic:** formed when **electrons are transferred** from a metal atom to a non-metal atom; example **sodium chloride** (magnesium oxide, calcium chloride) (1 mark).
- **Covalent:** formed when **electrons are shared** between non-metal atoms; example **water** (carbon(IV) oxide, hydrogen chloride) (1 mark).
- **Metallic:** formed by the **attraction between the positive metal ions and the free (mobile) electrons** around them; example **copper** (iron, aluminium) (1 mark).
- **Marking note: Each row earns its mark only when both the formation and a correct example are given.**

(b) [3 marks]

- (i) **Dry cells or a battery, connecting wires, a bulb in a holder (or an ammeter), two carbon electrodes and a beaker or dish** (1 mark for a workable list).
- (ii) **Connect the cell, the bulb and the two electrodes in series, leaving a gap between the electrodes**; place the dry salt in the dish and press the two electrodes into it, and observe the bulb (1 mark).
- (ii) **Clean the electrodes, dissolve the salt in water in the beaker, dip the two electrodes into the solution without letting them touch each other**, and again observe the bulb (1 mark).
- **Marking note: Accept an ammeter**

(c) [2 marks]

- In the **dry solid the ions are held firmly in fixed positions** in the crystal and cannot move, so no current flows (1 mark).
- When the salt **dissolves, the ions become free to move** and carry the charge through the solution, completing the circuit and lighting the bulb (1 mark).
- **Marking note: Do not award a mark for 'water conducts electricity'; the moving ions must be mentioned.**

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:**** 1: Diversity of Matter | ****Sub-strand:**** 1: Materials | ****Class:**** Basic 9 (JHS 3)

****Content Standard:**** ****B9/JHS3.1.1.1**** - Show an understanding of

Conversion and Conservation

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	13.1 Forms of Energy	Remember
2	C	13.1 Forms of Energy	Remember
3	C	13.1 Forms of Energy	Remember
4	B	13.1 Forms of Energy	Remember
5	B	13.1 Forms of Energy	Remember
6	C	13.1 Forms of Energy	Understand
7	B	13.2 Conversion of Energy from One Form to...	Remember
8	A	13.2 Conversion of Energy from One Form to...	Understand
9	A	13.2 Conversion of Energy from One Form to...	Understand
10	B	13.2 Conversion of Energy from One Form to...	Understand
11	D	13.2 Conversion of Energy from One Form to...	Understand
12	C	13.2 Conversion of Energy from One Form to...	Understand
13	D	13.2 Conversion of Energy from One Form to...	Understand
14	A	13.2 Conversion of Energy from One Form to...	Remember
15	B	13.2 Conversion of Energy from One Form to...	Understand
16	D	13.2 Conversion of Energy from One Form to...	Remember
17	D	13.3 The Law of Conservation of Energy	Remember
18	C	13.3 The Law of Conservation of Energy	Apply
19	A	13.3 The Law of Conservation of Energy	Understand
20	C	13.3 The Law of Conservation of Energy	Understand
21	B	13.3 The Law of Conservation of Energy	Analyse
22	C	13.3 The Law of Conservation of Energy	Understand
23	A	13.3 The Law of Conservation of Energy	Apply
24	A	13.4 Energy Conversion Compared with Energy...	Understand
25	B	13.4 Energy Conversion Compared with Energy...	Understand
26	D	13.4 Energy Conversion Compared with Energy...	Understand
27	B	13.4 Energy Conversion Compared with Energy...	Analyse
28	A	13.4 Energy Conversion Compared with Energy...	Understand
29	D	13.4 Energy Conversion Compared with Energy...	Evaluate
30	D	13.5 Application of Conversion and Conserv...	Apply
31	C	13.5 Application of Conversion and Conserv...	Understand
32	A	13.5 Application of Conversion and Conserv...	Understand
33	C	13.5 Application of Conversion and Conserv...	Remember
34	D	13.5 Application of Conversion and Conserv...	Understand
35	C	13.5 Application of Conversion and Conserv...	Analyse
36	A	13.6 Finding Opportunities to Conserve Ene...	Remember

Q	Ans	Curriculum Area	Cognitive Level
37	D	13.6 Finding Opportunities to Conserve Ene...	Apply
38	B	13.6 Finding Opportunities to Conserve Ene...	Evaluate
39	D	13.6 Finding Opportunities to Conserve Ene...	Evaluate
40	A	13.6 Finding Opportunities to Conserve Ene...	Apply

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 3 (7%) | Apply: 5 (12%) | Evaluate: 3 (7%) | Remember: 11 (27%) | Understand: 18 (45%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [3 marks]

- (i) **The change of energy from one form into another** (1 mark).
- (ii) **Energy can neither be created nor destroyed, but only changed from one form to another** (1 mark).
- (iii) **The joule (J)** (1 mark).
- **Marking note:** In (ii) the words 'created' and 'destroyed' (or an equivalent statement that the total energy stays the same) are required.*

(b) [3 marks]

- (i) **Light (solar) energy to electrical energy** (1 mark).
- (ii) **Light energy to chemical energy** (1 mark).
- (iii) **Potential energy to kinetic energy** (1 mark).
- **Marking note:** Both the starting form and the final form are needed in each answer.*

(c) [2 marks]

- **Chemical → electrical** (1 mark).
- **→ light energy (and some heat)** (1 mark).
- **Marking note:** A chain written in the wrong order earns no mark.*

(d) [2 marks]

- **Conversion:** a change of energy from one form to another, with a correct example (for example electrical energy to kinetic energy in a fan) (1 mark).
- **Conservation:** using energy carefully so that none is wasted, with a correct example (for example switching off lamps in an empty room) (1 mark).
- **Marking note:** Each mark needs both the meaning and the example.*

Question 2 [Total: 12 marks]

(a) [4 marks]

- A - **the water stored in the reservoir (lake)** (1 mark).
- B - **the dam (dam wall)** (1 mark).
- C - **the pipe that carries the water down (penstock)** (1 mark).
- E - **the generator** (1 mark).
- **Marking note:** Accept 'alternator' for E and 'water pipe' for C.*

(b) [4 marks]

- (i) **Potential energy (stored energy of position)** (1 mark).
- (ii) **Potential energy changing into kinetic energy** (1 mark).
- (iii) **Kinetic energy of the water changed into mechanical (turning) energy of the wheel** (1 mark).
- (iv) **Mechanical energy changed into electrical energy** (1 mark).
- **Marking note: Accept 'movement energy' for kinetic energy throughout.**

(c) [2 marks]

- **No energy is created or destroyed; it is only changed in form** (1 mark).
- The energy given up by the falling water **equals the electrical energy sent out plus the energy wasted as heat, sound and the motion of the water leaving** (1 mark).
- **Marking note: The second mark requires the idea that the wasted energy must be included in the account.**

(d) [2 marks]

- **The water is free and renewable, so the running cost is low**, while diesel must be bought and its price keeps rising (1 mark).
- **It gives off no smoke or carbon dioxide and little noise**, so it does not pollute the air (accept: the reservoir also serves for fishing, water supply and transport) (1 mark).
- **Marking note: Two distinct reasons are required.**

Question 3 [Total: 10 marks]**(a)** [3 marks]

- (i) **The law of conservation of energy** (1 mark).
- (ii) **$Q = 1\,000 - (600 + 50)$** (1 mark).
- (ii) **$Q = 350\text{ J}$ (unit required)** (1 mark).
- **Marking note: Award the method mark for the correct subtraction even if the arithmetic fails.**

(b) [3 marks]

- (i) **The 600 J of kinetic energy** (1 mark).
- (ii) **The sound energy (50 J) and the heat energy (350 J)** (1 mark).
- (iii) **$600/1\,000 \times 100 = 60\%$** (1 mark).
- **Marking note: In (ii) both wasted outputs are needed for the mark.**

(c) [2 marks]

- **Oil or grease the bearings and moving parts to reduce friction** (1 mark).
- **Tighten and service the machine regularly, keep it clean and cool, or fit a modern motor of the right size** (1 mark).
- **Marking note: Two distinct practical suggestions are required.**

(d) [2 marks]

- **Energy cannot be destroyed** - the law of conservation of energy (1 mark).
- The 400 J has been **changed into heat and sound that spread into the surroundings; it is wasted, not destroyed** (1 mark).
- **Marking note:** A candidate who merely repeats the law without applying it to the 400 J earns 1 mark.*

Question 4 [Total: 10 marks]**(a)** [3 marks]

- (i) **Light energy to electrical energy** (1 mark).
- (ii) **Electrical energy to chemical energy** (1 mark).
- (iii) **Chemical (then electrical) energy to light energy** (1 mark).
- **Marking note:** In (iii) accept 'electrical energy to light energy'.*

(b) [3 marks]

- (i) **Chemical → heat → kinetic → electrical energy** (1 mark).
- (ii) **Much of the heat escapes in the exhaust gases and in cooling the engine** (1 mark).
- (ii) **Energy is lost to friction, sound and vibration, and as heat in the wires** (1 mark).
- **Marking note:** In (i) accept the shorter chain 'chemical to electrical' only if heat or motion is mentioned somewhere in the answer.*

(c) [2 marks]

- **The sun is free and will not run out**, while fuels must be bought and are non-renewable (1 mark).
- **No smoke, carbon dioxide or noise is produced**, so the air stays clean (accept: it can serve places far from the grid) (1 mark).
- **Marking note:** Two distinct reasons are required.*

(d) [2 marks]

- **The solar home system**, because the energy from the sun is free and plentiful, there is no fuel to buy, and LED lamps need very little energy (1 mark).
- **Difficulty: the panel gives no energy at night or on cloudy days, so a battery must be bought, cared for and replaced** (accept: the first cost is high) (1 mark).
- **Marking note:** A candidate who chooses the generating set may earn both marks if the reason and the difficulty are sound and based on energy.*

Question 5 [Total: 10 marks]

(a) [4 marks]

- **List the appliances room by room with their power ratings and the hours they are used** (1 mark).
- **Read the meter regularly (for example at the same time each day for a week) to find the units used** (1 mark).
- **Note where energy is wasted** - lamps and fans left on in empty rooms, security lamps burning by day, old filament bulbs, faulty switches (1 mark).
- **Work out the energy and cost and write a report with a table of findings and recommendations** for the headteacher (1 mark).
- **Marking note:** Marks are for the steps of the survey; the order need not be exactly as in the sche

(b) [3 marks]

- Any one opportunity with its action, for example **lamps and fans left on in empty rooms - appoint a monitor to switch them off** (1 mark).
- A second, for example **security lamps burning by day - fit a timer or a light-sensitive switch** (1 mark).
- A third, for example **old filament bulbs - replace them with LED lamps** (accept: use daylight, service the air conditioners, repair faults) (1 mark).
- **Marking note:** Each mark needs both the opportunity and the action; a bare list of places earns no mark.*

(c) [3 marks]

- (i) Filament lamps use **$2 \text{ kW} \times 5 \text{ h} = 10 \text{ kWh}$** and the LED lamps **$0.2 \text{ kW} \times 5 \text{ h} = 1 \text{ kWh}$** (1 mark).
- (i) Saving = **$10 - 1 = 9 \text{ kWh}$ each day** (1 mark).
- (ii) **$9 \times 30 \times \text{GH}\text{¢}1.50 = \text{GH}\text{¢}405.00$** (1 mark).
- **Marking note:** Allow the candidate's own daily saving to be carried forward into (ii).*

Question 6 [Total: 8 marks]**(a) [4 marks]**

- **Apparatus** clearly named, for example a bicycle with a dynamo, a small lamp and connecting wires (1 mark).
- **What is done** - the wheel is turned (the pedals are worked) so that the dynamo is driven (1 mark).
- **Observation** - the lamp lights, and more brightly as the wheel turns faster (1 mark).
- **Energy conversion shown** - **kinetic (mechanical) energy $\rightarrow$ electrical energy $\rightarrow$ light energy** (1 mark).
- **Marking note:** Any correct demonstration of the candidate's own is accepted provided all four points are given.*

(b) [2 marks]

- ****Use only low-voltage cells or a dynamo, never the mains, and keep water away from the electrical parts**** (1 mark).
- ****Clamp the apparatus firmly and keep fingers, hair and clothing away from turning parts**** (accept: do not touch hot parts; keep visitors behind a line) (1 mark).
- ***Marking note: Two distinct rules are required.***

(c) [2 marks]

- ****Wasted energy is changed into heat and sound that spread out and cannot easily be used again**** (1 mark).
- ****The energy supplied must be paid for and comes largely from limited sources****, so waste costs money, uses up fuel and adds to pollution (1 mark).
- ***Marking note: A simple restatement of the law of conservation of energy, without the idea of usefulness or cost, earns no mark.***

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:** 4: Forces and Energy | ****Sub-strand:** 3: Conversion and Conservation of Energy | ****Class:** Basic 9 (JHS 3)******

****Content Standard:** ****B9/JHS3.4.3.1**** - S**

Crop Maturity

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	5.1 Crop Maturity and the Stages of Growth	Remember
2	D	5.1 Crop Maturity and the Stages of Growth	Remember
3	B	5.1 Crop Maturity and the Stages of Growth	Remember
4	D	5.1 Crop Maturity and the Stages of Growth	Remember
5	B	5.1 Crop Maturity and the Stages of Growth	Understand
6	A	5.1 Crop Maturity and the Stages of Growth	Remember
7	C	5.1 Crop Maturity and the Stages of Growth	Understand
8	A	5.1 Crop Maturity and the Stages of Growth	Analyse
9	C	5.2 Soils, Seed Beds and Their Effect on M...	Remember
10	B	5.2 Soils, Seed Beds and Their Effect on M...	Understand
11	D	5.2 Soils, Seed Beds and Their Effect on M...	Understand
12	D	5.2 Soils, Seed Beds and Their Effect on M...	Remember
13	C	5.2 Soils, Seed Beds and Their Effect on M...	Understand
14	D	5.2 Soils, Seed Beds and Their Effect on M...	Understand
15	D	5.2 Soils, Seed Beds and Their Effect on M...	Understand
16	B	5.2 Soils, Seed Beds and Their Effect on M...	Remember
17	B	5.2 Soils, Seed Beds and Their Effect on M...	Understand
18	C	5.3 Observing and Recording the Maturity o...	Apply
19	D	5.3 Observing and Recording the Maturity o...	Understand
20	A	5.3 Observing and Recording the Maturity o...	Apply
21	A	5.3 Observing and Recording the Maturity o...	Remember
22	C	5.3 Observing and Recording the Maturity o...	Understand
23	A	5.3 Observing and Recording the Maturity o...	Analyse
24	B	5.4 Uses of Crops at Different Maturity St...	Remember
25	B	5.4 Uses of Crops at Different Maturity St...	Remember
26	C	5.4 Uses of Crops at Different Maturity St...	Understand
27	B	5.4 Uses of Crops at Different Maturity St...	Remember
28	D	5.4 Uses of Crops at Different Maturity St...	Remember
29	B	5.4 Uses of Crops at Different Maturity St...	Understand
30	A	5.4 Uses of Crops at Different Maturity St...	Understand
31	A	5.4 Uses of Crops at Different Maturity St...	Analyse
32	C	5.4 Uses of Crops at Different Maturity St...	Analyse
33	B	5.5 Importance of Knowing the Maturity Sta...	Understand
34	D	5.5 Importance of Knowing the Maturity Sta...	Apply
35	A	5.5 Importance of Knowing the Maturity Sta...	Understand
36	A	5.5 Importance of Knowing the Maturity Sta...	Apply
37	D	5.5 Importance of Knowing the Maturity Sta...	Apply

Q	Ans	Curriculum Area	Cognitive Level
38	C	5.5 Importance of Knowing the Maturity Sta...	Understand
39	C	5.5 Importance of Knowing the Maturity Sta...	Apply
40	A	5.5 Importance of Knowing the Maturity Sta...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 6 (15%) | Evaluate: 1 (2%) | Remember: 13 (32%) | Understand: 16 (40%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [2 marks]

- (i) The **stage** at which the crop, or the part of it that is wanted, is fully developed and ready to be harvested (1 mark).
- (ii) The **husks and silk are dry and brown**, the leaves have turned yellow, and the **grains are hard** (1 mark).
- *Marking note: Accept 'the grain is hard and dry' alone in (ii).*

(b) [3 marks]

- (i) **Germination -> seedling -> vegetative growth -> flowering -> maturity** (2 marks for the whole order correct; 1 mark where only one pair is out of place).
- (ii) The plant **produces leaves, stems and roots and grows larger**, making and storing food, but it does not yet flower (1 mark).
- *Marking note: No mark in (i) where two or more pairs are out of order.*

(c) [3 marks]

- Early maturing - **okra, cowpea, garden egg, lettuce or other leafy vegetable** (1 mark).
- Medium maturing - **maize, rice, groundnut or tomato** (1 mark).
- Late maturing - **cassava, yam, plantain, oil palm or cocoa** (1 mark).
- *Marking note: Accept any locally grown crop placed in the right group; the grouping, not the particular crop, earns the mark.*

(d) [2 marks]

- She should **count back the ten weeks from the date she needs the maize** and sow at that time - about the **first or second week of October** (1 mark).
- She should also make sure the crop will have **enough water (rain or irrigation) during those ten weeks**, and may sow in batches so that some maize is ready even if the weather changes (1 mark).
- *Marking note: Accept 'early October' or 'about the beginning of October' for the first mark.*

Question 2 [Total: 12 marks]

(a) [3 marks]

- A - **flat bed** (1 mark).
- B - **ridges** (1 mark).
- C - **mounds** (heaps) (1 mark).
- *Marking note: Accept 'level bed' for A and 'heaps' for C.*

(b) [3 marks]

- A (flat bed) - **easy and quick to prepare** and suitable for small seeds sown close together on well-drained land (1 mark).
- B (ridges) - **give a deeper layer of loose soil and allow excess water to drain away**, and make weeding and earthing up easier (1 mark).
- C (mounds) - **give the deepest loose soil for large tubers such as yam** and drain very well on heavy or wet land (1 mark).
- **Marking note:** One advantage only for each bed; mark the first given.*

(c) [3 marks]

- (i) **C, the mounds** (accept B, the ridges, where the reason is correct) (1 mark).
- (i) **Reason:** the crop is **raised above the wet soil, so excess water drains away and air can reach the roots** (1 mark).
- (ii) **Yam** (accept cassava or sweet potato) (1 mark).
- **Marking note:** The reason must mention drainage or air for the roots.*

(d) [3 marks]

- A bed of **deep, loose, well-drained soil lets the roots spread quickly** and take in water, air and nutrients, so the crop grows fast and **matures earlier** (1 mark).
- A **hard, compacted or waterlogged bed keeps air from the roots**, growth is slowed and the crop **matures late** or gives a poor yield (1 mark).
- A well-prepared bed gives **even germination and even moisture**, so the plants grow at the same rate and the whole crop is ready at about the same time (1 mark).
- **Marking note:** Award the third mark for any other correct point, such as the effect of the manure worked into the

Question 3 [Total: 10 marks]**(a)** [3 marks]

- (i) **About 41 cm** (accept 40 to 42 cm) (1 mark).
- (ii) **Soil Q** (1 mark).
- (iii) **60 cm - 27 cm = 33 cm** (accept 32 to 34 cm) (1 mark).
- **Marking note:** The unit cm is required in (i) and (iii).*

(b) [2 marks]

- Soil Q probably contains **more plant nutrients and humus** than soil R (1 mark).
- Soil Q **holds enough water while still letting air reach the roots**, while soil R may be sandy and lose water and nutrients quickly, or be hard and airless (1 mark).
- **Marking note:** Accept 'soil Q was loam and soil R was sandy' with a reason.*

(c) [3 marks]

- (i) **Soil Q** (1 mark).
- (i) **Reason:** its plants are **growing fastest and are already much taller**, so they will reach flowering and grain filling before the others (1 mark).
- (ii) The **date of flowering** (or the date of harvest, the number of cobs per plant, or the weight of grain harvested) (1 mark).
- **Marking note:** Accept 'number of leaves' in (ii) only with the date recorded.*

(d) [2 marks]

- The **same variety of maize, sown on the same day, at the same spacing and depth** (1 mark).
- The **same amount of water, the same weeding and the same exposure to sunlight** on all three plots (1 mark).
- **Marking note:** Any two distinct factors earn the marks; 'everything' without examples earns nothing.*

Question 4 [Total: 10 marks]**(a)** [4 marks]

- **Mark out two plots of the same size on the same soil, side by side**, preparing ridges on one and a flat bed on the other (1 mark).
- **Sow the same variety of okro on the same day, at the same spacing and the same depth** on both plots (1 mark).
- **Water, weed and treat both plots in exactly the same way** throughout the study (1 mark).
- **Record the date of sowing and the date on which the plants on each plot flower** (and measure several plants, taking the average) (1 mark).
- **Marking note:** A plan with only one plot cannot earn more than 2 marks, since there is nothing to compare.*

(b) [3 marks]

- (i) The **height of the plants** each week (1 mark).
- (i) The **number of leaves, the date of flowering, the rainfall or watering, or any pest damage** (1 mark).
- (ii) Plants **vary from one to another**, so the **average of ten gives a more reliable result** (1 mark).
- **Marking note:** Accept 'to avoid being misled by one unusual plant' in (ii).*

(c) [3 marks]

- (i) **42 - 35 = 7 days** (1 mark).
- (ii) The **ridges gave deeper, looser and better-drained soil**, so the roots spread quickly, the plants took in more water, air and nutrients and grew faster (1 mark).
- (iii) **Farmers should sow okro on ridges**, especially on heavy or wet land, if they want an earlier crop (1 mark).

- *Marking note: In (iii) accept any sensible advice that follows from the result.*

Question 5 [Total: 10 marks]

(a) [4 marks]

- Green maize cob - **roasted or boiled and eaten on the cob** (1 mark).
- Dry maize grain - **milled into flour for kenkey, banku, tuo zaafi or porridge**, used as poultry feed or kept as seed (1 mark).
- Green plantain - **boiled for ampesi, pounded for fufu or sliced for plantain chips** (1 mark).
- Ripe plantain - **fried for kelewele or for 'red red', or eaten raw** (1 mark).
- *Marking note: Any one correct use for each row earns the mark.*

(b) [3 marks]

- (i) Green pods - **cooked and eaten as a fresh vegetable** (1 mark).
- (i) Dry seeds - **stored and cooked as beans (waakye, red red, koose)**, or kept as seed for the next season (1 mark).
- (ii) **Protein** (1 mark).
- *Marking note: Do not accept 'vitamins' in (ii).*

(c) [3 marks]

- (i) The **two stages are used for different dishes** and customers ask for each, so the **stage of maturity decides the use** to which the crop is put (1 mark).
- (ii) The **mature tuber has stored much more starch**, so it gives a **larger and better quality yield of gari** (1 mark).
- (iii) The grains are **still soft and watery, so they shrivel and break during milling** and both the quantity and the quality of the rice are poor (1 mark).
- *Marking note: Accept 'he gets less money for the crop' in (iii) with a reason.*

Question 6 [Total: 8 marks]

(a) [3 marks]

- It helps him to **choose crops that will mature within his growing season** and suit his soil (1 mark).
- It helps him to **decide when to sow and when to harvest**, so that the crop is ready for the market, the festival or the family's needs (1 mark).
- It helps him to **harvest each crop at the stage that suits the use intended**, and to **plan labour, storage and the use of the land (intercropping and rotation)** so that food is available at different times (1 mark).
- *Marking note: Maximum 3 marks.*

(b) [3 marks]

- **No, the advice should not be followed** (1 mark).
- **Reason 1:** the crop would **still be in the field when the rains stopped**, so it would suffer drought at the very time the grain is filling, and the yield would be poor or fail altogether (1 mark).
- **Reason 2:** he would **lose the seed, money and labour** put into the crop; an **early maturing variety that ripens within four months** would be the right choice (1 mark).
- **Marking note:** The first mark is for taking a clear position. A candidate who agrees with the advice cannot earn the reason marks.*

(c) [2 marks]

- A **short duration crop with its time**, for example **cowpea or okro**, ready about 2 to 3 months after sowing (1 mark).
- A **longer duration crop with its time**, for example **maize ready in about 3 to 4 months**, or **cassava**, which can be harvested from about 12 months and left in the ground until needed (1 mark).
- **Marking note:** The crop must be named together with a reasonable time; a crop without a time earns nothing.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 2: Cycles | **Sub-strand:** 3: Crop Production | **Class:** Basic 9 (JHS 3)

Content Standard: **B9/JHS**

Ecosystem

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	9.1 Ecosystem, Habitat and the Components ...	Remember
2	C	9.1 Ecosystem, Habitat and the Components ...	Remember
3	B	9.1 Ecosystem, Habitat and the Components ...	Understand
4	C	9.1 Ecosystem, Habitat and the Components ...	Remember
5	B	9.1 Ecosystem, Habitat and the Components ...	Understand
6	D	9.1 Ecosystem, Habitat and the Components ...	Remember
7	B	9.1 Ecosystem, Habitat and the Components ...	Apply
8	D	9.1 Ecosystem, Habitat and the Components ...	Remember
9	A	9.2 Interdependence: Producers, Consumers ...	Understand
10	B	9.2 Interdependence: Producers, Consumers ...	Remember
11	D	9.2 Interdependence: Producers, Consumers ...	Apply
12	C	9.2 Interdependence: Producers, Consumers ...	Understand
13	A	9.2 Interdependence: Producers, Consumers ...	Remember
14	D	9.2 Interdependence: Producers, Consumers ...	Remember
15	C	9.2 Interdependence: Producers, Consumers ...	Apply
16	C	9.2 Interdependence: Producers, Consumers ...	Understand
17	D	9.2 Interdependence: Producers, Consumers ...	Remember
18	C	9.3 Food Chains and Food Webs	Remember
19	C	9.3 Food Chains and Food Webs	Remember
20	A	9.3 Food Chains and Food Webs	Apply
21	D	9.3 Food Chains and Food Webs	Remember
22	A	9.3 Food Chains and Food Webs	Apply
23	D	9.3 Food Chains and Food Webs	Remember
24	A	9.3 Food Chains and Food Webs	Understand
25	C	9.3 Food Chains and Food Webs	Understand
26	B	9.3 Food Chains and Food Webs	Analyse
27	A	9.4 Interference with the Balance of an Ec...	Analyse
28	B	9.4 Interference with the Balance of an Ec...	Understand
29	A	9.4 Interference with the Balance of an Ec...	Understand
30	B	9.4 Interference with the Balance of an Ec...	Analyse
31	C	9.4 Interference with the Balance of an Ec...	Understand
32	D	9.4 Interference with the Balance of an Ec...	Remember
33	B	9.4 Interference with the Balance of an Ec...	Understand
34	A	9.4 Interference with the Balance of an Ec...	Analyse
35	D	9.5 Protecting the Balance of an Ecosystem	Understand
36	B	9.5 Protecting the Balance of an Ecosystem	Remember

Q	Ans	Curriculum Area	Cognitive Level
37	A	9.5 Protecting the Balance of an Ecosystem	Evaluate
38	B	9.5 Protecting the Balance of an Ecosystem	Understand
39	D	9.5 Protecting the Balance of an Ecosystem	Analyse
40	A	9.5 Protecting the Balance of an Ecosystem	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 5 (12%) | Evaluate: 2 (5%) | Remember: 15 (37%) | Understand: 13 (32%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a)** [2 marks]

- (i) A **community of living organisms together with the non-living surroundings in which they live and with which they interact** (1 mark).
- (ii) The **particular place in which an organism lives** (1 mark).
- **Marking note:** Accept 'the home of an organism' for (ii).*

(b) [3 marks]

- (i) Any **two living things**, for example **water lilies and tilapia** (or grass, frogs, birds, bacteria) (1 mark).
- (ii) Any **two non-living things**, for example **water and sunlight** (or soil, air, dissolved oxygen, temperature) (1 mark).
- (iii) **All the organisms of the same kind living in one area at the same time** (1 mark).
- **Marking note:** Both examples are needed in (i) and in (ii) for the mark.*

(c) [3 marks]

- A habitat is the **place where an organism lives**, while an ecosystem is the **whole community of organisms together with the non-living surroundings** (1 mark).
- A **habitat is part of an ecosystem and is smaller**; one ecosystem contains many habitats (1 mark).
- Example - habitat: **the mud at the bottom of the pond** (or the bark of a tree); ecosystem: **the whole pond** (or the grassland) (1 mark for both examples).
- **Marking note:** The third mark requires one example of each.*

(d) [2 marks]

- (i) A **population** (1 mark).
- (ii) An **ecosystem** (1 mark).
- **Marking note:** No mark where two answers are given for one part.*

Question 2 [Total: 12 marks]**(a)** [3 marks]

- (i) **Grass and the maize plant** (1 mark for both).
- (ii) A chain of **four organisms taken correctly from the web**, for example **grass -> mouse -> snake -> hawk** (1 mark).
- (ii) The **arrows point from the organism eaten to the organism that eats it** (1 mark).

- *Marking note: A chain containing a link that is not in the web earns no mark.*

(b) [3 marks]

- (i) **Grasshopper, mouse or weaver bird** (1 mark).
- (ii) **Lizard or snake** (1 mark).
- (iii) The **hawk** (1 mark).
- *Marking note: In (ii) accept the hawk only where the candidate states that it is feeding on the weaver bird.*

(c) [3 marks]

- (i) The number of snakes would **fall (decrease)** (1 mark).
- (i) **Reason:** they would lose one of their main foods and would have to depend on lizards alone, so some would **starve or move away** (1 mark).
- (ii) **Less maize** would be eaten, so the harvest would improve for a time (1 mark).
- *Marking note: Accept 'the number of lizards would also fall as snakes eat more of them' as an additional correct point.*

(d) [3 marks]

- Prediction: the **maize harvest would fall** (1 mark).
- **The predators that controlled the mice and the weaver birds would be gone, so their numbers would increase rapidly** (1 mark).
- The larger numbers of mice and birds would **eat more maize in the field and in store**, so the balance of the ecosystem is upset and the farmers lose more than they gain (1 mark).
- *Marking note: A prediction without justification earns 1 mark only.*

Question 3 [Total: 10 marks]

(a) [3 marks]

- (i) **Green plants make their own food by photosynthesis** and so bring the Sun's energy into the ecosystem (1 mark).
- (ii) Consumers **feed on plants or on other animals**, passing energy and materials along the food chain (1 mark).
- (iii) Decomposers **break down dead organisms and waste and release nutrients back into the soil and water** (1 mark).
- *Marking note: Accept 'recycle nutrients' for (iii).*

(b) [3 marks]

- (i) The **Sun** (1 mark).
- (ii) It means **'is eaten by'** and shows the direction in which food and energy pass (1 mark).
- (iii) **Bacteria** (accept fungi, mushrooms or moulds) (1 mark).
- *Marking note: Do not accept 'earthworm' alone in (iii) unless described as feeding on dead matter.*

(c) [4 marks]

- Much of the energy is **used for the life processes of the organism (movement, growth, keeping warm)** (1 mark).
- This energy is finally **lost to the surroundings as heat**, and some is lost in **waste and in parts that are not eaten** (1 mark).
- Only the energy **stored in the body can pass to the next level**, so there is **less energy at every step** (1 mark).
- Therefore **food chains are short, and there must be far more producers than primary consumers, and more primary consumers than predators** (1 mark).
- **Marking note: The fourth mark requires the link to numbers of organisms.**

Question 4 [Total: 10 marks]**(a) [3 marks]**

- (i) **About 520 kg** (accept 510 to 530 kg) (1 mark).
- (ii) **About 95 kg** (accept 90 to 100 kg) (1 mark).
- (iii) **520 kg - 95 kg = 425 kg** (accept 410 to 440 kg) (1 mark).
- **Marking note: The unit kg is required in all three parts.**

(b) [3 marks]

- (i) Before 2022 the catch **fell only slowly** (from about 520 kg to about 465 kg) (1 mark).
- (i) After 2022 it **fell steeply and steadily** to about 95 kg (1 mark).
- (ii) The **mining appears to have damaged the river, so that the fish population has collapsed** (1 mark).
- **Marking note: In (i) a candidate must distinguish the two periods.**

(c) [2 marks]

- The water is **filled with mud, which cuts off light from water plants and clogs the gills of fish** (1 mark).
- **Poisonous chemicals such as mercury enter the water**, killing organisms and passing along the food chain; breeding grounds on the banks are destroyed (1 mark).
- **Marking note: Two distinct ways are required.**

(d) [2 marks]

- **No (not completely)** - the graph only shows that the two things **happened at the same time**, which does not by itself prove cause (1 mark).
- Other evidence: **tests of the mud and chemicals in the water before and after 2022**, records of the number of fishermen and the nets used, or the catch from a **similar river with no mining** (1 mark).
- **Marking note: A candidate who simply agrees with the pupil earns no mark.**

Question 5**(a) [4 marks]**

- Bush burning - **destroys humus, soil organisms and the homes of animals, and leaves the soil bare to erosion** (1 mark).
- Cutting down a forest - **removes habitats, causes soil erosion and reduces rainfall in the area** (1 mark).
- Overhunting - **the species may disappear from the area, affecting all the organisms that depend on it** (1 mark).
- Heavy use of pesticide - **kills useful organisms as well as pests and builds up in the bodies of animals along the food chain** (1 mark).
- **Marking note: One correct effect for each row earns the mark.**

(b) [3 marks]

- The pesticide **washes into the lagoon and is taken in by the small plants and animals**, and is **not broken down or passed out** of their bodies (1 mark).
- Each fish **eats many of these small organisms and keeps all their pesticide**, and each bird eats many fish (1 mark).
- So the quantity **becomes greater at every step of the food chain**, until the birds at the top are harmed (1 mark).
- **Marking note: The idea of building up along the chain must be clear.**

(c) [3 marks]

- Prediction 1 with justification: the **grass dies because there is too little water for photosynthesis and growth**, so the producers and the food supply of the ecosystem are reduced (1 mark).
- Prediction 2 with justification: the **plant-eating animals starve, die or move away**, and their **predators decline too**, because each level depends on the one below (1 mark).
- A further correct prediction with justification - for example **water holes dry up so animals crowd together and disease spreads**, or **the bare soil is eroded when the rains return** (1 mark).
- **Marking note: A predic**

Question 6 [Total: 8 marks]**(a) [3 marks]**

- **Stop all mining in and along the river** and report illegal mining to the authorities (1 mark).
- **Replant trees and grass along the banks** to hold the soil and shade the water (1 mark).
- **Ban poisons, chemicals and small-mesh nets in fishing, agree on a closed season, and keep refuse out of the river** (1 mark).
- **Marking note: Maximum 3 marks; any three sensible measures are accepted.**

(b) [2 marks]

- **Every organism depends on others, so damage to one part upsets the whole system**, including the human beings in it (1 mark).
- People **depend on the ecosystem for food, water, fuel, medicines and income**, and a lost species cannot be brought back (1 mark).
- **Marking note: Two distinct reasons are required.**

(c) [3 marks]

- **Reason 1:** the work is **short lived** - when the river and the land are destroyed the mining ends and nothing is left for the community (1 mark).
- **Reason 2:** the damage **costs far more than the wages earned**: farming, fishing and clean drinking water are lost and mercury harms health (1 mark).
- A better way: **licensed and regulated mining with reclamation of the land**, or **farming, fish farming, tree planting, agro-processing or ecotourism** supported by the district assembly (1 mark).
- **Marking note: Candidates may argue for regulated mining; award the marks where the reason**

Electricity and Electr

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	12.1 Electric Circuits and Circuit Symbols	Remember
2	C	12.1 Electric Circuits and Circuit Symbols	Understand
3	C	12.1 Electric Circuits and Circuit Symbols	Remember
4	D	12.1 Electric Circuits and Circuit Symbols	Remember
5	B	12.1 Electric Circuits and Circuit Symbols	Remember
6	C	12.1 Electric Circuits and Circuit Symbols	Remember
7	D	12.1 Electric Circuits and Circuit Symbols	Understand
8	D	12.2 Series and Parallel Circuits	Remember
9	B	12.2 Series and Parallel Circuits	Understand
10	C	12.2 Series and Parallel Circuits	Understand
11	A	12.2 Series and Parallel Circuits	Remember
12	D	12.2 Series and Parallel Circuits	Remember
13	B	12.2 Series and Parallel Circuits	Understand
14	C	12.2 Series and Parallel Circuits	Apply
15	B	12.2 Series and Parallel Circuits	Analyse
16	B	12.2 Series and Parallel Circuits	Apply
17	C	12.3 Transformation of Electrical Energy	Remember
18	D	12.3 Transformation of Electrical Energy	Remember
19	A	12.3 Transformation of Electrical Energy	Remember
20	C	12.3 Transformation of Electrical Energy	Remember
21	D	12.3 Transformation of Electrical Energy	Understand
22	B	12.3 Transformation of Electrical Energy	Understand
23	A	12.4 Current, Voltage, Resistance and the ...	Remember
24	A	12.4 Current, Voltage, Resistance and the ...	Remember
25	B	12.4 Current, Voltage, Resistance and the ...	Apply
26	A	12.4 Current, Voltage, Resistance and the ...	Apply
27	B	12.4 Current, Voltage, Resistance and the ...	Apply
28	D	12.4 Current, Voltage, Resistance and the ...	Apply
29	D	12.4 Current, Voltage, Resistance and the ...	Apply
30	C	12.4 Current, Voltage, Resistance and the ...	Analyse
31	B	12.5 Diodes and LEDs: Forward Bias and Rev...	Remember
32	A	12.5 Diodes and LEDs: Forward Bias and Rev...	Remember
33	D	12.5 Diodes and LEDs: Forward Bias and Rev...	Understand
34	A	12.5 Diodes and LEDs: Forward Bias and Rev...	Understand
35	C	12.5 Diodes and LEDs: Forward Bias and Rev...	Analyse
36	A	12.5 Diodes and LEDs: Forward Bias and Rev...	Remember
37	C	12.6 Resistors and Capacitors in Electroni...	Understand

Q	Ans	Curriculum Area	Cognitive Level
38	B	12.6 Resistors and Capacitors in Electroni...	Remember
39	A	12.6 Resistors and Capacitors in Electroni...	Analyse
40	A	12.6 Resistors and Capacitors in Electroni...	Analyse

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 7 (17%) | Remember: 18 (45%) | Understand: 10 (25%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [3 marks]

- (i) **Electrical energy to light energy** (accept 'and heat') (1 mark).
- (ii) **Electrical energy to kinetic (mechanical) energy** (1 mark).
- (iii) **Electrical energy to heat energy** (1 mark).
- **Marking note:** The starting form, electrical energy, must be stated in at least one of the three answers for full credit.*

(b) [3 marks]

- A **complete (closed) circuit** drawn as a single loop (1 mark).
- **Correct symbols for the cell and the switch** (1 mark).
- **Two lamps drawn one after the other in the same path (in series), all parts labelled** (1 mark).
- **Marking note:** Accept clearly labelled drawings of the components in place of the standard symbols; a circuit with a gap in it loses the first mark.*

(c) [2 marks]

- **Arrangement or voltage:** in series the lamps are in one path and share the voltage; in parallel each lamp is across the supply and receives the full voltage (1 mark).
- **Effect of a break:** in series one broken lamp puts off all the lamps; in parallel the other lamps continue to work (accept the current comparison) (1 mark).
- **Marking note:** Each mark needs both sides of the comparison.*

(d) [2 marks]

- Each lamp **receives the full supply voltage and so lights normally, and each has its own switch** (1 mark).
- **A lamp that blows does not put out the other lamps**, since the other branches are still complete (1 mark).
- **Marking note:** A bare statement 'parallel is better' earns no mark.*

Question 2 [Total: 12 marks]

(a) [4 marks]

- W - **the battery (two cells)** (1 mark).
- Z - **the switch (open)** (1 mark).
- C - **the ammeter** (1 mark).
- D - **a lamp (bulb)** (1 mark).
- **Marking note:** Accept 'cells' for W; do not accept 'voltmeter' for C.*

(b) [3 marks]

- (i) **In parallel** (1 mark).
- (ii) **Lamp X continues to burn (stays on)** (1 mark).
- (ii) Because **its own branch is still a complete path from the battery**; only the branch carrying Y is broken (1 mark).
- **Marking note: 'It goes off' earns no mark, even with a good explanation.**

(c) [3 marks]

- (i) **Both X and Y go off** (1 mark).
- (i) Because **D is in series with the whole parallel section, so the only path from the battery is broken** (1 mark).
- (ii) **Electrical energy to light energy (and heat energy)** (1 mark).
- **Marking note: In (i) the reason must mention that D is in the main line (in series).**

(d) [2 marks]

- Correct formula and substitution: **$V = IR = 1.5 \times 6$** (1 mark).
- **$V = 9 \text{ V}$** (unit required) (1 mark).
- **Marking note: Award the first mark for the correct substitution even if the arithmetic fails.**

Question 3 [Total: 10 marks]**(a)** [3 marks]

- (i) **$V = IR$** (1 mark).
- (ii) Current: **ampere (A)** (1 mark).
- (ii) Resistance: **ohm (Ω)** (1 mark).
- **Marking note: Accept 'amp' for ampere.**

(b) [3 marks]

- (i) **$I = V/R = 6/3 = 2 \text{ A}$** (1 mark).
- (ii) Total resistance = **$3 + 3 = 6 \Omega$** (1 mark).
- (ii) **$I = 6/6 = 1 \text{ A}$** (unit required) (1 mark).
- **Marking note: A candidate who writes 1 A without showing the total resistance earns 1 mark only.**

(c) [2 marks]

- **$R = V/I = 12/0.5$** (1 mark).
- **$R = 24 \Omega$** (unit required) (1 mark).
- **Marking note: A candidate who writes 6Ω (having multiplied) earns no marks.**

(d) [2 marks]

- **No / the pupil is wrong** (1 mark).
- **Reason:** $I = V/R$, so at the same voltage doubling the resistance **halves the current** (1 mark).
 - **Marking note:** The reason mark requires the relation $I = V/R$ or a clear statement of the inverse relationship.*

Question 4 [Total: 10 marks]**(a)** [3 marks]

- J - **a resistor** (1 mark).
- K - **a light-emitting diode (LED)** (1 mark).
- L - **the cells (battery)** (1 mark).
 - **Marking note:** Accept 'battery' for L; 'diode' alone for K earns the mark only if 'light-emitting' or LED is written.*

(b) [3 marks]

- (i) **Forward biased** (1 mark).
- (ii) The **positive terminal of the cells is connected to the anode of the LED** (the broad side of the triangle), so current flows in the direction in which the triangle points (1 mark).
- (ii) **The LED conducts and gives out light**, which happens only when it is forward biased (1 mark).
 - **Marking note:** Accept an answer argued from the observation (the LED lights, so it must be forward biased) for the second mark.*

(c) [2 marks]

- **The LED would not light (goes off)** (1 mark).
- Because it is now **reverse biased**, and a diode conducts in one direction only, so practically no current flows (1 mark).
 - **Marking note:** 'It will burn out' earns no mark.*

(d) [2 marks]

- (i) **To limit the current so that the LED is not damaged** (1 mark).
- (ii) **$I = V/R = 1.2/60 = 0.02 \text{ A}$** (accept 20 mA; unit required) (1 mark).
 - **Marking note:** In (i) 'to increase the brightness' earns no mark.*

Question 5 [Total: 10 marks]**(a)** [3 marks]

- (i) **A component that allows current to flow in one direction only** (1 mark).
- (ii) **The positive of the supply is joined to the anode**, so the diode conducts (1 mark).
- (iii) **The positive of the supply is joined to the cathode**, so practically no

current flows (1 mark).

- *Marking note: Accept answers written in terms of 'the way the cells are turned round' if the direction of conduction is clear.*

(b) [3 marks]

- (i) ****Controls or limits the current**** (opposes the flow of current) (1 mark).
- (ii) ****Stores electric charge and releases it when needed**** (1 mark).
- (iii) ****Gives out light when current passes through it in the forward direction**** (1 mark).

- *Marking note: In (ii) 'stores electricity' is accepted.*

(c) [2 marks]

- ****LED lamps give the same light for much less electrical energy**** (a filament lamp wastes most of its energy as heat), so the bill falls (1 mark).
- ****They last much longer**** (accept: they run cool and are safer, or they suit a solar supply) (1 mark).

- *Marking note: Two distinct reasons are required.*

(d) [2 marks]

- ****The LED lights very brightly for a moment and is then damaged (burns out)****, because the current is not limited (1 mark).
- ****A resistor should be connected in series with the LED**** to limit the current to about 20 mA (1 mark).

- *Marking note: A candidate who says only 'it will spoil' without the idea of the large current earns 1 mark.*

Question 6 [Total: 8 marks]

(a) [4 marks]

- ****Set-up:**** the cell, switch, lamp and diode joined in series in a complete circuit (1 mark).
- ****Method:**** close the switch and note the lamp; then turn the diode round (or turn the cell round), leaving everything else unchanged, and note the lamp again (1 mark).
- ****Observation:**** the lamp lights in one connection only and does not light in the other (1 mark).
- ****Conclusion:**** the diode ****allows current in one direction only - it conducts when forward biased**** (1 mark).

- *Marking note: A group that turns the cell round instead of the diode gains the same marks.*

(b) [2 marks]

- ****Keep the switch open until the circuit has been checked, and never short-circuit the cell**** (1 mark).

- **Use only the low-voltage cells supplied (never the mains), and disconnect the cell when the work is finished** (accept: include a series resistor with an LED; make clean tight connections) (1 mark).

- **Marking note:** Two distinct precautions are required.*

(c) [2 marks]

- **The circuit is not complete** - a loose or broken connection, or the switch left open (1 mark).

- **The cell is flat, the lamp's filament is broken, or the diode itself is damaged** (1 mark).

- **Marking note:** Two different reasons are required; 'the diode is reverse biased' cannot explain a failure in both connections and earns no mark.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 2: Electricity and Electronics |
Class: Basic 9 (JHS 3)

Content Standard: **B9/JHS3.4.2.1** - Construct electrical circuits and illustrate how electrical

Energy

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	11.1 Ways of Conserving Energy	Remember
2	B	11.1 Ways of Conserving Energy	Remember
3	B	11.1 Ways of Conserving Energy	Understand
4	D	11.1 Ways of Conserving Energy	Understand
5	B	11.1 Ways of Conserving Energy	Understand
6	C	11.1 Ways of Conserving Energy	Understand
7	C	11.1 Ways of Conserving Energy	Apply
8	B	11.1 Ways of Conserving Energy	Evaluate
9	C	11.2 The Importance of Energy Conservation	Remember
10	A	11.2 The Importance of Energy Conservation	Understand
11	C	11.2 The Importance of Energy Conservation	Remember
12	B	11.2 The Importance of Energy Conservation	Understand
13	C	11.2 The Importance of Energy Conservation	Apply
14	D	11.2 The Importance of Energy Conservation	Evaluate
15	A	11.3 Calculating the Energy Consumed	Remember
16	A	11.3 Calculating the Energy Consumed	Remember
17	B	11.3 Calculating the Energy Consumed	Apply
18	D	11.3 Calculating the Energy Consumed	Apply
19	B	11.3 Calculating the Energy Consumed	Apply
20	A	11.3 Calculating the Energy Consumed	Apply
21	D	11.3 Calculating the Energy Consumed	Apply
22	B	11.3 Calculating the Energy Consumed	Apply
23	D	11.3 Calculating the Energy Consumed	Analyse
24	A	11.4 The Camera and the Formation of Images	Remember
25	C	11.4 The Camera and the Formation of Images	Remember
26	A	11.4 The Camera and the Formation of Images	Understand
27	B	11.4 The Camera and the Formation of Images	Remember
28	D	11.4 The Camera and the Formation of Images	Apply
29	D	11.5 Shadows: Umbra and Penumbra	Understand
30	B	11.5 Shadows: Umbra and Penumbra	Remember
31	A	11.5 Shadows: Umbra and Penumbra	Remember
32	C	11.5 Shadows: Umbra and Penumbra	Understand
33	D	11.6 Eclipses of the Sun and of the Moon	Understand
34	D	11.6 Eclipses of the Sun and of the Moon	Understand
35	C	11.6 Eclipses of the Sun and of the Moon	Analyse
36	A	11.7 Light Changing Path from One Medium t...	Remember

Q	Ans	Curriculum Area	Cognitive Level
37	C	11.7 Light Changing Path from One Medium t...	Understand
38	A	11.7 Light Changing Path from One Medium t...	Remember
39	A	11.7 Light Changing Path from One Medium t...	Understand
40	D	11.7 Light Changing Path from One Medium t...	Analyse

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 3 (7%) | Apply: 9 (22%) | Evaluate: 2 (5%) | Remember: 13 (32%) | Understand: 13 (32%)

Energy

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a)** [3 marks]

- (i) **Using energy carefully so that it is not wasted**, that is, doing the same work with less energy (1 mark).
- (ii) **Switch off the lamps, fan and television when they are not in use** (1 mark).
- (ii) **Iron all the clothes at one time (ironing in bulk)** or **fit energy-efficient lamps** or **use daylight during the day** (1 mark).
- **Marking note:** In (ii) the two ways must be different from each other.*

(b) [3 marks]

- (i) The iron is **heated up once instead of four times**, so less energy is used in heating it (1 mark).
- (ii) An LED **gives the same light for much less electrical energy**, while a filament bulb wastes most of its energy as heat (1 mark).
- (iii) The lid **keeps the heat and steam in the pot**, so the food cooks faster and less fuel is used (1 mark).
- **Marking note:** A bare statement such as 'it saves energy' earns no mark; a reason is required.*

(c) [2 marks]

- **It reduces the electricity bill and the money spent on fuel** (accept: it makes the supply serve more people) (1 mark).
- **It makes non-renewable fuels last longer and reduces pollution and the release of carbon dioxide** (1 mark).
- **Marking note:** Two distinct reasons are required; two ways of saying 'saves money' earn 1 mark only.*

(d) [2 marks]

- **No / I do not agree** (1 mark).
- **Reason:** the food would warm up and spoil, and the motor must then work hard to cool the whole cabinet again, so little energy is saved; **it is better to keep the door shut, defrost it and keep it away from the cooker** (1 mark).
- **Marking note:** A candidate who agrees cannot gain the reason mark, because the reason must support a sound judgement.*

Question 2

(a) [4 marks]

- A - **the object (the lighted candle)** (1 mark).
- B - **the pinhole / small opening** (1 mark).
- C - **the light-proof box (body of the camera)** (1 mark).
- E - **the image of the candle** (1 mark).
- **Marking note:** Accept 'aperture' for B and 'inverted image' for E.*

(b) [3 marks]

- (i) It is **real** (it is caught on a screen) (1 mark).
- (i) It is **inverted (upside down)**; accept 'smaller than the object' or 'laterally inverted' (1 mark).
- (ii) **Light travels in straight lines**, so rays from the top of the flame cross at the hole and reach the lower part of the screen, and rays from the foot reach the upper part (1 mark).
- **Marking note:** 'Clear' or 'bright' is not a property for the purpose of this question.*

(c) [3 marks]

- Correct relation: **height of image / height of object = length of box / distance of the object** (1 mark).
- Substitution: **height of image = $12 \times 15 / 60$** (1 mark).
- **Height of image = 3 cm** (unit required) (1 mark).
- **Marking note:** Award the first two marks for correct method even if the arithmetic is wrong; the final mark needs the unit cm.*

(d) [2 marks]

- (i) The picture becomes **larger** (1 mark).
- (ii) The picture becomes **brighter but blurred (not sharp)**, because overlapping patches of light reach the screen (1 mark).
- **Marking note:** In (i) accept 'bigger and less sharp'; the key idea is the increase in size.*

Question 3 [Total: 10 marks]**(a)** [2 marks]

- (i) **$P = IV$** (power = current $\times$ voltage) (1 mark).
- (ii) **$I = P/V = 1\,500/240 = 6.25\text{ A}$** (unit required) (1 mark).
- **Marking note:** Accept 6.3 A.*

(b) [4 marks]

- (i) **$0.24\text{ kW} \times 5\text{ h} = 1.2\text{ kWh}$** (1 mark).
- (ii) **$0.1\text{ kW} \times 4\text{ h} = 0.4\text{ kWh}$** (1 mark).
- (iii) **$1.5\text{ kW} \times 1\text{ h} = 1.5\text{ kWh}$** (1 mark).

- (iv) **$0.15 \text{ kW} \times 24 \text{ h} = 3.6 \text{ kWh}$** (1 mark).

- **Marking note:** A candidate who works in watt-hours and then divides by 1 000 gains full marks; the unit kWh must appear.*

(c) [2 marks]

- Energy in one day = **$1.2 + 0.4 + 1.5 + 3.6 = 6.7 \text{ kWh}$** , so in 30 days = **201 kWh** (1 mark).

- Cost = **$201 \times \text{GH}\text{¢}1.50 = \text{GH}\text{¢}301.50$** (1 mark).

- **Marking note:** Allow the candidate's own total from (b) to be carried forward; the second mark is for multiplying correctly by the tariff.*

(d) [2 marks]

- Any one sensible change with a reason drawn from the table, for example **fit LED lamps in place of the filament lamps, because the lamps run for five hours a day** (1 mark).

- A second change with a reason, for example **keep the refrigerator door closed, defrost it and move it away from the cooker, because it runs for 24 hours and uses the most energy (3.6 kWh a day)** (accept: iron in bulk, since the iron has the highest power rating) (1 mark).

- **Marking note:** The reason must refer to the power rating or the hours of use shown in the table.*

Question 4 [Total: 10 marks]

(a) [3 marks]

- (i) **The umbra (full or total shadow)** (1 mark).

- (ii) **The penumbra (partial shadow)** (1 mark).

- (iii) **A fully lit, bright part of the screen**, since light from the whole lamp reaches it (1 mark).

- **Marking note:** 'Dark part' alone is not accepted for (i); the name is required.*

(b) [3 marks]

- (i) **Light travels in straight lines and the ball is opaque**, so it cuts off the light from every part of the lamp (1 mark).

- (ii) **The lamp is broad, so light comes from many points on it** (1 mark).

- (ii) **At Q and R the ball blocks the light from some parts of the lamp but not from others**, so these regions receive some light (1 mark).

- **Marking note:** The idea of a broad source with many points is essential in (ii).*

(c) [2 marks]

- The shadow becomes **larger (both the umbra and the penumbra widen)** (1 mark).

- Because the rays grazing the ball **travel a greater distance to the screen and**

so spread out more** (1 mark).

- *Marking note: Accept 'the edges become less sharp' as part of the explanation.*

(d) [2 marks]

- **A sharp, completely dark shadow (umbra only), with no penumbra around it** (1 mark).

- Because a very small lamp is **a point source**: every point of the screen either sees the whole source or none of it (1 mark).

- *Marking note: A candidate who says the shadow disappears earns no mark.*

Question 5 [Total: 10 marks]

(a) [4 marks]

- (i) **The cutting off of the sun's light from one heavenly body by another**, so that it passes into a shadow (1 mark).

- (ii) **The moon is between the sun and the earth** (the three in a straight line), and the moon's shadow falls on the earth (1 mark).

- (iii) **The earth is between the sun and the moon**, and the earth's shadow falls on the moon (1 mark).

- (iv) Any one correct difference, for example **a solar eclipse is seen from only a small part of the earth and lasts a few minutes, while a lunar eclipse is seen from the whole night side and lasts much longer** (1 mark).

- *Marking

(b) [3 marks]

- (i) The **umbra of the moon's shadow covers only a small patch of the earth's surface** (1 mark).

- (i) **Only observers standing in that umbra see a total eclipse** (those in the penumbra see a partial one), and the patch moves quickly across the earth (1 mark).

- (ii) **Do not look at the sun directly; use a proper filter or view the image projected through a pinhole on to a card** (1 mark).

- *Marking note: Accept 'the moon is much smaller than the earth' as part of the first mark in (i).*

(c) [3 marks]

- The three bodies drawn **in a straight line in the order sun, moon, earth** (1 mark).

- **Straight rays drawn from the sun grazing the edges of the moon** and continuing to the earth (1 mark).

- The **shadow on the earth shown and labelled (umbra, and penumbra around it)**, with all three bodies named (1 mark).

- *Marking note: Marks are for the physics, not for the quality of the drawing; the sizes and distances need not be to scale.*

Question

(a) [4 marks]

- **Apparatus and setting up** - a coin placed at the bottom of an opaque cup and the eye moved until the coin is just out of sight (or a straight stick stood slanting in a beaker) (1 mark).
- **What is done** - water is poured gently into the cup without moving the head or the coin (1 mark).
- **Observation** - the coin comes into view (or the stick appears bent at the water surface) (1 mark).
- **Conclusion** - **light changes its path (is refracted) when it passes from water into air**, so it does not travel in one straight line through the two media (1 mark).
- **Marking note:** Any correct

(b) [2 marks]

- (i) **Refraction** (1 mark).
- (ii) **The speed of light changes in the new medium**, so a ray meeting the surface at an angle changes direction (1 mark).
- **Marking note:** Do not accept 'reflection' in (i).

(c) [2 marks]

- Light from the fish **is refracted as it leaves the water, so the fish appears nearer the surface (shallower) than it really is** (1 mark).
- He should **aim below the image he sees** (accept: stand almost directly above the fish) (1 mark).
- **Marking note:** The idea that the apparent position is higher than the real position is essential for the first mark.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 1: Energy | **Class:** Basic 9 (JHS 3)

Content Standard: **B9/JHS3.4.1.1** - Show understanding of the concept of conservation of energy

Force and Motion

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	14.1 Pressure: Meaning, Formula and Unit	Remember
2	B	14.1 Pressure: Meaning, Formula and Unit	Remember
3	A	14.1 Pressure: Meaning, Formula and Unit	Remember
4	A	14.1 Pressure: Meaning, Formula and Unit	Remember
5	D	14.1 Pressure: Meaning, Formula and Unit	Understand
6	D	14.1 Pressure: Meaning, Formula and Unit	Understand
7	B	14.1 Pressure: Meaning, Formula and Unit	Remember
8	A	14.1 Pressure: Meaning, Formula and Unit	Understand
9	D	14.2 Pressure in Everyday Life	Understand
10	B	14.2 Pressure in Everyday Life	Understand
11	D	14.2 Pressure in Everyday Life	Understand
12	D	14.2 Pressure in Everyday Life	Understand
13	A	14.2 Pressure in Everyday Life	Understand
14	B	14.2 Pressure in Everyday Life	Analyse
15	C	14.2 Pressure in Everyday Life	Understand
16	B	14.2 Pressure in Everyday Life	Remember
17	C	14.3 Calculations on Pressure	Apply
18	B	14.3 Calculations on Pressure	Apply
19	D	14.3 Calculations on Pressure	Apply
20	C	14.3 Calculations on Pressure	Apply
21	B	14.3 Calculations on Pressure	Apply
22	A	14.3 Calculations on Pressure	Analyse
23	A	14.4 Newton's Third Law: Action and Reaction	Remember
24	C	14.4 Newton's Third Law: Action and Reaction	Understand
25	C	14.4 Newton's Third Law: Action and Reaction	Apply
26	B	14.4 Newton's Third Law: Action and Reaction	Understand
27	D	14.4 Newton's Third Law: Action and Reaction	Understand
28	D	14.4 Newton's Third Law: Action and Reaction	Understand
29	C	14.4 Newton's Third Law: Action and Reaction	Analyse
30	C	14.4 Newton's Third Law: Action and Reaction	Evaluate
31	D	14.5 Newton's First and Second Laws and In...	Remember
32	A	14.5 Newton's First and Second Laws and In...	Remember
33	A	14.5 Newton's First and Second Laws and In...	Understand
34	D	14.5 Newton's First and Second Laws and In...	Remember
35	C	14.5 Newton's First and Second Laws and In...	Understand
36	A	14.6 Using the Relation $f = ma$, and Momentum	Apply
37	A	14.6 Using the Relation $f = ma$, and Momentum	Apply

Q	Ans	Curriculum Area	Cognitive Level
38	C	14.6 Using the Relation $f = ma$, and Momentum	Remember
39	C	14.6 Using the Relation $f = ma$, and Momentum	Analyse
40	B	14.6 Using the Relation $f = ma$, and Momentum	Analyse

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 8 (20%) | Evaluate: 1 (2%) | Remember: 11 (27%) | Understand: 15 (37%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a)** [3 marks]

- (i) **The force acting at right angles on each unit area of a surface** (1 mark).
- (ii) **Pressure = force ÷ area** (accept $P = F/A$) (1 mark).
- (iii) **The pascal (Pa)** (accept N/m^2) (1 mark).
- **Marking note:** In (i) 'force per unit area' is sufficient.*

(b) [3 marks]

- (i) The sharp edge has a **very small area**, so the same force gives a **large pressure** and cuts easily (1 mark).
- (ii) Wide tyres **spread the weight over a large area**, so the pressure on the soil is small and the tractor does not sink (1 mark).
- (iii) The wide foundation **spreads the load over a greater area of soil**, reducing the pressure on it (1 mark).
- **Marking note:** Each mark requires the area of contact to be mentioned.*

(c) [2 marks]

- Sucking **lowers the air pressure inside the straw** (1 mark).
- The **greater atmospheric pressure on the surface of the drink pushes the drink up the straw** (1 mark).
- **Marking note:** An answer that says the straw 'pulls' or 'sucks' the drink up, with no mention of pressure, earns no mark.*

(d) [2 marks]

- **Activity clearly described**, for example pressing a drawing pin between the thumb and the finger, or pressing a nail and a coin into clay with the same force (1 mark).
- **Observation given**, for example the point hurts (or sinks in) far more than the head, showing that the smaller area gives the greater pressure (1 mark).
- **Marking note:** The observation mark is lost if the candidate only names an activity.*

Question 2 [Total: 12 marks]**(a)** [4 marks]

- (i) **Area = $0.6 \times 0.4 = 0.24 \text{ m}^2$** (1 mark).
- (i) **Pressure = $480 \div 0.24 = 2\,000 \text{ Pa}$** (1 mark).
- (ii) **Area = $0.4 \times 0.2 = 0.08 \text{ m}^2$** (1 mark).
- (ii) **Pressure = $480 \div 0.08 = 6\,000 \text{ Pa}$** (1 mark).
- **Marking note:** The units m^2 and Pa are required; accept N/m^2 .*

(b) [3 marks]

- (i) **Position Q** (1 mark).
- (ii) The **force (weight) is the same in both positions** (1 mark).
- (ii) But in Q the **area of contact is smaller (0.08 m^2 against 0.24 m^2)**, and pressure = force $\div$ area, so the pressure is three times as great (1 mark).
 - **Marking note:** A candidate who says the block 'is heavier' in Q loses both explanation marks.*

(c) [3 marks]

- (i) **The block will sink into the ground** (1 mark).
- (ii) **Lay it on its largest face (position P)** - the larger area lowers the pressure (1 mark).
- (ii) **Stand it on a wide flat board or slab** - this spreads the weight over a greater area and lowers the pressure (1 mark).
 - **Marking note:** Each suggestion must be accompanied by the reason in terms of area.*

(d) [2 marks]

- **$4\,000 \text{ Pa}$** (1 mark).
- Because **the area is unchanged and the force is doubled, so the pressure is doubled** (1 mark).
 - **Marking note:** A correct long calculation ($960 \div 0.24$) also earns both marks.*

Question 3 [Total: 10 marks]**(a)** [3 marks]

- (i) **Pressure = $600 \div 0.3$** (1 mark).
- (i) **$= 2\,000 \text{ Pa}$ (unit required)** (1 mark).
- (ii) **Force = $5\,000 \times 0.2 = 1\,000 \text{ N}$** (1 mark).
 - **Marking note:** Award the first mark for the correct substitution even if the division fails.*

(b) [3 marks]

- (i) **Pressure = $500 \div 0.05$** (1 mark).
- (i) **$= 10\,000 \text{ Pa}$** (1 mark).
- (ii) **It doubles (to $20\,000 \text{ Pa}$)**, because the area is halved while the weight is unchanged (1 mark).
 - **Marking note:** In (ii) 'it increases' alone earns the mark only if the reason or the value is given.*

(c) [2 marks]

- The water leaves through a **much smaller area** and so moves faster (1 mark).
- The force of the jet acting on that **small area gives a much greater pressure**,

which drives the mud off (1 mark).

- *Marking note: The idea of the smaller area is essential.*

(d) [2 marks]

- **Yes - letting out a little air makes the tyre flatten and touch the sand over a larger area**, so the pressure on the sand is lower (1 mark).

- **But only a little, and the tyres must be pumped up again before he returns to the road**, since very soft tyres wear quickly and are unsafe (accept: he could reduce the load) (1 mark).

- *Marking note: A candidate who answers 'no' may earn the second mark for a sound safety argument, but not the first.*

Question 4 [Total: 10 marks]

(a) [3 marks]

- (i) **For every action there is an equal and opposite reaction** (1 mark).

- (ii) P - **the action: the balloon pushes the air backwards** (1 mark).

- (ii) Q - **the reaction: the air pushes the balloon forwards with an equal and opposite force** (1 mark).

- *Marking note: Accept the naming the other way round if the candidate makes clear which force acts on which body.*

(b) [3 marks]

- (i) The **balloon pushes the air backwards and the air pushes the balloon forwards with an equal and opposite force** (1 mark).

- (i) The forward force **acts on the balloon**, and friction on the string is small, so the balloon moves forward (1 mark).

- (ii) **It would move faster** (and empty sooner), since the forward force is greater (1 mark).

- *Marking note: In (i) the second mark depends on the candidate noting that the two forces act on different bodies.*

(c) [2 marks]

- One correct example with both forces named, for example **a swimmer pushes the water back; the water pushes the swimmer forward** (1 mark).

- A second correct example with both forces named, for example **a gun pushes the bullet forward; the bullet pushes the gun back (recoil)** (1 mark).

- *Marking note: An example named without its pair of forces earns no mark.*

(d) [2 marks]

- Forces **cancel only when they act on the same body** (1 mark).

- Here **the action acts on the air and the reaction on the balloon**, so the balloon has an unbalanced forward force and moves (1 mark).

- *Marking note: The idea of two different bodies is essential for the first mark.*

Question

(a) [3 marks]

- (i) **A body stays at rest or moves with uniform velocity in a straight line unless an external force acts on it** (1 mark).
- (ii) **The rate of change of momentum is proportional to the applied force and takes place in the direction of the force** (accept: the acceleration produced is proportional to the force and inversely proportional to the mass) (1 mark).
- (iii) **$f = ma$** (1 mark).
 - **Marking note:** Accept the second law stated in the acceleration form at this level.*

(b) [3 marks]

- (i) **$a = f/m = 120/40$** (1 mark).
- (i) **$a = 3 \text{ m/s}^2$** (unit required) (1 mark).
- (ii) **$f = 40 \times 5 = 200 \text{ N}$** (1 mark).
 - **Marking note:** Award the method mark wherever $f = ma$ is correctly rearranged.*

(c) [2 marks]

- (i) The passengers **were moving forward and, by their inertia, continued to move forward when the vehicle stopped** (1 mark).
- (ii) **Wearing a seat belt** (accept head rests, air bags, securing loads, driving at a moderate speed) (1 mark).
 - **Marking note:** The word inertia is not required if the idea is clear.*

(d) [2 marks]

- (i) **Momentum = mass $\times$ velocity** (1 mark).
- (ii) The loaded vehicle has **greater mass and so greater momentum, needing a larger force (or a longer time and distance) to stop** (1 mark).
 - **Marking note:** 'It is heavier' alone, without momentum or force, earns no mark in (ii).*

Question 6 [Total: 8 marks]**(a)** [4 marks]

- **Apparatus** named, for example two spring balances hooked together (or two trolleys and two pupils) (1 mark).
- **Method** - the two pupils pull in opposite directions and both balances are read at the same moment (1 mark).
- **Observation** - the two readings are always equal, however hard they pull (the trolleys move apart in opposite directions) (1 mark).
- **Conclusion** - **the two forces are equal in size and opposite in direction**, which justifies Newton's third law (1 mark).

- *Marking note: Any sound activity of the candidate's own design is accepted if all four points are given

(b) [2 marks]

- **Check the zero of each balance and read both at the same moment at eye level** (1 mark).

- **Keep the balances in line with the pull, do not exceed their range, and repeat with different pulls** (1 mark).

- *Marking note: Two distinct precautions are required.*

(c) [2 marks]

- **The prediction was wrong - both balances read the same** (1 mark).

- Because **the force of the first pupil on the second is equal and opposite to the force of the second on the first**; being stronger only decides how large that pair of forces becomes (1 mark).

- *Marking note: A candidate who says the prediction was right earns no mark, however the answer is argued.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 4: Force and Motion |
 Class: Basic 9 (JHS 3)

Content Standard: **B9/JHS3.4.4.1** - Demonstrate understanding of the conc

Human Health

ANSWER KEY**SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	17.1 Health and Disease and Their Relation...	Remember
2	C	17.1 Health and Disease and Their Relation...	Remember
3	C	17.1 Health and Disease and Their Relation...	Understand
4	A	17.1 Health and Disease and Their Relation...	Apply
5	B	17.1 Health and Disease and Their Relation...	Remember
6	B	17.1 Health and Disease and Their Relation...	Understand
7	A	17.2 Non-Communicable Diseases and Their R...	Remember
8	D	17.2 Non-Communicable Diseases and Their R...	Understand
9	D	17.2 Non-Communicable Diseases and Their R...	Understand
10	A	17.2 Non-Communicable Diseases and Their R...	Understand
11	C	17.2 Non-Communicable Diseases and Their R...	Remember
12	B	17.2 Non-Communicable Diseases and Their R...	Apply
13	A	17.2 Non-Communicable Diseases and Their R...	Understand
14	D	17.2 Non-Communicable Diseases and Their R...	Understand
15	B	17.3 Cancer: Causes, Signs and Prevention	Remember
16	D	17.3 Cancer: Causes, Signs and Prevention	Understand
17	C	17.3 Cancer: Causes, Signs and Prevention	Remember
18	C	17.3 Cancer: Causes, Signs and Prevention	Understand
19	B	17.3 Cancer: Causes, Signs and Prevention	Understand
20	B	17.3 Cancer: Causes, Signs and Prevention	Evaluate
21	C	17.4 Common Diseases in a Community	Remember
22	D	17.4 Common Diseases in a Community	Understand
23	C	17.4 Common Diseases in a Community	Remember
24	B	17.4 Common Diseases in a Community	Understand
25	C	17.4 Common Diseases in a Community	Analyse
26	B	17.4 Common Diseases in a Community	Understand
27	B	17.4 Common Diseases in a Community	Apply
28	D	17.4 Common Diseases in a Community	Evaluate
29	A	17.5 Fungal Diseases of Humans	Remember
30	A	17.5 Fungal Diseases of Humans	Understand
31	C	17.5 Fungal Diseases of Humans	Understand
32	A	17.5 Fungal Diseases of Humans	Remember
33	C	17.5 Fungal Diseases of Humans	Understand
34	D	17.5 Fungal Diseases of Humans	Apply
35	A	17.5 Fungal Diseases of Humans	Evaluate
36	D	17.6 Managing Disease: the Individual, the...	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	B	17.6 Managing Disease: the Individual, the...	Remember
38	D	17.6 Managing Disease: the Individual, the...	Understand
39	D	17.6 Managing Disease: the Individual, the...	Create
40	A	17.6 Managing Disease: the Individual, the...	Analyse

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 2 (5%) | Apply: 4 (10%) | Create: 1 (2%) | Evaluate: 3 (7%) | Remember: 12 (30%) | Understand: 18 (45%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [3 marks]

- (i) **A state of complete physical, mental and social well-being** (1 mark), **and not merely the absence of disease or infirmity** (1 mark).
- (ii) **Any condition that upsets the normal working of the body or mind** (1 mark).
 - **Marking note:** In (i) all three kinds of well-being are needed for the first mark.*

(b) [2 marks]

- **Disease lowers the level of health, and the removal or prevention of disease restores health** (1 mark).
- **Freedom from disease alone is not full health, since mental and social well-being are also needed** (1 mark).
 - **Marking note:** Accept any clear statement of the inverse relationship.*

(c) [3 marks]

- **Cholera, measles and tuberculosis** (2 marks for all three; 1 mark for any two).
- Mode of spread of any one named: **cholera - contaminated water or food**; **measles or tuberculosis - droplets in the air from a patient** (1 mark).
 - **Marking note:** A non-communicable disease named among the three loses the first mark.*

(d) [2 marks]

- **He is not completely healthy** (1 mark).
- Reason: **his mental and social well-being is poor, and health is more than the absence of disease** (1 mark).
 - **Marking note:** No mark for the judgement without the reason.*

Question 2 [Total: 12 marks]

(a) [3 marks]

- (i) **A disease that occurs in a large number of people in the community** (1 mark).
- (ii) **Malaria** (1 mark), **120 cases** (1 mark).
 - **Marking note:** Read the value from the bar; 120 only.*

(b) [4 marks]

- (i) $(120 \div 300) \times 100$ (1 mark) $= 40\%$ (1 mark).
- (ii) $40 + 20 = 60$ cases (1 mark); $(60 \div 300) \times 100 = 20\%$ (1 mark).
 - *Marking note: In (ii) the candidate must select high blood pressure and diabetes; selecting any communicable disease loses the first mark.*

(c) [3 marks]

- (i) Malaria (1 mark); the female Anopheles mosquito (accept mosquito) (1 mark).
- (ii) Drinking or eating water or food contaminated with faeces (accept dirty hands, flies on food, poor toilets) (1 mark).
 - *Marking note: Accept 'mosquito' alone for the insect.*

(d) [2 marks]

- Any TWO actions each with a reason: drain stagnant water and clear bush and gutters - removes the breeding places of the mosquito (1 mark); sleep under treated nets and screen windows - prevents biting (1 mark); report and treat fever early - removes the source of the parasite (1 mark). Maximum 2 marks.
 - *Marking note: No mark for the action without its reason.*

Question 3 [Total: 10 marks]**(a)** [2 marks]

- (i) Diseases that are not passed from one person to another (1 mark).
- (ii) Kwashiorkor (accept marasmus, anaemia, rickets, goitre, night blindness) (1 mark).
 - *Marking note: Any named deficiency disease is acceptable in (ii).*

(b) [3 marks]

- Any THREE of: too much salt in the food; fatty or fried food; sugary drinks; lack of physical exercise (sitting all day); being overweight (1 mark each).
 - *Marking note: The risk factor must come from the passage.*

(c) [3 marks]

- (i) Any TWO of: frequent passing of urine; great thirst; loss of weight in spite of eating; tiredness; blurred vision; sores that heal slowly (1 mark each).
- (ii) Stroke, heart failure, kidney damage or damage to the eyes (1 mark).
 - *Marking note: A cause written in place of a symptom earns no mark.*

(d) [2 marks]

- Any TWO of: reduce salt, fat and sugar and eat more fruits and vegetables; take regular exercise and reduce weight; avoid alcohol and tobacco; attend the clinic regularly and take prescribed medicines (1 mark each).

- *Marking note: The advice must be practical and related to her condition.*

Question 4 [Total: 10 marks]

(a) [2 marks]

- (i) **A disease in which body cells divide in an uncontrolled way to form a growth that may spread** (1 mark).
- (ii) Any ONE of **lung, breast, cervical, prostate, liver or skin cancer** (1 mark).
- *Marking note: The idea of uncontrolled growth of cells is essential in (i).*

(b) [3 marks]

- Any THREE of: **a lump or swelling**; **a sore that does not heal**; **unusual bleeding or discharge**; **a change in a mole or wart**; **a lasting cough or hoarseness**; **difficulty in swallowing**; **a change in bowel or bladder habit**; **unexplained loss of weight** (1 mark each).
- *Marking note: Three distinct signs are required.*

(c) [3 marks]

- (i) Tobacco smoke contains **chemicals that damage the cells of the lungs and air passages**; which then grow uncontrollably (1 mark).
- (ii) **Ultraviolet rays damage the skin cells**; which may grow out of control as skin cancer (1 mark).
- (iii) Mouldy grain carries **poisons (aflatoxins) from fungi that damage the liver** (1 mark).
- *Marking note: The idea of damage to cells must appear in (i) and (ii).*

(d) [2 marks]

- **At an early stage the growth is small and has not spread to other parts of the body** (1 mark).
- **It can then be removed or treated successfully, while a cancer that has spread is much harder to treat** (1 mark).
- *Marking note: The idea of spreading must appear.*

Question 5 [Total: 10 marks]

(a) [3 marks]

- (i) **Ringworm** (accept tinea or athlete's foot for the toes) (1 mark).
- (ii) **Fungi** (1 mark).
- (iii) **Candidiasis (thrush)**; **nail infection** or **fungal eye infection** (1 mark).
- *Marking note: Do not accept 'worm' or 'germ' for the group of organisms.*

(b) [4 marks]

- W - ****a shared comb carries infected hairs and skin scales from one scalp to another**** (1 mark).
- X - ****a shared towel rubs the fungus and skin scales on to another pupil's damp skin**** (1 mark).
- Y - ****a shared cap passes the fungus from head to head and keeps the scalp warm and sweaty**** (1 mark).
- Z - ****skin scales with the fungus are left on the wet floor and picked up by bare feet**** (1 mark).
- ***Marking note: Each mark needs the idea of transfer of the fungus from one person to another.***

(c) [3 marks]

- Any THREE of: ****no sharing of combs, towels, caps, clothing or footwear****; ****early treatment of infected pupils at the clinic****; ****daily cleaning of bathrooms and wearing slippers in them****; ****washing, sun-drying and ironing of clothing, towels and bedding****; ****keeping the body clean and dry, especially between the toes****; ****avoiding overcrowding and keeping the dormitory airy**** (1 mark each).
- ***Marking note: Three distinct measures; repeated ideas score once.***

Question 6 [Total: 8 marks]**(a) [3 marks]**

- (i) ****Personal cleanliness, no sharing of personal articles and early reporting for treatment**** (1 mark).
- (ii) ****Keeping the surroundings and public bathhouses clean, supporting education and encouraging treatment**** (1 mark).
- (iii) ****Providing clinics, medicines, trained workers, school screening and public health rules**** (1 mark).
- ***Marking note: The role must match the level named.***

(b) [3 marks]

- Any THREE of: ****a short bold heading naming the disease****; ****a clear drawing of the signs****; ****short simple points on how it spreads****; ****the steps to prevent it****; ****where to go for treatment****; ****large letters and few words, in a language the people read**** (1 mark each).
- ***Marking note: Three distinct features of a good poster.***

(c) [2 marks]

- Danger: ****the real cause is not treated, so the disease spreads, and sand and hot leaves wound the skin and may cause further infection or scarring**** (1 mark).
- Advice: ****ringworm is caused by a fungus spread by contact and shared articles, and it is cured with antifungal medicine from the clinic**** (1 mark).
- ***Marking note: The answer must name the fungus as the cause to earn the**

second mark.*

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:**** 5: Humans and the Environment | ****Sub-strand:**** 2: Human Health |
****Class:**** Basic 9 (JHS 3)

****Content Standard:**** ****B9/JHS3.5.2.1**** - Demonst

Life Cycle Grasshopper

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	4.1 The Grasshopper: Body Parts and Habitat	Remember
2	D	4.1 The Grasshopper: Body Parts and Habitat	Remember
3	A	4.1 The Grasshopper: Body Parts and Habitat	Remember
4	B	4.1 The Grasshopper: Body Parts and Habitat	Remember
5	C	4.1 The Grasshopper: Body Parts and Habitat	Understand
6	D	4.1 The Grasshopper: Body Parts and Habitat	Understand
7	A	4.1 The Grasshopper: Body Parts and Habitat	Understand
8	B	4.2 The Life Cycle of the Grasshopper: Egg...	Remember
9	A	4.2 The Life Cycle of the Grasshopper: Egg...	Remember
10	B	4.2 The Life Cycle of the Grasshopper: Egg...	Remember
11	B	4.2 The Life Cycle of the Grasshopper: Egg...	Understand
12	C	4.2 The Life Cycle of the Grasshopper: Egg...	Remember
13	C	4.2 The Life Cycle of the Grasshopper: Egg...	Understand
14	B	4.2 The Life Cycle of the Grasshopper: Egg...	Remember
15	C	4.2 The Life Cycle of the Grasshopper: Egg...	Understand
16	D	4.2 The Life Cycle of the Grasshopper: Egg...	Understand
17	C	4.2 The Life Cycle of the Grasshopper: Egg...	Remember
18	D	4.3 Incomplete and Complete Metamorphosis ...	Remember
19	A	4.3 Incomplete and Complete Metamorphosis ...	Understand
20	A	4.3 Incomplete and Complete Metamorphosis ...	Remember
21	B	4.3 Incomplete and Complete Metamorphosis ...	Understand
22	A	4.3 Incomplete and Complete Metamorphosis ...	Remember
23	A	4.3 Incomplete and Complete Metamorphosis ...	Understand
24	C	4.3 Incomplete and Complete Metamorphosis ...	Understand
25	A	4.3 Incomplete and Complete Metamorphosis ...	Apply
26	B	4.3 Incomplete and Complete Metamorphosis ...	Understand
27	C	4.4 Activities of the Grasshopper and Thei...	Remember
28	D	4.4 Activities of the Grasshopper and Thei...	Remember
29	D	4.4 Activities of the Grasshopper and Thei...	Understand
30	B	4.4 Activities of the Grasshopper and Thei...	Remember
31	D	4.4 Activities of the Grasshopper and Thei...	Understand
32	C	4.4 Activities of the Grasshopper and Thei...	Understand
33	C	4.4 Activities of the Grasshopper and Thei...	Apply
34	D	4.4 Activities of the Grasshopper and Thei...	Analyse
35	A	4.5 Controlling Grasshoppers on the Farm	Understand
36	D	4.5 Controlling Grasshoppers on the Farm	Understand
37	A	4.5 Controlling Grasshoppers on the Farm	Remember

Q	Ans	Curriculum Area	Cognitive Level
38	D	4.5 Controlling Grasshoppers on the Farm	Understand
39	B	4.5 Controlling Grasshoppers on the Farm	Apply
40	B	4.5 Controlling Grasshoppers on the Farm	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 1 (2%) | Apply: 3 (7%) | Evaluate: 1 (2%) | Remember: 17 (42%) | Understand: 18 (45%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [4 marks]

- P - **head** (1 mark).
- Q - **thorax** (1 mark).
- R - **abdomen** (1 mark).
- S - **wing** (1 mark).
- *Marking note: Accept 'fore wing' for S.*

(b) [2 marks]

- T (the **hind leg**) is used for **jumping** (and for pushing the insect into the air before flight) (1 mark).
- The **antennae** are used for **feeling and smelling** - for finding food and sensing danger (1 mark).
- *Marking note: Accept 'for walking as well as jumping' for the hind leg.*

(c) [2 marks]

- The body is divided into **three parts - head, thorax and abdomen** (1 mark).
- It has **three pairs of legs (six legs) joined to the thorax** (1 mark).
- It has **one pair of antennae on the head** and **wings on the thorax** (1 mark, maximum 2).
- *Marking note: Maximum 2 marks.*

(d) [2 marks]

- **Biting and chewing** mouth parts (1 mark).
- With them the insect **bites off and chews the leaves and young shoots**, leaving ragged holes, so that the plants cannot make enough food and the yield falls (1 mark).
- *Marking note: Do not accept 'piercing and sucking'.*

Question 2 [Total: 12 marks]

(a) [4 marks]

- J - **eggs** (an egg pod in the soil) (1 mark).
- K - **young nymph** (a nymph without wings) (1 mark).
- L - **older nymph**, with wing buds (1 mark).
- M - **adult** (the fully winged insect) (1 mark).
- *Marking note: Accept 'nymph' for both K and L only where the candidate shows the difference, for example 'young nymph' and 'late nymph with wing buds'. Do not accept 'larva' or 'pupa'.*

(b) [2 marks]

- (i) **In a pod in moist soil**, below the surface (1 mark).
- (ii) The soil and the pod **protect the eggs from drying up and from enemies** (and keep them moist until they hatch) (1 mark).
 - *Marking note: Accept 'so that the young have food and cover when they hatch'.*

(c) [3 marks]

- (i) **Moulting** (casting off the outer covering) (1 mark).
- (ii) The **hard outer covering (cuticle) cannot stretch as the insect grows**, so it must be cast off and a larger one formed (1 mark).
- (iii) **L is larger than K and has wing buds**, while K has no wings at all (1 mark).
 - *Marking note: Accept 'ecdysis' in (i).*

(d) [3 marks]

- (i) **Incomplete metamorphosis** (1 mark).
- (ii) Because there is **no pupal stage**: the young stage (nymph) **looks like the adult** and only grows larger and develops wings at each moult (1 mark).
- (iii) The **housefly**, the **mosquito** or the **butterfly** (1 mark).
 - *Marking note: In (ii) either the absence of the pupa or the resemblance of the nymph to the adult earns the mark; both together are not required.*

Question 3 [Total: 10 marks]**(a)** [4 marks]

- Number of stages - **three** / **four** (1 mark for both).
- Name of the young stage - **nymph** / **larva** (1 mark for both).
- Resting (pupal) stage - **no** / **yes** (1 mark for both).
- Example - **grasshopper** (or cockroach) / **housefly** (or mosquito or butterfly) (1 mark for both).
 - *Marking note: Each row earns its mark only when both columns are correct.*

(b) [3 marks]

- The **nymph looks like the adult**, while the **larva is quite unlike the adult** in shape (1 mark).
- The nymph usually **feeds on the same food and lives in the same place as the adult**; the larva often feeds on different food and lives elsewhere (1 mark).
- The nymph **changes gradually into the adult by moulting**, while the larva must first change into a **pupa** (1 mark).
- The nymph **develops wing buds that grow into wings**, while the larva has no wing buds at all (1 mark, maximum 3).
 - *Marking note: Maximum 3 marks; each point must compare the two.*

(c) [3 marks]

- (i) Jar 1 - **complete metamorphosis** (1 mark).
- (i) Jar 2 - **incomplete metamorphosis** (1 mark).
- (ii) In Jar 1 the young was **worm-like and quite unlike the adult**, and it passed through a **still, resting stage inside a case (a pupa)** before the adult came out (1 mark).
 - **Marking note:** In (ii) either the pupal stage or the difference between the young and the adult earns the mark.*

Question 4 [Total: 10 marks]**(a)** [3 marks]

- It **feeds on grasses, weeds and the leaves of crops** (1 mark).
- It **jumps and flies from plant to plant** in the field (1 mark).
- The female **lays her eggs in a pod in the soil** (1 mark).
- The nymph **moults several times as it grows**, and in large numbers the insects **move together in swarms** (1 mark, maximum 3).
- **Marking note:** Maximum 3 marks.*

(b) [3 marks]

- They **eat the leaves, shoots and even the grain of crops**, so the yield falls (1 mark).
- A **swarm can clear a whole field in a short time**, causing food shortage and loss of income for the family (1 mark).
- They **destroy the grass on which cattle, sheep and goats feed**, and farmers must **spend money on insecticides and labour** to control them (1 mark, maximum 3).
- **Marking note:** Maximum 3 marks.*

(c) [2 marks]

- They are **eaten as food rich in protein** in some communities, and are sold for income (1 mark).
- They serve as **food for birds, lizards, frogs and other animals** in the food chain (1 mark).
- Their droppings and dead bodies **return nutrients to the soil**, and they help to keep weeds down (1 mark, maximum 2).
- **Marking note:** Maximum 2 marks.*

(d) [2 marks]

- **No** - a clear position is taken (1 mark).
- **Reason:** grasshoppers are **part of the food chain** - birds, lizards and even people feed on them - so destroying them completely would **upset the balance of nature**; the aim should be to **keep their numbers low** at the time the crops are young (1 mark).
- **Marking note:** A candidate who answers 'yes' can still earn the second mark

only with a well-argued reason; the expected answer is 'no'.*

Question 5 [Total: 10 marks]

(a) [3 marks]

- **Cultural control** - for example **ploughing the land to expose and destroy the egg pods**, clean weeding or early planting (1 mark).
- **Biological control** - for example **protecting the birds, lizards and frogs** that feed on grasshoppers (1 mark).
- **Chemical control** - for example **spraying carefully with an approved insecticide**; or **manual control** - hand-picking the insects (1 mark).
- **Marking note:** The method and its example must agree; a correct example with the wrong method name earns no mark.*

(b) [3 marks]

- Ploughing **turns up the egg pods buried in the soil** (1 mark).
- The exposed eggs are **dried by the sun or eaten by birds and ants**, so **fewer nymphs hatch** (1 mark).
- Clean weeding **removes the grasses and weeds on which the nymphs and adults feed, hide and lay their eggs**, so fewer insects survive near the crop (1 mark).
- **Marking note:** A candidate who writes only 'it kills them' without linking to the eggs or the feeding plants earns at most 1 mark.*

(c) [2 marks]

- They **kill useful insects as well**, such as bees and the birds' and insects' natural enemies of the pest (1 mark).
- They **pollute the soil, water and crop and may harm the farmer or the person who eats the food**; they are also **costly**, and the pests may become **resistant** to them in time (1 mark).
- **Marking note:** Maximum 2 marks.*

(d) [2 marks]

- One clear, correct piece of advice, for example **'Plough your land early; the sun will destroy the eggs hidden in the soil'** (1 mark).
- A second clear, correct piece of advice, for example **'Keep the farm free of tall weeds'** or **'Protect the birds and lizards; they eat the pests for you'** (1 mark).
- **Marking note:** The advice must be short, addressed to farmers and scientifically correct.*

Question 6

(a) [4 marks]

- Three stages drawn, in pencil and large enough to be seen: **egg** (or egg pod), nymph and adult (1 mark).
- **Egg** and **nymph** correctly labelled (1 mark).
- **Adult** correctly labelled (1 mark).
- **Arrows** showing the correct order, egg → nymph → adult and back to egg (1 mark).
- **Marking note:** A candidate who inserts a pupa loses the mark for the stages. Accept a straight line of stages with an arrow returning to the egg.*

(b) [4 marks]

- (i) **Plot B** (1 mark).
- (i) Rise = $31 - 14 = 17$ grasshoppers (1 mark; the working or the figure 17 is enough).
- (ii) Plot B **kept its weeds**, which provided food, shelter and places to lay eggs, while plot A was weeded clean and offered none of these (1 mark).
- (iii) **Keeping a farm free of weeds** reduces the number of grasshoppers on it (1 mark).
- **Marking note:** In (i) award the second mark for 17 even where the candidate does not show the subtraction.*

CURRICULUM ASSESSMENT BLUEPRINT**Strand:** 2: Cycles | **Sub-strand:** 2: Life Cycle of Organisms | **Class:** B

Manure Preparation

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	10.1 Manure and the Types of Manure	Remember
2	B	10.1 Manure and the Types of Manure	Remember
3	D	10.1 Manure and the Types of Manure	Understand
4	A	10.1 Manure and the Types of Manure	Remember
5	B	10.1 Manure and the Types of Manure	Remember
6	C	10.1 Manure and the Types of Manure	Remember
7	C	10.1 Manure and the Types of Manure	Understand
8	B	10.1 Manure and the Types of Manure	Understand
9	D	10.2 Plant Wastes and Animal Wastes Used i...	Remember
10	C	10.2 Plant Wastes and Animal Wastes Used i...	Remember
11	B	10.2 Plant Wastes and Animal Wastes Used i...	Remember
12	A	10.2 Plant Wastes and Animal Wastes Used i...	Remember
13	C	10.2 Plant Wastes and Animal Wastes Used i...	Remember
14	B	10.2 Plant Wastes and Animal Wastes Used i...	Understand
15	B	10.2 Plant Wastes and Animal Wastes Used i...	Understand
16	C	10.2 Plant Wastes and Animal Wastes Used i...	Understand
17	B	10.2 Plant Wastes and Animal Wastes Used i...	Apply
18	B	10.3 Choosing Manure for Different Soils a...	Understand
19	A	10.3 Choosing Manure for Different Soils a...	Understand
20	A	10.3 Choosing Manure for Different Soils a...	Understand
21	D	10.3 Choosing Manure for Different Soils a...	Apply
22	C	10.3 Choosing Manure for Different Soils a...	Apply
23	C	10.3 Choosing Manure for Different Soils a...	Apply
24	A	10.3 Choosing Manure for Different Soils a...	Analyse
25	A	10.4 Preparing Compost: Method and Conditions	Remember
26	C	10.4 Preparing Compost: Method and Conditions	Understand
27	A	10.4 Preparing Compost: Method and Conditions	Remember
28	D	10.4 Preparing Compost: Method and Conditions	Understand
29	C	10.4 Preparing Compost: Method and Conditions	Understand
30	C	10.4 Preparing Compost: Method and Conditions	Understand
31	B	10.4 Preparing Compost: Method and Conditions	Understand
32	D	10.4 Preparing Compost: Method and Conditions	Remember
33	D	10.4 Preparing Compost: Method and Conditions	Analyse
34	A	10.5 Practical Work, Application and Safety	Apply
35	D	10.5 Practical Work, Application and Safety	Remember
36	D	10.5 Practical Work, Application and Safety	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	A	10.5 Practical Work, Application and Safety	Apply
38	D	10.5 Practical Work, Application and Safety	Understand
39	A	10.5 Practical Work, Application and Safety	Analyse
40	D	10.5 Practical Work, Application and Safety	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 3 (7%) | Apply: 6 (15%) | Evaluate: 1 (2%) | Remember: 14 (35%) | Understand: 16 (40%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [2 marks]

- (i) **Material of plant or animal origin added to the soil to supply nutrients and improve it** (1 mark).
- (ii) Manure is made from **plant and animal waste and adds humus, improving the structure and the water-holding of the soil**, while fertiliser is manufactured and supplies nutrients only (1 mark).
- **Marking note: Accept 'manure acts slowly, fertiliser acts quickly' in (ii).**

(b) [3 marks]

- **Farmyard manure** (1 mark).
- **Compost** (1 mark).
- **Green manure** (accept poultry manure, liquid manure or treated night soil) (1 mark).
- **Marking note: Do not accept the names of inorganic fertilisers.**

(c) [3 marks]

- (i) Two plant wastes, for example **dry leaves and kitchen vegetable peelings** (accept weeds, maize stalks, cocoa pod husks, rice husk) (1 mark).
- (i) Two animal wastes, for example **poultry droppings and cow dung** (accept pig dung, goat droppings, bone meal) (1 mark).
- (ii) **Nitrogen** (1 mark).
- **Marking note: Both examples are needed in each half of (i).**

(d) [2 marks]

- A sandy soil **loses water and nutrients quickly**, since the particles are large and the soil is poor in humus (1 mark).
- The **humus in the manure holds water and nutrients in the topsoil** where the roots can reach them, and feeds the soil organisms (1 mark).
- **Marking note: A general answer such as 'manure is good for soil' earns 1 mark only.**

Question 2 [Total: 12 marks]

(a) [4 marks]

- A - **coarse woody material: twigs, chopped maize stalks or stems** (1 mark).
- B - **green plant waste: leaves, weeds, grass or kitchen peelings** (1 mark).
- C - **animal waste: cow dung or poultry droppings** (1 mark).
- E - **a cover of grass, palm fronds or old sacks** (1 mark).

- *Marking note: Accept any sensible material for each layer, provided it is placed correctly.*

(b) [3 marks]

- (i) It **introduces the micro-organisms (bacteria and fungi) that cause the waste to decay** (1 mark).
- (ii) So that the **different materials are well mixed and rot evenly** (1 mark).
- (iii) It **keeps the heap moist and warm and protects it from heavy rain and the hot Sun** (1 mark).
- *Marking note: Accept 'to keep flies away' as a second use in (iii).*

(c) [3 marks]

- (i) To **test the condition inside the heap** from time to time (1 mark).
- (ii) The heap is **working well: the organisms are active (it is warm) and there is enough moisture** (1 mark).
- (iii) To **add water when the heap becomes dry**, since the organisms of decay need moisture (1 mark).
- *Marking note: In (ii) accept 'decomposition is going on well'.*

(d) [2 marks]

- The **micro-organisms respire as they break down the waste and release heat**, which is trapped inside the covered heap (1 mark).
- The heat **destroys many weed seeds, insect eggs and disease-causing organisms** (accept: it speeds up the rotting) (1 mark).
- *Marking note: 'The Sun heats the heap' earns no mark.*

Question 3 [Total: 10 marks]

(a) [4 marks]

- **Collect and sort the waste**, removing plastics, glass, metal and stones, and **chop large materials into small pieces** (1 mark).
- **Choose a shady site away from classrooms and from any well or stream** and prepare the heap or pit (1 mark).
- **Build the heap in layers** - coarse material, green plant waste, animal dung, a thin layer of topsoil - **watering each layer** (1 mark).
- **Cover the heap and turn it every two or three weeks, watering when dry**, until the compost is ready (1 mark).
- *Marking note: Steps given out of order lose one mark overall.*

(b) [3 marks]

- **Plastics such as water sachets and bags** (1 mark).
- **Broken glass or metal (tins, nails)** (1 mark).
- **Stones, rubber, batteries or chemicals** (accept badly diseased plants) (1 mark).
- *Marking note: Maximum 3 marks.*

(c) [3 marks]

- (i) It is **dark brown or black and crumbly**, and the original materials can no longer be recognised (1 mark).
- (i) It **smells like fresh earth, not foul**, and the heap has cooled and shrunk (1 mark).
- (ii) About **two to three months** (accept 6 to 12 weeks) (1 mark).
- **Marking note:** Accept 'about three months' in (ii).*

Question 4 [Total: 10 marks]**(a) [3 marks]**

- (i) **18 kg** (1 mark).
- (ii) **7 kg** (1 mark).
- (iii) **18 kg - 7 kg = 11 kg** (1 mark).
- **Marking note:** The unit kg is required in all three parts.*

(b) [3 marks]

- (i) **Well-rotted poultry manure (plot Y)** (1 mark).
- (ii) Plot W **received no manure**, so the soil could not supply enough nutrients, especially nitrogen (1 mark).
- (iii) **Adding organic manure raises the yield of maize**, and in this trial poultry manure gave the biggest increase (1 mark).
- **Marking note:** In (iii) a conclusion that goes beyond the data earns no mark.*

(c) [2 marks]

- The **same variety of maize, sown on the same day at the same spacing**, on plots of the same size and soil (1 mark).
- The **same weeding, watering, pest control and time of harvest** (1 mark).
- **Marking note:** Two distinct factors are required.*

(d) [2 marks]

- Advise him to **compost the stalks, weeds and kitchen waste and use the compost on his field** (1 mark).
- **Reason:** plot Z shows that **compost gave almost as good a yield as the bought manure**, the materials cost nothing, and the compost improves the soil for later seasons (1 mark).
- **Marking note:** The reason must refer to the data or to the benefit to the soil.*

Question 5 [Total: 10 marks]**(a) [4 marks]**

- Sandy soil - **work in plenty of well-rotted manure or compost**, because the humus holds water and nutrients (1 mark).

- Heavy clay - **work in manure to open up the soil so that air and water move through it** and the soil is easier to work (1 mark).
- Heavy rains - **work the manure into the soil rather than leave it on the surface, where it would be washed away** (1 mark).
- Fresh poultry droppings - **rot or compost them first, because fresh droppings are strong enough to scorch young plants** (1 mark).
- **Marking note: Each mark needs both the action and the reason.**

(b) [3 marks]

- (i) So that it has time to **mix with the soil and break down, releasing its nutrients when the crop needs them** (1 mark).
- (ii) It is **too strong and may scorch the plants, and it may carry weed seeds and disease-causing organisms** (1 mark).
- (iii) The **humus added each year makes the soil crumbly and open, so it holds water and is easier to dig** (1 mark).
- **Marking note: Accept 'it attracts flies' as part of the answer in (ii).**

(c) [3 marks]

- The practice is **poor**: burning **destroys the organic matter that would have formed humus and sends the nitrogen into the air** (1 mark).
- The fire also **kills useful soil organisms and leaves the soil bare to erosion**; the ash gives some potash but **no humus** (1 mark).
- Advice: he should **compost the residues or plough them into the soil** and apply the compost to the field (1 mark).
- **Marking note: A candidate who defends burning cannot earn the first two marks.**

Question 6 [Total: 8 marks]

(a) [3 marks]

- A **suitable shaded site away from classrooms, the kitchen and any well or stream** (1 mark).
- **Separate containers or a sorting point** so that waste that rots is kept apart from plastics, glass and metal (1 mark).
- **Tools (cutlass, hoe, fork, watering can, gloves) and a duty roster with a record book** for turning and watering (1 mark).
- **Marking note: Maximum 3 marks; any three sensible arrangements are accepted.**

(b) [2 marks]

- **Wear gloves or use a tool, and wash the hands with soap and water afterwards and before eating** (1 mark).
- **Do not handle broken glass, sharp metal or chemicals; keep the heap away from drinking water and keep it covered** (accept: never use untreated night soil on vegetables eaten raw) (1 mark).

- *Marking note: Two distinct rules are required.*

(c) [3 marks]

- **The waste becomes free manure that raises yields and saves the money spent on fertiliser** (1 mark).
- **The compound and gutters are kept clean**, so there are fewer flies, mosquitoes, smells and blocked drains (1 mark).
- **Burning pollutes the air and destroys humus, and dumping pollutes soil and water**, while compost improves the soil season after season (1 mark).
- *Marking note: Three distinct points are needed; repetition of one idea earns 1 mark.*

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:** 3: Systems | **Sub-strand:** 4: Farming Systems | **Class:** Basic 9 (JHS 3)**

*

Nitrogen Cycle

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	3.1 Nitrogen in Nature and the Idea of a C...	Remember
2	B	3.1 Nitrogen in Nature and the Idea of a C...	Understand
3	C	3.1 Nitrogen in Nature and the Idea of a C...	Remember
4	A	3.1 Nitrogen in Nature and the Idea of a C...	Understand
5	D	3.1 Nitrogen in Nature and the Idea of a C...	Remember
6	D	3.1 Nitrogen in Nature and the Idea of a C...	Remember
7	C	3.1 Nitrogen in Nature and the Idea of a C...	Analyse
8	A	3.2 Nitrogen Fixation and Nitrification	Remember
9	C	3.2 Nitrogen Fixation and Nitrification	Remember
10	B	3.2 Nitrogen Fixation and Nitrification	Remember
11	A	3.2 Nitrogen Fixation and Nitrification	Remember
12	B	3.2 Nitrogen Fixation and Nitrification	Understand
13	D	3.2 Nitrogen Fixation and Nitrification	Remember
14	A	3.2 Nitrogen Fixation and Nitrification	Remember
15	A	3.2 Nitrogen Fixation and Nitrification	Understand
16	C	3.2 Nitrogen Fixation and Nitrification	Apply
17	D	3.3 Assimilation, Ammonification and Denit...	Remember
18	A	3.3 Assimilation, Ammonification and Denit...	Understand
19	B	3.3 Assimilation, Ammonification and Denit...	Remember
20	D	3.3 Assimilation, Ammonification and Denit...	Remember
21	B	3.3 Assimilation, Ammonification and Denit...	Remember
22	D	3.3 Assimilation, Ammonification and Denit...	Remember
23	C	3.3 Assimilation, Ammonification and Denit...	Understand
24	B	3.3 Assimilation, Ammonification and Denit...	Remember
25	A	3.3 Assimilation, Ammonification and Denit...	Apply
26	B	3.4 Importance of Nitrogen and of the Nitr...	Understand
27	B	3.4 Importance of Nitrogen and of the Nitr...	Remember
28	C	3.4 Importance of Nitrogen and of the Nitr...	Understand
29	A	3.4 Importance of Nitrogen and of the Nitr...	Apply
30	D	3.4 Importance of Nitrogen and of the Nitr...	Understand
31	C	3.4 Importance of Nitrogen and of the Nitr...	Understand
32	D	3.4 Importance of Nitrogen and of the Nitr...	Evaluate
33	A	3.4 Importance of Nitrogen and of the Nitr...	Analyse
34	D	3.5 Interference with the Nitrogen Cycle a...	Remember
35	D	3.5 Interference with the Nitrogen Cycle a...	Understand
36	C	3.5 Interference with the Nitrogen Cycle a...	Understand
37	B	3.5 Interference with the Nitrogen Cycle a...	Apply

Q	Ans	Curriculum Area	Cognitive Level
38	B	3.5 Interference with the Nitrogen Cycle a...	Analyse
39	A	3.5 Interference with the Nitrogen Cycle a...	Analyse
40	C	3.5 Interference with the Nitrogen Cycle a...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 4 (10%) | Evaluate: 2 (5%) | Remember: 18 (45%) | Understand: 12 (30%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a) [2 marks]**

- (i) About **78%** (accept 78 to 79 per cent, or 'about four-fifths') (1 mark).
- (ii) Nitrogen gas is **unreactive (inert)**, so the plant cannot take it in through its roots or leaves in that form; it must first be changed into nitrates (1 mark).
- **Marking note:** Do not award (ii) for 'there is too little nitrogen in the air'.

(b) [3 marks]

- (i) As **nitrates dissolved in the soil water** (accept ammonium compounds) (1 mark).
- (ii) For **making proteins**, which are needed for **growth and the repair of worn-out parts** (1 mark).
- (ii) For making other nitrogen compounds such as **chlorophyll, enzymes and the material of the nucleus** (1 mark).
- **Marking note:** Do not award a mark for 'to give the plant energy'.

(c) [3 marks]

- Nitrogen passes **from the air into the soil** as compounds (fixation) and **from the soil into plants and then animals** (1 mark).
- It is **returned to the soil** when plants and animals die or pass out waste, and **returned to the air** by denitrifying bacteria (1 mark).
- The **same series of changes is then repeated**, so the nitrogen is used over and over again and the amount in nature stays roughly the same (1 mark).
- **Marking note:** The idea of returning to the starting point and beginning again must be present for the third mark.

(d) [2 marks]

- (i) **Nitrogen** (1 mark).
- (ii) **Apply well-rotted manure or compost**, **apply a nitrogen fertiliser (sulphate of ammonia, urea, NPK)**, or **grow a leguminous crop such as cowpea or groundnut on the plot** (1 mark).
- **Marking note:** Accept any one correct remedy in (ii).

Question 2 [Total: 12 marks]**(a) [4 marks]**

- 1 - **nitrogen fixation** (1 mark).
- 2 - **assimilation** (accept 'absorption of nitrates by plants and their use in making protein') (1 mark).
- 5 - **nitrification** (1 mark).

- 6 - **denitrification** (1 mark).

- *Marking note: Accept 'fixation' for 1. Do not accept 'nitrification' for 1 or 'nitrogen fixation' for 5.*

(b) [3 marks]

- (i) **Ammonification** (accept 'decay' or 'decomposition' with the formation of ammonium compounds) (1 mark).

- (ii) **Decomposers - bacteria and fungi** (1 mark).

- (iii) By the farmer **adding manure, compost or nitrogen fertiliser**, or by **rain washing in the nitrates formed by lightning** (1 mark).

- *Marking note: Accept 'putrefying bacteria' in (ii).*

(c) [3 marks]

- **Nitrogen gas is fixed** into compounds and reaches the soil as **nitrates** (by nitrogen-fixing bacteria, by lightning, or after nitrification) (1 mark).

- The **grass absorbs the nitrates** through its roots and uses them to make **plant protein** (1 mark).

- The **cow feeds on the grass**, digests the plant protein and uses it to build **animal protein** in its own body (1 mark).

- *Marking note: The three stages must be in the right order. A candidate who leaves out the plant earns at most 1 mark.*

(d) [2 marks]

- **Ammonium compounds would not be changed into nitrates**, so the nitrate in the soil would fall (1 mark).

- The plants would be **short of nitrogen**, so their leaves would **turn yellow**, growth would be stunted and yields would fall (1 mark).

- *Marking note: A candidate who writes only 'the plants would die' without a reason earns 1 mark.*

Question 3 [Total: 10 marks]

(a) [2 marks]

- (i) **Root nodules** (1 mark).

- (ii) **Nitrogen-fixing bacteria** (accept Rhizobium) (1 mark).

- *Marking note: Do not award (ii) for 'bacteria' alone without 'nitrogen-fixing'.*

(b) [3 marks]

- The bacteria inside the nodules **fix nitrogen from the air into compounds (nitrates) that the plant can use** (1 mark).

- The plant in turn **supplies the bacteria with food and shelter**, so both gain from the relationship (1 mark).

- When the plant dies or its remains are ploughed in, the **nitrogen compounds are left in the soil**, making it more fertile for the next crop (1 mark).

- *Marking note: Accept 'symbiosis' or 'mutualism' as the name of the relationship, but the mark is for the explanation.*

(c) [3 marks]

- **Mark out two similar plots** on the same soil, side by side (1 mark).
- **Grow a leguminous crop (cowpea or groundnut) on plot 1** and leave plot 2 without a legume; at the end of the season **dig the legume remains into plot 1**, then **plant maize on both plots and treat them in exactly the same way** (1 mark).
- **Expected result:** the maize on the plot that carried the legume grows **taller and greener and gives a better yield**, showing that the legume has enriched the soil with nitrogen (1 mark).

- *Marking note: A plan without a control plot earns at most 1 mark. Accept pots in p

(d) [2 marks]

- **Advice: he should not burn them**; he should **plough or dig the remains into the soil, or use them to make compost** (1 mark).
- **Reason:** burning **destroys the organic matter, drives off the nitrogen and kills the soil organisms**, while digging the remains in **returns nitrogen compounds and humus to the soil** for the next crop (1 mark).

- *Marking note: The advice must be clear; a candidate who merely lists the effects of burning without advising earns 1 mark.*

Question 4 [Total: 10 marks]

(a) [4 marks]

- Nitrogen fixation - **nitrogen gas is changed into nitrogen compounds that plants can use** - **nitrogen-fixing bacteria** (in root nodules or free in the soil) **or lightning** (1 mark).
- Nitrification - **ammonium compounds are changed into nitrates** - **nitrifying bacteria** (1 mark).
- Ammonification - **the protein in dead plants, dead animals and animal waste is broken down into ammonium compounds** - **decomposers: bacteria and fungi** (1 mark).
- Denitrification - **nitrates in the soil are changed back into nitrogen gas, which escapes into the air** - **denitrifying bacteria** (1

(b) [3 marks]

- The **energy of the lightning flash makes nitrogen and oxygen of the air combine**, forming oxides of nitrogen (1 mark).
- These oxides **dissolve in the rain water** (forming weak acids) (1 mark).
- The rain **carries them into the soil, where they form nitrates** that the plants can absorb (1 mark).
- *Marking note: Accept 'nitric acid' for the dissolved product.*

(c) [3 marks]

- (i) **Denitrification** (1 mark).
- (ii) In the **waterlogged soil there is little air**, so **denitrifying bacteria change the nitrates into nitrogen gas**, which escapes; the crop is therefore short of nitrogen (1 mark).
- (iii) **Drain the field** (dig channels or make ridges and furrows) so that air can enter the soil, and apply the fertiliser after draining (1 mark).
 - **Marking note:** In (ii) the loss of nitrate as nitrogen gas must be stated for the mark.*

Question 5 [Total: 10 marks]**(a)** [3 marks]

- **Grow leguminous crops** (cowpea, groundnut, soya bean) or include them in a crop rotation (1 mark).
- **Apply well-rotted farmyard manure or compost** (1 mark).
- **Plough in crop remains or green manure** (1 mark).
- **Apply a nitrogen fertiliser** such as sulphate of ammonia, urea or NPK (1 mark, maximum 3).
- **Marking note:** Maximum 3 marks.*

(b) [3 marks]

- Different crops **take different nutrients, and in different amounts**, so changing the crop prevents one nutrient from being used up (1 mark).
- The **leguminous crop carries nitrogen-fixing bacteria in its root nodules**, which add nitrogen compounds to the soil (1 mark).
- When the **legume remains are ploughed in**, the nitrogen compounds and humus are left in the soil for the crop that follows (1 mark).
 - **Marking note:** Accept 'it also helps to control pests and diseases' as an extra point only after the three above.*

(c) [2 marks]

- **Water plants and algae grow very quickly and cover the surface** of the water (1 mark).
- When they die and decay, the **oxygen dissolved in the water is used up**, so fish and other water animals die and the water becomes unfit for domestic use (1 mark).
 - **Marking note:** Accept 'the river becomes green and smells bad' as the first point.*

(d) [2 marks]

- **Organic manure:** it is **cheap and available on the farm**, and it **adds humus**, which improves the structure, the water-holding and the life of the soil (1 mark).

- **Inorganic fertiliser:** it supplies a **known, concentrated amount of nitrogen** that acts quickly, and it is easy to carry and to apply (1 mark).
- **Marking note:** One advantage of each is required; two advantages of only one kind earn 1 mark.

Question 6 [Total: 8 marks]

(a) [3 marks]

- (i) Burning **destroys the plant cover, the humus and the decomposers and nitrifying bacteria in the topsoil**, and the nitrogen in the burnt material escapes into the air (1 mark).
- (ii) Heavy rain **washes the dissolved nitrates down through the soil, beyond the reach of the roots**, so less nitrogen is available to the crop (1 mark).
- (iii) Clearing the legumes **removes the nitrogen-fixing bacteria of the root nodules**, so less nitrogen from the air is fixed into the soil (1 mark).
- **Marking note:** Mark each of the three parts on its own.

(b) [3 marks]

- **Yields will fall season after season**, because the humus and the nitrogen compounds in the topsoil are destroyed by the fire and are not replaced (1 mark).
- The **bare soil will be eroded by rain and wind and its nitrates leached away**, so the topsoil becomes thin and poor (1 mark).
- The **soil organisms (decomposers and nitrifying bacteria) will die out**, so decay and nitrification slow down and the farmer must spend more on fertiliser, or abandon the land (1 mark).
- **Marking note:** Each prediction earns its mark only when a reason is given with it.

(c) [2 marks]

- A clear, correct message on **not burning the bush**, for example 'Do not burn the bush - fire burns away the food in your soil' (1 mark).
- A clear, correct message on a **good practice**, for example 'Plant cowpea and groundnut - they feed your soil with nitrogen' or 'Cover your land after harvest; bare soil loses its riches to the rain' (1 mark).
- **Marking note:** The message must be short, addressed to the community and scientifically correct. Do not award a mark for a message that merely repeats the question.

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 2: Cycles | **Sub-stra**

Non Planetary Bodies

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	8.1 The Solar System and Its Members	Remember
2	D	8.1 The Solar System and Its Members	Remember
3	D	8.1 The Solar System and Its Members	Remember
4	A	8.1 The Solar System and Its Members	Remember
5	D	8.1 The Solar System and Its Members	Remember
6	D	8.1 The Solar System and Its Members	Remember
7	A	8.1 The Solar System and Its Members	Understand
8	A	8.1 The Solar System and Its Members	Remember
9	C	8.2 Asteroids and the Asteroid Belt	Remember
10	C	8.2 Asteroids and the Asteroid Belt	Remember
11	A	8.2 Asteroids and the Asteroid Belt	Understand
12	C	8.2 Asteroids and the Asteroid Belt	Understand
13	A	8.2 Asteroids and the Asteroid Belt	Remember
14	D	8.2 Asteroids and the Asteroid Belt	Analyse
15	C	8.2 Asteroids and the Asteroid Belt	Understand
16	B	8.2 Asteroids and the Asteroid Belt	Remember
17	D	8.3 Comets and Their Movement	Remember
18	C	8.3 Comets and Their Movement	Remember
19	B	8.3 Comets and Their Movement	Remember
20	A	8.3 Comets and Their Movement	Remember
21	D	8.3 Comets and Their Movement	Understand
22	A	8.3 Comets and Their Movement	Understand
23	A	8.3 Comets and Their Movement	Understand
24	D	8.3 Comets and Their Movement	Remember
25	C	8.3 Comets and Their Movement	Analyse
26	B	8.4 Meteoroids, Meteors and Meteorites	Remember
27	B	8.4 Meteoroids, Meteors and Meteorites	Remember
28	C	8.4 Meteoroids, Meteors and Meteorites	Remember
29	B	8.4 Meteoroids, Meteors and Meteorites	Understand
30	B	8.4 Meteoroids, Meteors and Meteorites	Remember
31	D	8.4 Meteoroids, Meteors and Meteorites	Understand
32	B	8.4 Meteoroids, Meteors and Meteorites	Analyse
33	B	8.4 Meteoroids, Meteors and Meteorites	Understand
34	D	8.5 Comparing the Non-Planetary Bodies and...	Understand
35	A	8.5 Comparing the Non-Planetary Bodies and...	Analyse
36	C	8.5 Comparing the Non-Planetary Bodies and...	Analyse

Q	Ans	Curriculum Area	Cognitive Level
37	C	8.5 Comparing the Non-Planetary Bodies and...	Remember
38	A	8.5 Comparing the Non-Planetary Bodies and...	Apply
39	B	8.5 Comparing the Non-Planetary Bodies and...	Understand
40	B	8.5 Comparing the Non-Planetary Bodies and...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 1 (2%) | Evaluate: 1 (2%) | Remember: 21 (52%) | Understand: 12 (30%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [2 marks]

- (i) The **Sun** together with all the bodies that move round it (planets, satellites, asteroids, comets and meteoroids) (1 mark).
- (ii) The **gravitational pull (gravity)** of the Sun (1 mark).
- **Marking note:** Accept 'gravity' alone in (ii).

(b) [3 marks]

- **Asteroids** (1 mark).
- **Comets** (1 mark).
- **Meteoroids** (accept natural satellites or moons, or dwarf planets) (1 mark).
- **Marking note:** Do not accept 'stars' or the name of a planet.

(c) [3 marks]

- (i) A **planet** moves round the Sun, while a **natural satellite** moves round a planet (1 mark).
- (ii) Planet - **the Earth** (accept any of the eight planets) (1 mark).
- (ii) Natural satellite - **the Moon** (accept a named moon of another planet) (1 mark).
- **Marking note:** A difference in size alone earns no mark.

(d) [2 marks]

- A planet must **move in its own orbit round the Sun**; the Moon **moves round the Earth** (1 mark).
- The Moon goes round the Sun **only because it is carried along by the Earth**, so it is classed as a **natural satellite** (1 mark).
- **Marking note:** The idea that the Moon orbits a planet is essential.

Question 2 [Total: 12 marks]

(a) [4 marks]

- A - the **Sun** (1 mark).
- C - the **asteroid belt** (a belt of asteroids, small rocky bodies) (1 mark).
- D - a **comet** (1 mark).
- E - the **orbit (path)** of the comet (1 mark).
- **Marking note:** Accept 'asteroid' for C where the candidate shows it is the belt.

(b) [3 marks]

- (i) **E is a long, narrow ellipse, while F is nearly circular** (1 mark).
- (i) The **distance of D from the Sun changes greatly along E**, and one journey round E takes far longer than one journey round F (1 mark).
- (ii) At the **part of the path nearest to the Sun** (1 mark).
- **Marking note: Accept 'at perihelion' or 'when closest to the Sun' in (ii).***

(c) [3 marks]

- (i) Between the orbits of **Mars and Jupiter** (1 mark).
- (ii) They are **small, rocky (or metallic) bodies of irregular shape**, much smaller than a planet (1 mark).
- (ii) They **move round the Sun in the same direction as the planets, in nearly circular orbits in about the same plane** (1 mark).
- **Marking note: Either of the last two points earns the second mark.***

(d) [2 marks]

- It would travel in towards the inner solar system; a **small body would burn up in the atmosphere and be seen as a meteor** (1 mark).
- A **large one could strike the ground as a meteorite, forming a crater, destroying life around it and throwing dust into the air** (1 mark).
- **Marking note: Either outcome, correctly explained, earns both marks only where the two cases are distinguished.***

Question 3 [Total: 10 marks]**(a)** [3 marks]

- P - the **nucleus** (1 mark).
- Q - the **coma** (1 mark).
- R - the **tail** (1 mark).
- **Marking note: Accept 'head' for Q only where the candidate writes 'coma (head)'.***

(b) [3 marks]

- (i) **Position 3** (1 mark).
- (ii) There the comet is **heated most strongly, so much more of its ice turns into gas and dust is released** (1 mark).
- (ii) The gas and dust are **pushed far out by the radiation and the stream of particles coming from the Sun**, giving a long, bright tail (1 mark).
- **Marking note: Accept 'solar wind' for the stream of particles.***

(c) [2 marks]

- The tail points **away from the Sun** (1 mark).
- **Reason:** the gas and dust are **pushed outwards by the Sun's radiation and solar wind**, so the tail always lies on the side away from the Sun, whatever the direction of travel (1 mark).

- *Marking note: 'Behind the comet' earns no mark; the tail is not controlled by the direction of travel.*

(d) [2 marks]

- The orbit is a **very long ellipse**, so the comet is close to the Sun, with a bright tail, for only a short part of the journey (1 mark).
- For the rest of the time it is **far from the Sun**, with no tail and too faint to be seen, and one complete orbit takes many years (1 mark).
- *Marking note: Accept a reference to Halley's comet returning after about 76 years as supporting detail.*

Question 4 [Total: 10 marks]

(a) [3 marks]

- (i) A **small piece of rock or metal moving through space** (1 mark).
- (ii) The **streak of light seen when such a body burns up in the Earth's atmosphere** (1 mark).
- (iii) A **piece that survives the burning and lands on the surface of the Earth** (1 mark).
- *Marking note: The three terms must be distinguished; two identical definitions earn one mark only.*

(b) [3 marks]

- (i) **Friction between the fast-moving body and the air** heats it until it glows (1 mark).
- (ii) They are **small and burn up completely** in the atmosphere before reaching the ground (1 mark).
- (iii) **Many meteors seen in a short time**, when the Earth passes through the **trail of dust left by a comet** (1 mark).
- *Marking note: Accept 'air resistance' for friction in (i).*

(c) [4 marks]

- (i) It lies in a **fresh crater or hollow with soil thrown out around it** (1 mark).
- (i) It is **unusually heavy, rich in iron (a magnet may stick to it) and has a dark melted crust**, unlike the rocks of the district (1 mark).
- (ii) A **large crater is formed and buildings, farms and living things nearby are destroyed** by the shock and heat (1 mark).
- (ii) **Dust is thrown high into the air, cutting off sunlight and affecting the weather and climate** (accept an earth tremor) (1 mark).
- *Marking note: In (i) 'people said so' is not an observation and earns no mark.*

Question 5**(a)** [4 marks]

- Asteroid - made of **rock and metal** (1 mark).
- Comet - made of **ice, frozen gases and dust** (1 mark).
- Asteroid - orbit **nearly circular**, mostly within the belt (1 mark).
- Comet - orbit a **long, narrow ellipse** (1 mark).
- **Marking note:** Marks are for the paired comparison; one column alone earns the marks in that column only.*

(b) [3 marks]

- They **all move round the Sun** in closed orbits, returning to the same part of the path (1 mark).
- They are **all held in their paths by the gravitational pull of the Sun** (1 mark).
- Each **moves faster when nearer to the Sun**, and most travel round the Sun in the same direction (1 mark).
- **Marking note:** Maximum 3 marks.*

(c) [3 marks]

- **No, the claim is not correct** (1 mark).
- **Reason 1:** a comet is a **natural body moving on a fixed orbit round the Sun**, and its return can be predicted **years ahead**, so it is not caused by, and does not cause, events on the Earth (1 mark).
- **Reason 2:** **records of many past comets show that no disaster followed them**; the belief is a superstition that observation and record keeping disprove (1 mark).
- **Marking note:** A candidate who agrees with the claim cannot earn the reason marks. This part examines the competence of identifying and disproving misconceptions.*

Question 6 [Total: 8 marks]**(a)** [3 marks]

- **Find out the date, the time and the part of the sky** from a reliable source such as an astronomy website or the Ghana Astronomy Centre (1 mark).
- **Choose a dark, open place away from street lights**, with permission and adult supervision, on a clear night (1 mark).
- **Record the observation:** sketch the comet and write down the date, the time, the direction and anything used, such as binoculars (1 mark).
- **Marking note:** Accept 'use a sky map or phone application' for the first mark.*

(b) [2 marks]

- **Who published it** - whether the author or organisation is a recognised scientific body, and **how recent the page is** (1 mark).
- **Whether the same information is given by other reliable sources**, and

whether the page offers evidence rather than advertisements or rumours (1 mark).

- *Marking note: Two distinct checks are required.*

(c) [3 marks]

- They **carry material left from the formation of the solar system**, so they tell us how the Sun, the Earth and the planets began (1 mark).

- **Watching the bodies that come near the Earth gives warning of a possible collision** in good time (1 mark).

- The study **replaces superstition with evidence** and **interests young people in science, mathematics and careers in technology** (1 mark).

- *Marking note: Accept 'comets may have brought water to the early Earth' for any one mark.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 3: Systems | **Sub-strand:** 2: The Solar System | **

Science and Industry

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	18.1 The Concept of Industry and Types of ...	Remember
2	A	18.1 The Concept of Industry and Types of ...	Understand
3	D	18.1 The Concept of Industry and Types of ...	Remember
4	B	18.1 The Concept of Industry and Types of ...	Understand
5	C	18.1 The Concept of Industry and Types of ...	Remember
6	B	18.1 The Concept of Industry and Types of ...	Understand
7	D	18.2 Industries and Their Products in Ghana	Remember
8	D	18.2 Industries and Their Products in Ghana	Remember
9	B	18.2 Industries and Their Products in Ghana	Remember
10	D	18.2 Industries and Their Products in Ghana	Remember
11	A	18.2 Industries and Their Products in Ghana	Remember
12	D	18.2 Industries and Their Products in Ghana	Understand
13	D	18.2 Industries and Their Products in Ghana	Analyse
14	A	18.3 Scientific Processes in Modern Indust...	Remember
15	C	18.3 Scientific Processes in Modern Indust...	Understand
16	B	18.3 Scientific Processes in Modern Indust...	Remember
17	D	18.3 Scientific Processes in Modern Indust...	Understand
18	C	18.3 Scientific Processes in Modern Indust...	Understand
19	C	18.3 Scientific Processes in Modern Indust...	Remember
20	A	18.3 Scientific Processes in Modern Indust...	Analyse
21	C	18.3 Scientific Processes in Modern Indust...	Understand
22	A	18.4 Indigenous Industries of Ghana	Remember
23	D	18.4 Indigenous Industries of Ghana	Understand
24	A	18.4 Indigenous Industries of Ghana	Remember
25	C	18.4 Indigenous Industries of Ghana	Remember
26	D	18.4 Indigenous Industries of Ghana	Remember
27	C	18.4 Indigenous Industries of Ghana	Understand
28	A	18.4 Indigenous Industries of Ghana	Understand
29	B	18.4 Indigenous Industries of Ghana	Analyse
30	B	18.5 The Science in Indigenous Processes a...	Understand
31	C	18.5 The Science in Indigenous Processes a...	Understand
32	A	18.5 The Science in Indigenous Processes a...	Analyse
33	B	18.5 The Science in Indigenous Processes a...	Understand
34	A	18.5 The Science in Indigenous Processes a...	Understand
35	B	18.5 The Science in Indigenous Processes a...	Analyse
36	B	18.6 Technology in Indigenous and Modern I...	Understand
37	C	18.6 Technology in Indigenous and Modern I...	Understand

Q	Ans	Curriculum Area	Cognitive Level
38	D	18.6 Technology in Indigenous and Modern I...	Analyse
39	A	18.6 Technology in Indigenous and Modern I...	Evaluate
40	B	18.6 Technology in Indigenous and Modern I...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Evaluate: 2 (5%) | Remember: 15 (37%) | Understand: 17 (42%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [3 marks]

- (i) **A group of firms producing the same kind of goods or services** (1 mark).
- (ii) **Primary (extractive), secondary (manufacturing) and tertiary (service) industries** (2 marks for all three; 1 mark for any two).
- **Marking note: Accept 'extractive, manufacturing and service' in (ii).**

(b) [4 marks]

- (i) **Cocoa buying shed / cocoa growing** (accept the sawmill's timber felling) (1 mark).
- (ii) Any TWO of **bakery**, **palm oil mill**, **kente weaving**, **blacksmith's shop**, **sawmill** (1 mark each).
- (iii) **The taxi station** (transport) (1 mark).
- **Marking note: A firm may be placed only once; mark each part on its own.**

(c) [3 marks]

- Order: **harvesting the fruits → boiling and pounding → separating (skimming) the oil from the water and fibre → packaging and marketing** (2 marks for the full correct order; 1 mark if one pair is out of place).
- Scientific separation: **at the skimming stage, because oil is less dense than water and floats on it** (1 mark).
- **Marking note: Accept 'pressing' for pounding and 'selling' for marketing.**

Question 2 [Total: 12 marks]

(a) [3 marks]

- (i) Raw material: **cassava roots** (1 mark); product: **gari** (1 mark).
- (ii) **B** (1 mark).
- **Marking note: Only the letter is needed in (ii).**

(b) [4 marks]

- (i) **The pressing squeezes out the liquid** (1 mark); **the three days allow fermentation, which sours the mash and helps to remove the harmful (cyanide-producing) substance** (1 mark).
- (ii) **The heat drives off the remaining water as steam and cooks the starch** (1 mark); **the dry product does not support micro-organisms, so it keeps long** (1 mark).
- **Marking note: Accept 'poisonous substance' for the cyanide-producing compound.**

(c) [3 marks]

- (i) **Fermentation** (1 mark).
- (ii) Any TWO of: **a sour smell or taste develops**; **bubbles of gas escape**; **liquid drips from the bags**; **the mash becomes warm**; **the colour changes** (1 mark each).
- **Marking note:** The observation must be something that can be seen, smelt, tasted or felt.*

(d) [2 marks]

- Any TWO improvements each with a reason: **a motorised grater** - faster and gives an even mash (1 mark); **a screw or hydraulic press** - the same pressing force on every bag (1 mark); **a mechanical roaster or a fixed roasting time** - every batch roasted alike (1 mark); **better hygiene in the shed** - a safer product of steady quality (1 mark). Maximum 2 marks.
- **Marking note:** No mark for an improvement without a reason.*

Question 3 [Total: 10 marks]**(a)** [4 marks]

- P - **the channel (and gate) that lets sea water into the site** (1 mark).
- Q - **the reservoir pan holding the sea water** (1 mark).
- R - **the shallow crystallising pan in which the salt forms** (1 mark).
- S - **the heap of harvested salt** (1 mark).
- **Marking note:** Accept 'storage pan' for Q and 'salt pan' for R.*

(b) [4 marks]

- (i) **Evaporation** (1 mark) and **crystallisation** (1 mark).
- (ii) **A wide, shallow pan gives a large surface area for evaporation** (1 mark); **the sun supplies heat and the wind carries away the vapour, so evaporation is faster** (1 mark).
- **Marking note:** 'Boiling' is wrong; no boiling takes place in the pans.*

(c) [2 marks]

- **Dissolve the dirty salt in water and filter to remove the sand and rubbish** (1 mark).
- **Evaporate the filtrate to obtain clean salt crystals** (1 mark).
- **Marking note:** Both steps are needed; one step alone earns one mark.*

Question 4 [Total: 10 marks]**(a)** [2 marks]

- (i) **An industry that uses local materials, simple tools and skills handed down in the community** (1 mark).

- (ii) Any ONE of ****kente weaving, pottery, blacksmithing, basketry, shea butter extraction, gari processing, salt winning, fish smoking, adinkra printing**** (1 mark).

- ***Marking note: Do not accept a modern factory in (ii).***

(b) [4 marks]

- (i) ****To dissolve the alkali (potash) out of the ash**** (1 mark); ****the insoluble matter is left behind and strained off**** (1 mark).

- (ii) ****Saponification**** (1 mark); ****soap (and glycerol) is formed**** (1 mark).

- ***Marking note: Accept 'lye' for the alkaline solution.***

(c) [2 marks]

- Test: ****dip red litmus paper (or add a local plant indicator) into the solution**** (1 mark).

- Result: ****the red litmus turns blue**** (accept the colour change of the local indicator) (1 mark).

- ***Marking note: A test without the expected result earns one mark only.***

(d) [2 marks]

- Any TWO precautions each with a reason: ****wear gloves and avoid splashes**** - the alkali burns the skin (1 mark); ****use a long ladle and keep the face away**** - the hot mixture spits (1 mark); ****work in the open air**** - the smoke and fumes irritate the chest and eyes (1 mark); ****keep children away from the fire**** - to prevent burns (1 mark). Maximum 2 marks.

- ***Marking note: No mark for a precaution without its reason.***

Question 5 [Total: 10 marks]

(a) [4 marks]

- (i) ****Removes suspended solid particles (mud, grit)**** (1 mark).

- (ii) ****Absorbs substances that give taste, smell and colour**** (1 mark).

- (iii) ****Kills bacteria and other germs**** (1 mark).

- (iv) ****Keeps out dust, germs and insects (and shows the bottle has not been opened)**** (1 mark).

- ***Marking note: Do not accept 'to make the water pure' alone in (i) or (ii).***

(b) [4 marks]

- (i) ****Fermentation**** (1 mark).

- (ii) ****Carbon dioxide**** (1 mark); ****it forms bubbles that make the dough rise and give the bread a light, spongy texture**** (1 mark).

- (iii) ****The yeast is most active in warmth; in the cold it works too slowly**** (1 mark).

- ***Marking note: Accept 'the yeast respire' for the process in (i) only if fermentation is named.***

(c) [2 marks]

- Any TWO of: **the yeast was dead, too old or forgotten**; **the water was too hot and killed the yeast**; **the place was too cold**; **no sugar was added, so the yeast had no food**; **too much salt was added**; **the dough was not left long enough** (1 mark each).

- *Marking note: Two distinct reasons are required.*

Question 6 [Total: 8 marks]**(a)** [4 marks]

- Tools - weaver: **a simple wooden loom worked by hand and foot** (1 mark); factory: **power-driven machines** (1 mark).

- Quantity - weaver: **small output, a few cloths at a time** (1 mark); factory: **large output, many rolls a day** (1 mark).

- *Marking note: The table must be copied; four correct entries earn the marks.*

(b) [2 marks]

- Any TWO of: **little money or power needed to start and run**; **uses local materials and keeps local skills and culture alive**; **provides work in the village and trains apprentices**; **each cloth is unique and made to order**; **little waste and no industrial pollution** (1 mark each).

- *Marking note: Two distinct advantages are required.*

(c) [2 marks]

- Any TWO of: **an improved or partly mechanised loom to weave faster while keeping his own designs**; **factory-spun, fast-dyed yarn of even quality**; **simple testing of colour fastness and strength**; **a mobile phone and the internet for orders and sales**; **electric lighting in the shed** (1 mark each).

- *Marking note: The suggestion must keep the weaving itself in the hands of the weaver; 'buy power looms and close the shed' earns no mark.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 5: Humans and the Environment | **Sub-strand:** 3: Science and Industry | **Class:** Basic

Specialised Cells

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	2.1 Cells, Tissues and the Meaning of Spec...	Remember
2	D	2.1 Cells, Tissues and the Meaning of Spec...	Remember
3	C	2.1 Cells, Tissues and the Meaning of Spec...	Remember
4	D	2.1 Cells, Tissues and the Meaning of Spec...	Remember
5	D	2.1 Cells, Tissues and the Meaning of Spec...	Understand
6	B	2.1 Cells, Tissues and the Meaning of Spec...	Understand
7	A	2.1 Cells, Tissues and the Meaning of Spec...	Analyse
8	C	2.2 How Specialised Cells Are Formed in Pl...	Remember
9	C	2.2 How Specialised Cells Are Formed in Pl...	Remember
10	D	2.2 How Specialised Cells Are Formed in Pl...	Understand
11	A	2.2 How Specialised Cells Are Formed in Pl...	Understand
12	C	2.2 How Specialised Cells Are Formed in Pl...	Understand
13	A	2.2 How Specialised Cells Are Formed in Pl...	Analyse
14	C	2.3 Specialised Cells of Dicotyledonous Pl...	Remember
15	A	2.3 Specialised Cells of Dicotyledonous Pl...	Understand
16	C	2.3 Specialised Cells of Dicotyledonous Pl...	Remember
17	B	2.3 Specialised Cells of Dicotyledonous Pl...	Remember
18	B	2.3 Specialised Cells of Dicotyledonous Pl...	Understand
19	C	2.3 Specialised Cells of Dicotyledonous Pl...	Remember
20	A	2.3 Specialised Cells of Dicotyledonous Pl...	Remember
21	D	2.3 Specialised Cells of Dicotyledonous Pl...	Understand
22	A	2.3 Specialised Cells of Dicotyledonous Pl...	Understand
23	B	2.3 Specialised Cells of Dicotyledonous Pl...	Understand
24	C	2.3 Specialised Cells of Dicotyledonous Pl...	Analyse
25	B	2.4 Specialised Cells of the Human Body	Remember
26	D	2.4 Specialised Cells of the Human Body	Remember
27	B	2.4 Specialised Cells of the Human Body	Remember
28	A	2.4 Specialised Cells of the Human Body	Understand
29	D	2.4 Specialised Cells of the Human Body	Remember
30	A	2.4 Specialised Cells of the Human Body	Remember
31	B	2.4 Specialised Cells of the Human Body	Understand
32	C	2.4 Specialised Cells of the Human Body	Remember
33	B	2.4 Specialised Cells of the Human Body	Remember
34	B	2.4 Specialised Cells of the Human Body	Understand
35	D	2.5 Adaptation of Specialised Cells to The...	Understand
36	C	2.5 Adaptation of Specialised Cells to The...	Understand
37	A	2.5 Adaptation of Specialised Cells to The...	Apply

Q	Ans	Curriculum Area	Cognitive Level
38	D	2.5 Adaptation of Specialised Cells to The...	Analyse
39	D	2.5 Adaptation of Specialised Cells to The...	Apply
40	A	2.5 Adaptation of Specialised Cells to The...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 2 (5%) | Evaluate: 1 (2%) | Remember: 18 (45%) | Understand: 15 (37%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a)** [2 marks]

- **Specialised cell** - a cell whose shape and structure have changed so that it can carry out one particular job very well (1 mark).
- **Tissue** - a group of similar cells that work together to carry out the same job (1 mark).
- **Marking note:** Accept 'a cell adapted to perform a special function'. Do not award the tissue mark for 'a part of the body' alone.*

(b) [3 marks]

- (i) **Cell -> tissue -> organ -> organ system (-> organism)** (1 mark; the order must be correct).
- (ii) Tissue: **muscle tissue**, **blood**, **nervous tissue** or **bone** (1 mark).
- (ii) Organ: **heart**, **lung**, **stomach**, **kidney**, **eye** or **brain** (1 mark).
- **Marking note:** In (ii) do not accept 'skin cell' as a tissue or 'blood cell' as an organ.*

(c) [3 marks]

- **Unspecialised cells divide** (by cell division) to produce new cells (1 mark).
- The new cells then **differentiate**, that is they change their shape, size and contents so that each can carry out one particular job (1 mark).
- In the plant this happens in the **meristems** - the tips of roots and shoots and the cambium*; in the human being the dividing cells are the **stem cells**, and similar specialised cells then group together to form tissues (1 mark).
- **Marking note:** The word differentiation, or a clear description of it, is needed for the second mark.*

(d) [2 marks]

- Each kind of cell carries out its own job **more efficiently**, because its structure suits that job (1 mark).
- The **work of the body is shared out (division of labour)**, so many different activities go on at the same time (1 mark).
- It allows the organism to **grow larger and more complex** and to survive in its surroundings (1 mark, maximum 2).
- **Marking note:** Maximum 2 marks.*

Question

(a) [3 marks]

- A - **epidermal cell** (1 mark).
- B - **guard cells** (a pair of guard cells with a stoma between them) (1 mark).
- C - **xylem vessel** (xylem) (1 mark).
 - **Marking note:** In B accept 'guard cell'. Do not award the mark for 'stoma' alone, since the stoma is the opening and not the cell.*

(b) [3 marks]

- A - **protects the tissues beneath and reduces the loss of water** from the leaf (1 mark).
- B - **opens and closes the stoma**, controlling the movement of gases into and out of the leaf and the loss of water vapour (1 mark).
- C - **carries water and dissolved mineral salts from the root to the rest of the plant** (and helps to support the plant) (1 mark).
 - **Marking note:** Only one function is required for each cell; mark the first one given.*

(c) [3 marks]

- It is a **long, narrow, hollow tube**, so water flows through it easily (1 mark).
- It has **no cross walls**, since the end walls have broken down, so the water column is not interrupted (1 mark).
- It is made of **dead, empty cells with no cytoplasm or nucleus**, leaving the whole space for water (1 mark).
- Its **walls are thickened with lignin**, which strengthens the tube and supports the plant (1 mark, maximum 3).
 - **Marking note:** Maximum 3 marks, one for each correct feature.*

(d) [3 marks]

- **Water passes into the guard cells by osmosis** and they become **turgid** (1 mark).
- Because the **inner wall is thicker and less elastic than the outer wall**, the turgid cells **curve away from each other** and the stoma opens (1 mark).
- On a hot afternoon when water is short, the guard cells **lose water, become flaccid and close the stoma**, so that less water vapour is lost from the leaf (1 mark).
 - **Marking note:** The second mark depends on the idea of the cells curving apart; a candidate who writes only 'they open' does not earn it.*

Question 3**(a)** [2 marks]

- The cambium is a **layer of thin-walled cells that keep dividing** (a meristem) (1 mark).
- It lies **between the xylem and the phloem** in the vascular bundle of the stem and the root (1 mark).
- *Marking note: Accept 'inside the bark, between the wood and the phloem'.*

(b) [3 marks]

- Epidermal cell - **forms a protective covering over the plant and reduces the loss of water** (1 mark).
- Cambium cell - **divides to produce new xylem and phloem cells, so that the stem grows thicker** (1 mark).
- Xylem vessel - **carries water and dissolved mineral salts upwards and gives support to the plant** (1 mark).
- *Marking note: Mark each row on its own. A function given for the wrong cell scores nothing.*

(c) [3 marks]

- **Water (and dissolved mineral salts) cannot reach the leaves**, so the leaves **wilt** and later dry up (1 mark).
- **Photosynthesis is reduced**, because water and minerals are short, so the tree makes less food and **grows poorly** (1 mark).
- The tree **loses support and may die** if the blockage is not cleared (1 mark).
- *Marking note: Accept 'the pods or fruits are small or do not form' as an alternative third point.*

(d) [2 marks]

- The cocoa tree, a **dicotyledonous plant**, has a **cambium** whose cells keep dividing and add **new xylem and phloem** each season, so the stem becomes thicker (1 mark).
- The maize plant, a **monocotyledon**, has **no cambium** in its vascular bundles, so no new xylem and phloem are formed and the stem does not thicken in the same way (1 mark).
- *Marking note: Both halves of the comparison are needed for full marks; one correct half earns 1 mark.*

Question 4 [Total: 10 marks]**(a)** [4 marks]

- P - **nerve cell** (neurone) (1 mark).
- Q - **red blood cell** (erythrocyte) (1 mark).
- R - **muscle cell** (muscle fibre) (1 mark).
- S - **sperm cell** (spermatozoon) (1 mark).

- *Marking note: Accept 'neurone' for P and 'muscle fibre' for R. Do not accept 'blood' alone for Q.*

(b) [3 marks]

- P - **carries messages (nerve impulses)** from one part of the body to another, for example from the brain to a muscle (1 mark).
- Q - **carries oxygen** from the lungs to all the cells of the body (1 mark).
- S - **fertilises the egg cell (ovum)** of the female (1 mark).
- *Marking note: Accept 'carries carbon dioxide away' as part of the answer for Q.*

(c) [3 marks]

- (i) It has **no nucleus**, so there is more room for the **haemoglobin** that carries the oxygen (1 mark).
- (i) Its **biconcave disc shape** gives a large surface area** for taking in and giving out oxygen; it is also small and flexible enough to pass through narrow capillaries (1 mark).
- (ii) It is **very long**, so one cell can carry a message a long way, and it has **fine branched endings** for passing the message to the next cell (1 mark).
- *Marking note: In (i) award one mark for each distinct feature with its reason. A feature without a reason earns the mark only where the reason is

Question 5 [Total: 10 marks]

(a) [3 marks]

- The red blood cell has **no nucleus** when mature; the white blood cell **has a nucleus** (1 mark).
- The red blood cell contains **haemoglobin and is red**; the white blood cell has **no haemoglobin and is colourless** (1 mark).
- The red blood cell is a **biconcave disc of fixed shape**; the white blood cell can **change its shape** to engulf germs (1 mark).
- Red blood cells are **far more numerous** than white blood cells, and they carry oxygen while white cells defend the body (1 mark, maximum 3).
- *Marking note: Maximum 3 marks. Each difference must compare the two cells.*

(b) [3 marks]

- (i) With **fewer red blood cells there is less haemoglobin**, so **less oxygen is carried** to the cells of her body (1 mark).
- (i) With less oxygen, **less energy is released during respiration** in the cells, so she feels weak and tired (1 mark).
- (ii) A named iron-rich food - **kontomire (cocoynam leaves), liver, beans, dark green leafy vegetables, fish or meat** - together with the mineral **iron** (1 mark).

- *Marking note: In (ii) the food and the mineral are both needed for the mark. Accept any locally sold iron-rich food.*

(c) [2 marks]

- Muscle cells ****contract (shorten) and so bring about movement**** of the parts of the body (1 mark).
- Nerve cells ****carry messages (impulses)**** from the sense organs to the brain and from the brain to the muscles (1 mark).
- *Marking note: Accept 'cause movement' for the muscle cell.*

(d) [2 marks]

- The ****mitochondria are the parts of the cell where respiration takes place and energy is released**** (1 mark).
- A muscle cell ****needs a large supply of energy in order to contract again and again****, so it contains many mitochondria (1 mark).
- *Marking note: The link between respiration, energy and contraction must be clear for the second mark.*

Question 6 [Total: 8 marks]

(a) [4 marks]

- Correct outline: an ****oval or pear-shaped head, a short mid-piece and a long tail****, drawn large and in clear pencil lines (1 mark).
- ****Head**** correctly labelled (1 mark; the label line must touch the head).
- ****Nucleus**** shown inside the head and labelled (1 mark).
- ****Mid-piece**** and ****tail**** correctly labelled (1 mark for both).
- *Marking note: Do not award marks for a small, sketchy or shaded drawing. Labels written without label lines may still earn the marks where the part meant is clear.*

(b) [2 marks]

- It has a ****long tail that lashes from side to side****, so that the cell can swim towards the egg (1 mark).
- Its ****mid-piece contains many mitochondria****, which release the energy needed for swimming (1 mark).
- It is ****small and streamlined****, and the ****tip of its head carries enzymes**** that help it to enter the egg (1 mark, maximum 2).
- *Marking note: Maximum 2 marks.*

(c) [2 marks]

- ****No, the pupil is wrong**** (1 mark).
- A reason supported by an example: the cells ****differ in shape and structure because each is suited to a different job**** - for example, a ****red blood cell has no nucleus and is a biconcave disc so that it can carry oxygen****, while a ****nerve cell is long and branched so that it can carry messages**** (1 mark).

- *Marking note: The first mark is for taking a clear position; it is lost where the candidate agrees with the statement. The second mark requires a named cell and the job it does.*

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:** 1: Diversity**

Understanding the Environment

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	20.1 Plant Parts and Their Functions	Remember
2	A	20.1 Plant Parts and Their Functions	Understand
3	C	20.1 Plant Parts and Their Functions	Remember
4	C	20.1 Plant Parts and Their Functions	Remember
5	A	20.1 Plant Parts and Their Functions	Understand
6	D	20.1 Plant Parts and Their Functions	Analyse
7	D	20.2 Plant Parts Used for Planting and Pro...	Remember
8	A	20.2 Plant Parts Used for Planting and Pro...	Remember
9	A	20.2 Plant Parts Used for Planting and Pro...	Remember
10	D	20.2 Plant Parts Used for Planting and Pro...	Remember
11	A	20.2 Plant Parts Used for Planting and Pro...	Remember
12	D	20.2 Plant Parts Used for Planting and Pro...	Understand
13	D	20.2 Plant Parts Used for Planting and Pro...	Analyse
14	C	20.3 Other Agricultural Uses of Plant Parts	Remember
15	B	20.3 Other Agricultural Uses of Plant Parts	Understand
16	C	20.3 Other Agricultural Uses of Plant Parts	Remember
17	A	20.3 Other Agricultural Uses of Plant Parts	Understand
18	D	20.3 Other Agricultural Uses of Plant Parts	Understand
19	B	20.3 Other Agricultural Uses of Plant Parts	Remember
20	B	20.3 Other Agricultural Uses of Plant Parts	Understand
21	A	20.3 Other Agricultural Uses of Plant Parts	Understand
22	B	20.3 Other Agricultural Uses of Plant Parts	Evaluate
23	C	20.4 Non-Agricultural Uses of Plant Parts	Remember
24	D	20.4 Non-Agricultural Uses of Plant Parts	Understand
25	A	20.4 Non-Agricultural Uses of Plant Parts	Remember
26	C	20.4 Non-Agricultural Uses of Plant Parts	Understand
27	B	20.4 Non-Agricultural Uses of Plant Parts	Analyse
28	D	20.4 Non-Agricultural Uses of Plant Parts	Understand
29	B	20.4 Non-Agricultural Uses of Plant Parts	Remember
30	A	20.4 Non-Agricultural Uses of Plant Parts	Understand
31	B	20.5 Making Agricultural Materials from Pl...	Apply
32	D	20.5 Making Agricultural Materials from Pl...	Apply
33	B	20.5 Making Agricultural Materials from Pl...	Apply
34	B	20.5 Making Agricultural Materials from Pl...	Apply
35	D	20.5 Making Agricultural Materials from Pl...	Understand
36	B	20.5 Making Agricultural Materials from Pl...	Evaluate
37	C	20.6 Making Non-Agricultural Materials and...	Apply

Q	Ans	Curriculum Area	Cognitive Level
38	A	20.6 Making Non-Agricultural Materials and...	Understand
39	C	20.6 Making Non-Agricultural Materials and...	Evaluate
40	C	20.6 Making Non-Agricultural Materials and...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 3 (7%) | Apply: 5 (12%) | Evaluate: 4 (10%) | Remember: 14 (35%) | Understand: 14 (35%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [5 marks]

- A - **root** (1 mark).
- B - **stem** (1 mark).
- C - **leaf** (1 mark).
- D - **flower** (1 mark).
- E - **fruit** (1 mark).
- *Marking note: One mark for each part; no half marks.*

(b) [3 marks]

- (i) **Absorbs water and mineral salts / anchors the plant** (1 mark).
- (ii) **Supports the plant and conducts water and food** (1 mark).
- (iii) **Makes food by photosynthesis** (1 mark).
- *Marking note: One correct function in each case is enough.*

(c) [2 marks]

- Method: **planting stem cuttings** (1 mark).
- Advantage: **the new plants are exactly like the parent and mature earlier** (accept: no seed need be bought) (1 mark).
- *Marking note: 'Cutting' alone is accepted for the method.*

Question 2 [Total: 12 marks]

(a) [4 marks]

- Cassava - **stem cuttings** (1 mark).
- Yam - **setts cut from the tuber** (1 mark).
- Plantain - **suckers** (1 mark).
- Maize - **seeds (grains)** (1 mark).
- *Marking note: The table must be copied; four correct entries earn the marks.*

(b) [4 marks]

- (i) **Maize stover and groundnut haulms** (1 mark); **dried and fed to cattle, sheep and goats, especially in the dry season** (1 mark).
- (ii) **Mucuna (a leguminous plant)** (accept leaves and crop residues for compost) (1 mark); **grown and dug into the soil as green manure, adding nitrogen and organic matter** (1 mark).
- *Marking note: The material must come from the list given.*

(c) [4 marks]

- Any FOUR, each naming a material and its use: **odum timber - building, furniture or carving**; **bamboo or palm fronds - roofing, fencing, baskets, mats or brooms**; **bark or leaves - herbal medicine**; **palm fronds - ceremonies and festivals**; **wood - charcoal for cooking**; **leaves or bark - dye for cloth** (1 mark each).
- **Marking note: No mark for a use given without the material, or the reverse.**

Question 3 [Total: 10 marks]**(a) [4 marks]**

- W - **planting (propagation) of a new cassava crop** (1 mark).
- X - **mulching the soil round the plant** (1 mark).
- Y - **housing birds or small animals, or carrying produce** (1 mark).
- Z - **a natural pesticide for controlling insect pests** (1 mark).
- **Marking note: The purpose, not merely a description of the picture, is required.**

(b) [4 marks]

- (i) Any TWO of: **reduces loss of water from the soil**; **checks the growth of weeds**; **keeps the soil cool**; **protects the soil from heavy rain and erosion**; **adds humus as it rots** (1 mark each).
- (ii) Any TWO of: **cheap and made from local material**; **leaves little harmful residue on the crop**; **safer for the farmer, the consumer and useful insects**; **always available without buying** (1 mark each).
- **Marking note: Two distinct points are needed in each part.**

(c) [2 marks]

- **Crush or pound the neem leaves and soak them in water overnight** (1 mark).
- **Strain the liquid through a cloth before pouring it into the sprayer** (accept adding a little soap to make it stick) (1 mark).
- **Marking note: The straining step is essential for the second mark.**

Question 4 [Total: 10 marks]**(a) [4 marks]**

- Roofing - **grass stems and palm fronds (leaves)** (1 mark).
- Stool - **the trunk (woody stem) of a tree** (1 mark).
- Basket - **stems, leaf strips or fibres of raffia, cane or palm** (1 mark).
- Herbal medicine - **bark, leaves or roots** (1 mark).
- **Marking note: The table must be copied; four correct entries earn the marks.**

(b) [3 marks]

- Any THREE of: **cheap and locally available**; **easily worked with simple tools and local skills**; **keep buildings cool**; **easily repaired or replaced**; **provide work and income locally**; **renewable and they rot away harmlessly** (1 mark each).

- **Marking note:** Three distinct reasons; repeated ideas score once.*

(c) [3 marks]

- (i) **They are harvested faster than they grow back - whole trees felled, bark ringed, young plants cut** (accept bush fires or clearing of land) (1 mark).

- (ii) Any TWO of: **take only mature parts and never ring the bark**; **plant and care for new seedlings or keep a protected grove**; **protect the area from bush fire**; **agree community rules or a closed season** (1 mark each).

- **Marking note:** A measure must act on the cause given in (i).*

Question 5 [Total: 10 marks]**(a)** [3 marks]

- (i) **Banana or plantain leaves** (accept palm leaf strips) (1 mark); **softened, rolled into a cup, tied at the side and filled with soil** (1 mark).

- (ii) **It rots away, so the seedling can be planted without disturbing the roots and no plastic is left in the soil** (accept: it costs nothing) (1 mark).

- **Marking note:** Any workable description of making the pot earns the second mark.*

(b) [4 marks]

- (i) Any ONE of **raffia palm, oil palm frond, sisal, baobab or kapok bark, banana or plantain stem** (1 mark).

- (ii) **Strip or beat out the long fibres** (1 mark); **soak, wash and dry them in the sun** (1 mark); **twist two or three strands together, adding fibre to lengthen the rope** (1 mark).

- **Marking note:** The three steps must be in a sensible order.*

(c) [3 marks]

- (i) **Hang a known load, heavier than the frame requires, from the rope** (1 mark); **it passes if it holds the load without breaking or stretching much** (1 mark).

- (ii) **Remake it with more or thicker strands, twist it more tightly or use a stronger fibre, and test it again** (1 mark).

- **Marking note:** A test without a stated result earns one mark only.*

Question

(a) [3 marks]

- (i) Any ONE of **raffia, straw, palm leaf, screw pine leaf** (1 mark).
- (ii) Any TWO, each with a reason: **split into strips of even width** - the weave is regular (1 mark); **dried in the sun** - the material becomes tough and will not rot or shrink (1 mark); **soaked or softened** - the strips bend without cracking (1 mark). Maximum 2 marks.
- **Marking note: No mark for a step given without its reason.**

(b) [3 marks]

- Any THREE of: **the plant and part used and its source**; **the steps followed, with a labelled drawing**; **the tools used and the time taken**; **the difficulties met and how they were solved**; **how the article performed when it was used**; **the cost and how the material will be replaced** (1 mark each).
- **Marking note: Three distinct items are required.**

(c) [2 marks]

- Reason: **felling a young tree kills it before it has produced seed, and the whole tree is lost for one small article** (1 mark).
- Instead: **take small patches of bark from a mature tree, or use fallen branches, dry fronds and leaves, and plant a seedling to replace what is taken** (1 mark).
- **Marking note: The idea of sustainable harvesting must appear in the second mark.**

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 5: Humans and the Environment | **Sub-strand:** 5: Understanding the Environment | **Class:** Basic 9 (JHS 3)

Content Standard: **B9/JHS3.5.5.1** - Demonstra

Waste Management

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	16.1 Types and Sources of Waste	Remember
2	D	16.1 Types and Sources of Waste	Understand
3	B	16.1 Types and Sources of Waste	Remember
4	A	16.1 Types and Sources of Waste	Understand
5	B	16.1 Types and Sources of Waste	Remember
6	C	16.1 Types and Sources of Waste	Understand
7	B	16.1 Types and Sources of Waste	Remember
8	D	16.1 Types and Sources of Waste	Remember
9	D	16.2 Impact of Waste on the Environment	Understand
10	A	16.2 Impact of Waste on the Environment	Understand
11	C	16.2 Impact of Waste on the Environment	Understand
12	C	16.2 Impact of Waste on the Environment	Understand
13	B	16.2 Impact of Waste on the Environment	Understand
14	A	16.2 Impact of Waste on the Environment	Analyse
15	B	16.2 Impact of Waste on the Environment	Understand
16	A	16.2 Impact of Waste on the Environment	Analyse
17	C	16.3 Scientific Methods of Waste Management	Remember
18	A	16.3 Scientific Methods of Waste Management	Understand
19	B	16.3 Scientific Methods of Waste Management	Remember
20	C	16.3 Scientific Methods of Waste Management	Understand
21	D	16.3 Scientific Methods of Waste Management	Apply
22	A	16.3 Scientific Methods of Waste Management	Apply
23	C	16.3 Scientific Methods of Waste Management	Remember
24	D	16.3 Scientific Methods of Waste Management	Apply
25	B	16.4 Scientific Principles Behind the Methods	Remember
26	C	16.4 Scientific Principles Behind the Methods	Understand
27	D	16.4 Scientific Principles Behind the Methods	Analyse
28	A	16.4 Scientific Principles Behind the Methods	Analyse
29	D	16.4 Scientific Principles Behind the Methods	Understand
30	C	16.4 Scientific Principles Behind the Methods	Analyse
31	D	16.5 Innovative Waste Management for Susta...	Apply
32	C	16.5 Innovative Waste Management for Susta...	Understand
33	A	16.5 Innovative Waste Management for Susta...	Remember
34	B	16.5 Innovative Waste Management for Susta...	Understand
35	A	16.5 Innovative Waste Management for Susta...	Apply
36	B	16.6 Waste Audits and Waste Management Pra...	Remember
37	B	16.6 Waste Audits and Waste Management Pra...	Apply

Q	Ans	Curriculum Area	Cognitive Level
38	A	16.6 Waste Audits and Waste Management Pra...	Evaluate
39	D	16.6 Waste Audits and Waste Management Pra...	Evaluate
40	D	16.6 Waste Audits and Waste Management Pra...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 6 (15%) | Evaluate: 3 (7%) | Remember: 11 (27%) | Understand: 15 (37%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [2 marks]

- (i) **Any material that is no longer wanted or needed by the person or place that produced it** (1 mark).
- (ii) Biodegradable waste **is broken down by micro-organisms**, non-biodegradable waste **is not and remains in the environment** (1 mark).
- **Marking note:** In (ii) the comparison must be made; one half earns no mark.*

(b) [4 marks]

- (i) **Cassava or plantain peels** (accept used exercise books) (1 mark).
- (ii) **Water sachets** (1 mark).
- (iii) **Dirty water from the canteen** (1 mark).
- (iv) **Old mobile phone batteries** (1 mark).
- **Marking note:** The example must come from the list given in the scenario.*

(c) [4 marks]

- Any FOUR of: **the gutter is choked and the area floods when it rains**; **stagnant water breeds mosquitoes which spread malaria**; **flies and rats are attracted and carry germs to food, spreading cholera and typhoid**; **a foul smell pollutes the air**; **liquid from the waste pollutes the stream and the underground water**; **plastics are carried into the water and choke fish and other animals** (1 mark each).
- **Marking note:** Each mark requires a stated effect; 'it is dirty' alone earns nothing.*

Question 2 [Total: 12 marks]

(a) [5 marks]

- P - **the lining (liner) of the pit** (accept clay or plastic lining) (1 mark).
- Q - **a layer of waste (refuse)** (1 mark).
- R - **the soil cover over the layer of waste** (1 mark).
- S - **the gas vent pipe** (accept pipe that lets out the gas) (1 mark).
- T - **the pipe that collects the liquid (leachate) from the waste** (1 mark).
- **Marking note:** One mark for each part; no half marks.*

(b) [4 marks]

- (i) **Stops the liquid from the waste from reaching the underground water** (1 mark).
- (ii) **Keeps off flies, rats and smell and stops the waste being blown about** (1 mark).

- (iii) **Lets the gas produced by decay escape safely** (1 mark).
- (iv) **Carries away the liquid collected at the base for treatment** (1 mark).
- **Marking note: Any one correct purpose earns the mark in each case.**

(c) [3 marks]

- Any THREE of: **the waste is covered, so flies, rats and smell are controlled**; **the lining and drain keep the liquid out of the water supply**; **the gas is let out safely instead of building up**; **the waste is not blown about or burnt in the open, so the air is cleaner**; **the land can be used again when the site is full** (1 mark each).
- **Marking note: The reason must contrast with the open dump.**

Question 3 [Total: 10 marks]

(a) [2 marks]

- (i) **Collecting, sorting, weighing and recording the waste produced by a place in order to find out its kinds and quantities** (1 mark).
- (ii) **60 kg** ($24 + 16 + 12 + 5 + 3$) (1 mark).
- **Marking note: In (ii) the unit kg is required.**

(b) [4 marks]

- (i) **$(12 \div 60) \times 100$** (1 mark) **= 20%** (1 mark).
- (ii) **$60 \div 6$** (1 mark) **= 10 kg per day** (1 mark).
- **Marking note: Award the method mark for correct working even if the arithmetic is wrong; the unit or % sign is needed for the final mark.**

(c) [2 marks]

- **Food and plant waste** (1 mark).
- Reason: it is **organic (biodegradable)** and so can be broken down by micro-organisms into manure (accept: it forms the largest part of the waste) (1 mark).
- **Marking note: The reason must refer to decay or biodegradability.**

(d) [2 marks]

- Any TWO of: **start a compost pit for the food and plant waste and use the manure on the school garden**; **place separate labelled bins and send the paper, plastics, tins and glass for recycling**; **reduce plastics by using refillable bottles instead of sachets**; **repeat the audit later to check whether the changes have worked** (1 mark each).
- **Marking note: The suggestion must be supported by the figures; 'keep the school clean' earns no mark.**

Question 4

(a) [3 marks]

- (i) **The controlled rotting of plant and animal waste by micro-organisms to form manure (compost)** (1 mark).
- (ii) Any TWO of **kitchen waste (peels)**, **grass cuttings**, **dry leaves**, **poultry manure** (1 mark each).
- **Marking note: The materials must be taken from the list given.**

(b) [4 marks]

- (i) **Bacteria and fungi** (accept micro-organisms, with earthworms and termites) (1 mark).
- (ii) **To let air (oxygen) in so that the decomposers work faster** (1 mark).
- (iii) **The micro-organisms need moisture to live and to break down the material** (1 mark).
- (iv) **Heat is released as the micro-organisms respire while breaking down the waste** (1 mark).
- **Marking note: In (iv) 'the sun heats it' is wrong and earns no mark.**

(c) [3 marks]

- (i) **Water and tight packing drove the air out, so there was no oxygen** (1 mark); **decay then took place without oxygen (anaerobic decay) and foul gases were produced** (1 mark).
- (ii) **Turn the heap and add dry material to let air in** (accept: cover the heap against rain, or make the heap less tightly packed) (1 mark).
- **Marking note: The idea of lack of air (oxygen) must appear in (i).**

Question 5 [Total: 10 marks]**(a)** [3 marks]

- (i) **Recycling** (1 mark).
- (ii) Any TWO of: **removes plastic waste from the environment, so there is less choking of gutters and less flooding**; **creates jobs and income**; **provides a useful and cheaper product**; **saves raw materials and energy** (1 mark each).
- **Marking note: Two distinct benefits are required.**

(b) [4 marks]

- (i) **Methane** (1 mark).
- (ii) **Bacteria break down the waste inside the closed digester where there is no air (anaerobic decay)** (1 mark), **and the gas released collects above the slurry and is piped off** (1 mark).
- (iii) **The slurry is used as manure (fertiliser) on the farm or garden** (1 mark).
- **Marking note: Accept 'biogas' for methane in (i) only if methane is named.**

(c) [3 marks]

- Meets present needs ****without damaging the environment for future generations**** (1 mark).
- Recycling ****saves raw materials, reduces lasting plastic waste and creates jobs**** (1 mark).
- The digester ****gives clean fuel so fewer trees are cut and less smoke is produced, and the slurry returns nutrients to the soil**** (1 mark).
- ***Marking note: The idea of the future generation must appear for the first mark.***

Question 6 [Total: 8 marks]**(a) [2 marks]**

- Any TWO of: ****regular door-to-door collection by waste companies****; ****communal containers in markets and lorry parks****; ****engineered landfill and treatment plants in some cities****; ****companies that buy plastics, scrap metal and e-waste for recycling****; ****compost plants that treat market waste****; ****clean-up days and sanitation by-laws**** (1 mark each).
- ***Marking note: The feature must be a practice, not a wish.***

(b) [3 marks]

- Any THREE of: ****the waste is not separated, so recyclable and compostable material is lost****; ****one tipping site fills up quickly and may pollute water and air****; ****households that cannot pay the fee dump waste in gutters, streams or open spaces****; ****a monthly clean-up is too seldom for waste produced daily, and the heaps collected are often not carried away****; ****the by-laws are not enforced**** (1 mark each).
- ***Marking note: Three distinct weaknesses; repeated ideas score once.***

(c) [3 marks]

- Any THREE improvements, each with a reason: ****separate labelled bins in every home**** - so recyclable material is not buried (1 mark); ****a community compost site or biogas digester**** - so food waste becomes manure or fuel and the landfill lasts longer (1 mark); ****buying-back of sorted plastics and scrap**** - so people gain by separating (1 mark); ****continuous education and enforcement of the by-laws**** - so that dumping in gutters stops (1 mark). Maximum 3 marks.
- ***Marking note: No mark for an improvement given without a reason.***

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:** 5: Human**

Integrated Science Glossary — Basic 9

This glossary contains key terms and definitions that appear throughout this book. Use it as a quick reference while studying and preparing for examinations.

Absorption

The process by which substances are taken into cells or tissues.

Acid Rain

Rain with a pH below 5.6, caused by pollution from burning fossil fuels.

Activation Energy

The minimum energy required to start a chemical reaction.

Amplitude

The maximum displacement of a wave from its rest position.

Antibiotic

A substance that kills or inhibits the growth of bacteria.

Atom

The smallest unit of an element that retains its chemical properties.

Atomic Mass

The mass of an atom, measured in atomic mass units (amu).

Atomic Number

The number of protons in the nucleus of an atom.

Barometer

An instrument used to measure atmospheric pressure.

Biomass

Organic material from plants and animals used as fuel.

Biotechnology

The use of living organisms or their products to make or modify useful products.

Capacitor

A device that stores electric charge.

Carbon Cycle

The circulation of carbon through the atmosphere, oceans, and living organisms.

Cell Division

The process by which a cell divides to form two or more cells.

Chemical Bond

The force that holds atoms together in a compound.

Chemical Change

A change that produces new substances with different properties.

Climate Change

Long-term changes in global or regional climate patterns.

Combustion

A chemical reaction between a fuel and oxygen that releases energy.

Conservation of Energy

The principle that energy cannot be created or destroyed, only transformed.

Conservation of Mass

The principle that mass is neither created nor destroyed in a chemical reaction.

Constant

A quantity that does not change in an experiment.

Control

A standard of comparison in an experiment.

Covalent Bond

A chemical bond formed by sharing electrons between atoms.

Decomposition

The breakdown of a complex substance into simpler substances.

Deforestation

The clearing of forests for other uses such as agriculture or development.

Diode

An electronic component that allows current to flow in only one direction.

DNA

Deoxyribonucleic acid, the molecule that carries genetic information.

Doppler Effect

The change in frequency of a wave as the source and observer move relative to each other.

Ecology

The study of interactions between organisms and their environment.

Electromagnetic Induction

The production of electricity by moving a conductor through a magnetic field.

Electromagnet

A magnet created by passing electric current through a coil of wire.

Electroplating

The process of coating an object with a thin layer of metal using electricity.

Emission

The release of substances into the atmosphere.

Endothermic

A chemical reaction that absorbs energy from its surroundings.

Energy Efficiency

The ratio of useful energy output to total energy input.

Environmental Impact

The effect that human activities have on the environment.

Equilibrium

A state of balance where opposing forces or processes are equal.

Exothermic

A chemical reaction that releases energy to its surroundings.

Fertilization

The fusion of male and female gametes to form a zygote.

Fossil Fuel

Fuel formed from the remains of ancient plants and animals.

Galvanometer

An instrument used to detect and measure small electric currents.

Gene

A unit of heredity that determines a specific characteristic.

Genetic Engineering

The direct manipulation of an organism's genes using biotechnology.

Geothermal Energy

Energy from heat within the Earth.

Greenhouse Gas

A gas that traps heat in the atmosphere, such as CO₂, CH₄, or N₂O.

Half-life

The time taken for half the atoms in a radioactive sample to decay.

Heredity

The transmission of characteristics from parents to offspring.

Hormone

A chemical messenger produced by glands and transported in the blood.

Hydrocarbon

A compound containing only hydrogen and carbon atoms.

Hydroelectric Power

Electricity generated using the energy of falling or flowing water.

Hypothesis

A testable explanation for an observation or phenomenon.

Indicator

A substance that changes color to show whether a solution is acidic or basic.

Inertia

The tendency of an object to resist changes in its state of motion.

Infrared

Electromagnetic radiation with wavelengths longer than visible light.

Innovation

The introduction of new ideas, methods, or products.

Ion

An atom or molecule with a net electric charge.

Ionic Bond

A chemical bond formed by the transfer of electrons between atoms.

Isotope

Atoms of the same element with different numbers of neutrons.

Joule

The unit of energy. One joule equals one newton-meter.

Junction

A point where two or more conductors meet in an electrical circuit.

Kinetic Energy

The energy an object possesses due to its motion.

Laser

Light Amplification by Stimulated Emission of Radiation.

Latitude

The angular distance north or south of the equator.

Lens

A transparent object that refracts light to form an image.

Longitude

The angular distance east or west of the prime meridian.

Magnetic Pole

The region at each end of a magnet where the magnetic force is strongest.

Magnification

The ratio of the size of an image to the size of the object.

Mass Number

The total number of protons and neutrons in an atom's nucleus.

Meiosis

Cell division that produces gametes with half the normal number of chromosomes.

Metamorphic Rock

Rock that has been changed by heat and pressure.

Mitosis

Cell division that produces two identical daughter cells.

Momentum

The product of an object's mass and velocity.

Mutation

A change in the DNA sequence of a gene or chromosome.

Natural Selection

The process by which organisms better adapted to their environment tend to survive and reproduce.

Neutralization

The reaction between an acid and a base to form a salt and water.

Neutron

An uncharged particle in the nucleus of an atom.

Newton's First Law

An object at rest stays at rest, and an object in motion stays in motion, unless acted on by a force.

Newton's Second Law

Force equals mass times acceleration ($F = ma$).

Newton's Third Law

For every action, there is an equal and opposite reaction.

Nuclear Energy

Energy released by nuclear reactions such as fission or fusion.

Nucleus

The central part of an atom containing protons and neutrons.

Ohm's Law

The current through a conductor is directly proportional to the voltage ($V = IR$).

Organ

A group of tissues that work together to perform a specific function.

Organic Compound

A compound containing carbon, usually bonded to hydrogen.

Osmosis

The movement of water molecules through a selectively permeable membrane.

Oxidation-Reduction

A chemical reaction involving the transfer of electrons.

Parallel Circuit

A circuit in which components are connected across the same voltage.

Period

The time for one complete cycle of a wave or oscillation.

Periodic Table

A table of elements arranged by atomic number and chemical properties.

pH

A measure of the acidity or basicity of a solution, from 0 to 14.

Phenotype

The observable characteristics of an organism.

Photosynthesis

The process by which plants convert light energy into chemical energy.

Physical Change

A change in form or appearance without changing chemical composition.

Plasma

The liquid component of blood in which cells are suspended.

Plate Tectonics

The theory that Earth's outer shell is divided into plates that move over the mantle.

Pollutant

A substance that contaminates the environment.

Polypeptide

A chain of amino acids that forms a protein.

Potential Energy

Energy stored due to an object's position or state.

Power

The rate at which work is done or energy is transferred.

Precipitation

Water falling from clouds as rain, snow, sleet, or hail.

Predator

An animal that hunts and eats other animals.

Pressure

Force per unit area, measured in pascals (Pa).

Prey

An animal that is hunted and eaten by another animal.

Prism

A transparent object that separates white light into its component colors.

Proton

A positively charged particle in the nucleus of an atom.

Radioactivity

The spontaneous emission of radiation from unstable atomic nuclei.

Recycling

The process of converting waste into reusable material.

Reduction

A chemical reaction involving the gain of electrons.

Reflection

The bouncing back of light or other waves from a surface.

Refraction

The bending of light as it passes from one medium to another.

Renewable Energy

Energy from sources that are naturally replenished.

Reproduction

The biological process by which organisms produce offspring.

Resistance

The opposition to the flow of electric current.

Resistor

A component that resists the flow of electric current.

Resonance

The amplification of vibrations when the frequency matches the natural frequency.

Satellite

An object that orbits a planet or other celestial body.

Scientific Method

A systematic approach to investigating phenomena and acquiring knowledge.

Semiconductor

A material with electrical conductivity between that of a conductor and an insulator.

Series Circuit

A circuit in which components are connected end-to-end.

Solar Cell

A device that converts sunlight directly into electricity.

Solar System

The sun and all the objects that orbit it, including planets, moons, and asteroids.

Solute

The substance dissolved in a solution.

Solvent

The substance that dissolves the solute in a solution.

Spectrum

The range of colors produced when light is dispersed.

Speed of Light

The speed at which light travels in a vacuum, approximately 300,000 km/s.

Stem Cell

An undifferentiated cell that can develop into specialized cell types.

Sustainable Development

Development that meets present needs without compromising future generations.

Symbiosis

A close relationship between two different species.

Technology

The application of scientific knowledge for practical purposes.

Thermal Equilibrium

The state when two objects in contact reach the same temperature.

Thermodynamics

The study of heat and its relationship to energy and work.

Tidal Energy

Energy generated from the movement of ocean tides.

Transformer

A device that changes the voltage of alternating current.

Transistor

A semiconductor device used to amplify or switch electronic signals.

Ultraviolet

Electromagnetic radiation with wavelengths shorter than visible light.

Variable

A factor that can change in an experiment.

Velocity

Speed in a specific direction.

Voltage

The electrical potential difference between two points.

Watt

The unit of power. One watt equals one joule per second.

Wave

A disturbance that transfers energy through a medium or space.

Wind Energy

Energy generated from the movement of air.

Work

The transfer of energy when a force moves an object.

Curriculum Reference

This book aligns with the following curriculum standards:

- Ghana Education Service (GES)
Standards-Based Curriculum for Junior High Schools (2019)
- National Council for Curriculum and Assessment (NaCCA)
Curriculum Framework for Basic Education
- West African Examinations Council (WAEC)
BECE Syllabus for Integrated Science / Mathematics

The content has been reviewed to ensure alignment with:

- Strand and sub-strand organization
- Learning indicators and objectives
- Cognitive levels (Knowledge, Understanding, Application, Analysis)
- Core competencies (Critical Thinking, Problem Solving, Communication)

Teachers and students should use this book in conjunction with the official curriculum documents available from the Ghana Education Service.

For more information:

Ghana Education Service: www.ges.gov.gh

NaCCA: www.nacca.gov.gh

WAEC: www.waec.org

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Beacon Educational Consult is a Ghanaian educational publishing company dedicated to producing high-quality learning materials for Junior High School students across Ghana.

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This image shows a full page of blank, lined paper. It features approximately 20 evenly spaced horizontal grey lines across its entire width, providing a guide for writing. The paper itself is a clean, off-white color, and there are no margins, text, or other markings present.

This image shows a full page of blank, lined paper. It features approximately 20 evenly spaced horizontal grey lines across the entire width of the page, typical of notebook or school paper. There are no margins, text, or other markings present.